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Technical Report No. 22

Process Simulation for Aseptically Filled Products

Revised 2025

PDA Parenteral Drug Association

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Revised 2025

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Table of Contents

1.0 Introduction **1**
1.1 Purpose 1
1.1 Scope 2
2.0 Glossary **3**
2.1 Abbreviations 8
3.0 Aseptic Process Simulation Concepts and Principles **9**
3.1 Prerequisites 9
3.2 Number and Frequency of Simulations 9
3.3 Worst-Case Scenarios 10
3.4 Risk Assessment for Aseptic Process Simulation Design 11
3.5 Ongoing Evaluation 12
4.0 Overview of Aseptic Process Simulation for Aseptically Produced Products **14**
4.1 General Requirements 14
4.1.1 Scope of Aseptic Process Simulation 14
4.1.2 Design of Aseptic Process Simulation 14
4.1.3 Interventions 15
4.1.4 Sterile Bulk Aseptic Process Simulation 21
4.2 Aseptic Process Simulation Design with Regard to Products 21
4.2.1 Solutions 22
4.2.2 Lyophilized Products 22
4.2.3 Suspensions 24
4.2.4 Semisolids—Ointments, Creams, and Gels 25
4.2.5 Powders 26
4.2.6 Other Dosage Forms and Drug/Device Combinations 30
4.3 Aseptic Process Simulation Design Based on Aseptic Processing Technologies 32

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4.3.1 Form-Fill-Seal and Blow-Fill-Seal 32
4.3.2 Restricted Access Barrier Systems 33
4.3.3 Isolator Technology 33
4.3.4 Closed Systems 34
4.3.5 Single-Use Systems Technologies 35
4.3.6 Pre-Use Post-Sterilization Integrity Test Assembly 35
4.3.7 Considerations for Advanced Therapy Medicinal Products 35

5.0 Aseptic Process Simulation Documentation 38
5.1 Process Definition 38
5.2 Protocol and Procedure Preparation 38
5.3 Aseptic Process Simulation Execution Record 39
5.4 Final Report 40
5.5 Aseptic Process Simulation Observation 40
5.5.1 Video Recording of Aseptic Process Simulations 41

6.0 Environmental Monitoring 43

7.0 Elements of Aseptic Process Simulations 44
7.1 Facility and Filling Machine Considerations 44
7.2 Equipment Setup 44
7.3 Media Selection and Preparation 45
7.4 Inert Gassing 45
7.5 Container-Closure Configuration 46
7.5.1 Inclusion of Multiple Container-Closure Sizes 46
7.5.2 Pigmented or Limited Observation Containers 48
7.5.3 Components from Varying Suppliers and of Different Designs and Compositions 49
7.6 Filling Speed 49
7.7 Fill Volume 49
7.8 Interventions 49
7.9 Duration of Aseptic Process Simulations 50
7.9.1 Aseptic Process Validation Limitations 51
7.9.2 Human Performance and Fatigue 51
7.9.3 Environmental and Equipment Considerations 51
7.9.4 Qualification of Maximum Hold Times 51

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7.10 Number of Units 52
7.11 Means to Achieve Full or Longer Duration APS Fills 52
7.12 Campaign Operations 54
7.13 Pre-Incubation Container Inspection 54
7.14 Incubation Conditions 55
7.15 Post-Incubation Inspection 55
7.16 Unit Accountability and Reconciliation 56
7.17 Growth Promotion/Positive Control 56
7.17.1 Sample Collection 57
7.17.2 Method Criteria 57
7.17.3 Acceptance Criteria 57
7.18 Post-Simulation Cleaning 57

8.0 Personnel Qualification 58
8.1 Qualification Level 1 59
8.2 Qualification Level 2 60
8.3 Qualification Level 3 60
8.4 Ongoing Qualification 60

9.0 Aseptic Processing Simulation Acceptance Criteria 61
9.1 Recommendations 61
9.2 Invalidating an Aseptic Process Simulation Run 62
9.3 Aborting or Discontinuing an Aseptic Process Simulation 62

10.0 Considerations for Investigation 64

11.0 The Use of Aseptic Process Simulation in Ongoing Process Evaluation 67
11.1 The Role of Aseptic Process Simulation in Ongoing Process Evaluation 67
11.2 Periodic (Time-Driven) Ongoing Aseptic Process Simulation Assessment 67
11.3 Event-Driven and For-Cause Ongoing Aseptic Process Simulation Assessment 68
11.4 Other Considerations 68

12.0 References 69

13.0 Appendix 1: Media and Placebo Considerations 71
13.1 Selection of Placebo or Surrogate Material 71
13.2 Sterilization of Placebo or Surrogate Material 71

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13.3 Growth-Promotion Testing of the Placebo or Surrogate Material 71

13.4 Solubility Testing of the Placebo or Surrogate Material 72

13.5 Media Preparation and Sterilization 72

14.0 Appendix 2: Aseptic Process Simulation Performance Sequence 74

15.0 Appendix 3: Practical Examples of Using Risk Assessment in Aseptic Process Simulation..... 81

15.1 Example 1: Applying Risk-Based Aseptic Process Simulation Frequency 81

15.2 Example 2: Intervention Simulation Frequency Based on Application of Intervention Risk Evaluation & Management Model 86

15.3 Example 3: Use of QRM to Define Inclusion and Frequency of Interventions During Aseptic Process Simulation 90

15.4 Example 4: Use of Quality Risk Management to Define Inclusion and Frequency of Interventions during Aseptic Process Simulation 94

15.5 Example 5: Aseptic Process Simulation Design for Lines Producing Single Dosage Form with Different Fill Volume 98

15.6 Example 6: Aseptic Process Simulation Design for Lines Producing Different Products or Dosage Forms 99

15.7 Example 7: Campaign Strategy Example for an Isolator Line 102

15.8 Example 8: Design of the Aseptic Process Simulation for an Advanced Therapy Medicinal Product 105

16.0 Appendices References 114

17.0 Relevant Vendor and Supplier Resources 115

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Figures and Tables Index

Table 4.2.5.1-1 Advantages and Disadvantages of Liquid Medium Filled by Powder Filling Equipment **28**

Table 4.2.5.2-1 Advantages and Disadvantages of Liquid Media Filled on Dry-Powder Filler with Supplementary Liquid-Fill Capability **29**

Table 4.2.5.3-1 Advantages and Disadvantages of On-Line Liquid Media Fill Followed by On-Line Placebo Powder Fill **29**

Table 4.2.5.4-1 On-Line Placebo Powder Fill Followed by On-Line Liquid Media Fill **30**

Table 4.2.5.5-1 Advantages and Disadvantages of Dry Media Filled as a Powder and Reconstituted to a Liquid Form After the Fill **30**

Table 7.5.1-1 Potential Risks Associated with Various Container-Closure Characteristics **47**

Figure 8.0-1 APS Personnel Qualification Criteria **59**

Table 14.0-1 Aseptic Process Simulation Performance Sequence Checklist **74**

Table 15.1-1 Applying Risk-Based APS Frequency **83**

Table 15.2-1 Intervention Simulation Frequency-Based on Application of Intervention Risk Evaluation & Management Model **87**

Table 15.3-1 Use of Intervention Risk Criteria with Scores to Define the Overall Intervention Risk Level (Example 3) **91**

Table 15.3-2 Recommended Frequencies and Number of Interventions for Each Risk Level (Example 3) **92**

Table 15.3-3 Risk Assessment of Possible Interventions for Vial-Filling in Restricted Access Barrier Systems without Built-In In-Process Control (Example 3) **92**

Table 15.4-1 Use of Intervention Risk Criteria with Scores to Define the Overall Intervention Risk Level (Example 4) **95**

Table 15.4-2 Recommended Frequencies and Number of Interventions for Each Risk Level (Example 4) **96**

Table 15.4-3 Risk Assessment of Possible Interventions for a Vial-Filling in Open Restricted Access Barrier Systems without Built-In In-Process Control (Example 4) **96**

Table 15.5-1 APS Design for Lines Producing Different Products/Dosage Forms (No Worst-Case Vial) **98**

Table 15.5-2 Aseptic Process Simulation Design for Lines Producing Different Products/Dosage Forms (Worst-Case Vial) **98**

Table 15.6-1 Assessing Number of Aseptic Process Simulation Requirements for a Line Used to Fill Different Dosage Forms **100**

Figure 15.7-1 Initial APS Validation: Conservative Approach **103**

Figure 15.7-2 Initial APS Validation: Risk-Based Approach **103**

Figure 15.7-3 Periodic APS Revalidation Setup Once the Full-Validation Criteria are Met **105**

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Figure 15.7-4 Process Flow Diagram of the Major Unit Operations of the Manufacturing Process **106**

Figure 15.7-5 Aseptic Risk Evaluation Model – Steps in the Risk Assessment Process **107**

Table 15.7-1 Definition of the Aseptic Manipulation Factors **108**

Table 15.7-2 Identification of the Risk Ranking for Each of the Factors **108**

Table 15.7-3 Preliminary Matrix Used in Determination of Initial Risk Class **109**

Table 15.7-4 Secondary Matrix for Determination of the Overall Risk Score **109**

Table 15.7-5 Action Level Table based on Overall Risk Score **110**

Table 15.7-6 Aseptic Risk Evaluation Method for the GMCT-A Process (Example: Unit Operation of Viral Vector Transduction) **111**

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1.0 Introduction

This document replaces the previous revision of PDA *Technical Report No. 22: Process Simulation Testing for Aseptically Filled Products* published in 2011 and represents a significant update of the content. The intent of the current effort is to provide updates that reflect changes in aseptic processing that have occurred within the global industry and regulatory landscape and, specifically, to support a manufacturing site's holistic contamination control strategy (CCS). The authors have attempted to address the subject as fully as possible, recognizing the notable contributions made by other organizations, regulators, compendia, and individuals who have worked in this area. In addition, the report provides guidance on and supports sound risk-based approaches in the design and execution of aseptic process simulation (APS).

Insights from the discussion between industry experts and regulatory agencies on the latest revision of the European Union (EU) GMP *Annex 1: Manufacture of Sterile Medicinal Products* (issued August 25, 2022)[1] concerning APS have been considered in this document to present the current thinking and guidance on this topic.

1.1 Purpose

PDA *Technical Report No. 22: Process Simulation Testing for Aseptically Filled Products (Revised 2025)* (TR 22) should be considered a guide; it is not intended to establish any mandatory or implied standard. Readers should recognize that there may be additional requirements imposed because of new or localized regulatory expectations that are not included in this document. This technical report does not provide a universally appropriate template for the execution of APS studies. Each company must determine the appropriate rationale and approaches applicable to their unique operations. While this document presents an approach that can be widely used, it is not possible to ensure that it meets every specific requirement from every country. Companies need to ensure that any country-specific requirements are addressed for their products.

The recurring theme in this report is the consideration of risk to product sterility and patient safety as criteria for the design of APS studies. Regulatory authorities have issued guidelines and regulations for aseptic process study design, and companies should be aware of and account for these requirements when planning their studies. The use of relative risk and scientific evaluation can provide useful information to help inform design decisions and are essential elements to improving understanding of the aseptic process and its capabilities. Validation of an aseptic process is an exhaustive process that includes at a minimum appropriate process design, equipment qualification, adherence to an adequate pharmaceutical quality system, well-defined process controls, process monitoring and data evaluation, personnel training and includes the APS. The APS alone should not be considered as the primary means to validate the aseptic

[1] The PIC/S GMP Annex 1 is identical to the EU Annex 1 and, hereafter, EU Annex 1 means EU-PIC/S Annex 1.

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process or aspects of the aseptic process; it is simply a final demonstration of the sum of the controls. The effectiveness of the aseptic process should be determined through process design, adherence to the pharmaceutical quality system and process controls, training, and evaluation of monitoring data ***(1)***.

1.1 Scope

This technical report addresses process capability assessment for aseptic processing. Such assessments consist of one or more APS during pharmaceutical and biopharmaceutical aseptic formulation and filling activities of products for human and veterinary use. Aseptic operations required in the preparation of sterile bulk materials and biotechnology inoculums and feed materials are not a part of this document.

While the focus of this document is on aseptic processing in the pharmaceutical and biopharmaceutical industry, application of the concepts and principles to the preparation of sterile medical devices and diagnostics may be appropriate. Readers are encouraged to review PDA's *Points to Consider No 1: Aseptic Processing (Revised 2023)* for a more detailed discussion on aseptic processing related topics that could have an impact on the APS exercise.

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2.0 Glossary

Definitions have been provided to help clarify the concepts discussed in this document. While some definitions used may vary among companies, geographic location, etc., the definitions described below are consistently used within this document. Where a definition is based on another published source, the source is cited.

Action Limit (Environmental Monitoring)

An established relevant measure (e.g., microbial or airborne particle limits) that, when exceeded, should trigger appropriate investigation and corrective action based on the investigation.

Aerobic Microorganism

A microorganism that will grow primarily in the presence of oxygen.

Note: For the purpose of this technical report, this definition encompasses facultative anaerobes.

Alert Level (Environmental Monitoring)

Established microbial or total particulate values giving early warning of potential drift from normal operating conditions, which is not necessarily grounds for definitive corrective action but typically requires follow-up action to address the potential problem. Alert levels are established based on routine and qualification trend data and are periodically reviewed. The alert level can be based on a number of parameters including adverse trends, individual excursions above a set value, and repeat events.

Anaerobic Microorganism

A microorganism that will grow only in the absence of oxygen.

Aseptic Filling

The part of aseptic processing where a presterilized or aseptically manufactured product is filled and sealed into sterile containers in an environment of suitable quality, such as a cleanroom (with appropriate physical separation), a restricted access barrier system (RABS), isolator, or equivalent.

Aseptic Processing

Handling of sterile products and materials in a controlled environment, in which the air supply, facility, materials, equipment and personnel are regulated to prevent microbial, endotoxin/pyrogen and particulate contamination ***(1)***.

Note: Manual aseptic processing is when aseptic processing is performed manually by an operator.

Aseptic Processing Area (APA)

A controlled environment, consisting of several zones, in which the air supply, facility, materials, equipment, and personnel are regulated to control microbial and particulate contamination.

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Aseptic Process Simulation (APS)

A simulation of the entire aseptic manufacturing process in order to demonstrate the capability of the process to assure product sterility. Includes all aseptic operations associated with routine manufacturing, e.g., equipment assembly, formulation, filling, lyophilization, and sealing processes, as necessary.

Note: Also referred to as "Media Fill".

Bioburden

The number of viable microorganisms associated with a specific item such as personnel, manufacturing environments (air and surfaces), equipment, product packaging, raw materials (including water), in-process materials, or finished products.

Bracket

A demonstration of process performance at two different values of a given parameter (e.g., extremes of container closure size, line speed, etc.) allowing the use of any values of that parameter falling within this range.

Campaign

Campaigns are the manufacture of a series of batches of the same product (or product family) in sequence in a given period, with strict adherence to established and validated control measures but without the need for cleaning, disinfection, or decontamination of the Grade A environment.

Change Control

A formal program that describes evaluation and actions to be taken if a change to facilities, materials, equipment, and/or processes used in the fabrication, packaging, and testing of drugs, is proposed or completed that may affect the operation of the quality or support systems.

Cleaning

A process for removing contamination (e.g., product residues or disinfectant residues).

Closed System

A system used for the manufacture of sterile products or the handling of sterile materials that maintains physical separation and microbiological integrity, required to isolate and prevent exposure of the product, materials, or interior surfaces from the surrounding environment and operators, throughout the use of the system.

Colony Forming Unit (CFU)

A microbiological term that describes a single detectable colony that originates from one or more microorganisms. Colony forming units are typically expressed as CFU per ml for liquid samples, CFU per m^3 for active air samples, CFU/time exposure for passive air samples and CFU per sample for touch samples captured on a solid medium (whether settle or contact plates).

Compounding

A process in which a bulk drug substance is combined with one or more excipients and/or another bulk drug substance to produce a drug product.

Note: Commonly referred to as "formulation".

Contaminated Unit

Unit filled in an APS that exhibits presence of microbial growth after incubation.

Contamination Control Strategy (CCS)

A summary of the processes and measures for identification, assessment, control, and monitoring of contamination risks that include microorganisms, pyrogens/endotoxins, and foreign particulates derived from current product and process understanding, that assures process performance and product quality.

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Contamination Rate

The percentage of units filled in an aseptic process simulation that are positive for microbial growth after incubation.

Critical Intervention

An intervention (corrective or inherent) into the critical zone ***(1)***.

Critical Zone

A location within aseptic processing area in which product and critical surfaces are exposed to the environment.

Disinfection

The process by which the reduction of the number of microorganisms is achieved by the irreversible action of a product on their structure or metabolism, to a level deemed to be appropriate for a defined purpose.

Dynamic (Operational) Environmental Monitoring

The condition where production operations are actively occurring as defined by the individual firm. Personnel may or may not be present at this time ***(2)***.

Environmental Monitoring Program

Describes the processes and activities that need to take place to characterize and monitor the quality of the environment.

Facility Isolates

Suitably representative microorganisms of a site that are frequently recovered through environmental monitoring within the classified zone/areas, including Grade A and Grade B areas, personnel monitoring; utilities; in process bioburden, or positive sterility test results.

First Air

Refers to air exiting a HEPA filter in a unidirectional airstream, that is essentially particle free and has not been interrupted (potential to add contamination) prior to reaching exposed product and product contact surfaces.

Grade A Air Supply

Air which is passed through a filter qualified as capable of producing grade A total particle quality air, but where there is no requirement to perform continuous total particle monitoring or meet grade A viable monitoring limits. Specifically used for the protection of fully stoppered vials where the cap has not yet been crimped ***(1)***.

Growth Promotion Test

A test performed on the growth media to demonstrate that it will support microbial growth for the selected challenge microorganisms.

Integrity Test

Test to determine the integrity of a system supposed to be integral, e.g. sterilizing grade filters (incl. PUPSIT), SUS, container-closure systems, barrier system gloves.

Intervention

Activities, manipulations, and tasks in Grade A during aseptic operations.

Corrective Intervention

An intervention that is performed to correct or adjust an aseptic process during its execution. Examples include activities such as: clearing component misfeed, adjusting sensors, and replacing equipment components. These may not occur at a set frequency in the routine aseptic process.

Inherent Intervention

An intervention that is an integral part of the aseptic process and is required for setup or routine operation and/or monitoring, for example, aseptic assembly, container replenishment, or environmental sampling.

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Inherent interventions are required by batch record, procedure, or work instruction for the proper conduct of the aseptic process.

Intervention Risk Evaluation & Management Model (IREM)

Risk assessment method used to identify, evaluate, and rank aseptic process interventions in an objective and logical manner.

ISO 5

Environmental operating conditions defined in *ISO 14644-1:2015 Cleanrooms and Associated Controlled Environments — Part 1: Classification of air Cleanliness by Particle Concentration* ***(3)***.

Note: For total particulates, International Organization for Standardization (ISO) 5 is roughly comparable to Grade A (total particulates limits) as defined in European Commission: *Manufacture of Sterile Medicinal Products, EurdraLex-Volume4-EU Guidelines for Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use*, 2022. The ISO classification as described in ISO 14644-1:2015 is not applicable for viable particulate measures.

Isolator

A contained, decontaminated environment meeting Grade A/ISO 5 conditions used for aseptic process manufacturing, that provides an uncompromised, continuous isolation of its interior from the external environment. Once decontaminated by a validated cycle, an isolator prevents the microbiological contamination of sterile products and product contact surfaces of the interior by enclosures and the supply of continuous, controlled overpressure of HEPA-filtered air ***(4)***.

Note: Isolators are typically categorized as being "closed" or "open".

Closed Isolator Systems

Excludes external contamination from the isolator's interior by transfer material via aseptic connection to auxiliary equipment, rather than using openings to the surrounding environment. Closed systems remain sealed throughout operations ***(4)***.

Open Isolator Systems

Designed to allow for the continuous or semicontinuous ingress and/or egress of materials during operations through one or more openings. Openings are engineered (e.g., using continuous overpressure) to exclude the entry of external contamination into the isolator ***(4)***.

Matrix

Describes the key variables that need to be considered in the APS program, it captures all products produced on a line and clearly shows how the overall APS program challenges each of those key variables.

Media Fill

See Aseptic Processing Simulation.

Observer

Individuals designated by the Quality Unit to observe, record, and address the performance and activities of the APS and routine aseptic operations.

Process Validation

The collection and evaluation of data, from the process design stage through commercial production, which establishes scientific evidence that a process is capable of consistently delivering quality product ***(5)***.

Quality Unit

A group organized within an organization to promote quality in general practice ***(6)***.

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Restricted Access Barrier System (RABS)

System that provides an enclosed, but not fully sealed, environment meeting defined air quality conditions (for aseptic processing grade A), and using a rigid-wall enclosure and integrated gloves to separate its interior from the surrounding cleanroom environment. The inner surfaces of the RABS are disinfected and decontaminated with a sporicidal agent. Operators use gloves, half-suits, rapid transfer ports (RTPs), and other integrated transfer ports to perform manipulations or convey materials to the interior of the RABS. Depending on the design, doors are rarely opened, and only under strictly pre-defined conditions ***(1)***.

Closed RABS (cRABS)

A RABS for which the barrier prevents air exchange with the surrounding environment. Closed RABS uses a dedicated unidirectional air flow unit (UDAF) system for air supply. Unlike a closed isolator, a closed RABS typically has mouseholes and commonly has no automated biodecontamination system.

Open RABS (oRABS)

A RABS for which the barrier permits air exchange with the surrounding environment. Passive Open RABS uses the room HVAC for supply. Active Open RABS uses a dedicated unidirectional air flow (UDAF) system for air supply but exchanges the exhaust with the room.

Note: Although Open RABS cannot be automatically be biodecontaminated, there are some installations where the cleanroom, and all the equipment it contains including the open RABS, is biodecontaminated, but this is a special case.

Sampling Frequency

Established period for collecting samples.

Shift

Scheduled periods of work, staffed by alternating groups of workers.

Single-Use Systems (SUS)

Systems in which product-contact components are used only once to replace reusable equipment such as stainless-steel transfer lines or bulk containers. SUS covered in this document are those that are used in the manufacturing processes of sterile products and are typically made up of disposable components such as bags, filters, tubing, connectors, storage bottles, and sensors.

Sterile

A condition of being free from living, viable microorganisms.

Sterility Test

Test performed to determine the presence or absence of viable microorganisms in a representative set of finished product samples.

Sterilization

A process used to render a system free of viable microorganisms with a specified probability ***(7)***.

Worst-Case

A set of conditions encompassing upper and lower processing limits and circumstances, within standard operating parameters, that pose the greatest chance of process or product failure (when compared with ideal conditions). Such conditions have the highest potential to, but do not necessarily always, result in product or process failure.

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2.1 Abbreviations

APA	Aseptic Processing Area
APS	Aseptic Processing Simulation
ATMP	Advanced Therapy Medicinal Product
BFS	Blow-Fill-Seal
BSC	Biosafety Cabinet
CAPA	Corrective and Preventive Actions
CCS	Contamination Control Strategy
CIP	Clean-in-Place
EM	Environmental Monitoring
FFS	Form-Fill-Seal
HVAC	Heating Ventilation and Air Conditioning
OOS	Out-of-Specification
QRM	Quality Risk Management
RABS	Restricted Access Barrier System
RTP	Rapid Transfer Port
SCDM*	Soybean-Casein Digest Medium
SIP	Sterilize-in-Place
SOP	Standard Operating Procedure
SUS	Single Use Systems
TSB*	Tryptic Soy Broth
WFI	Water for Injection

* **Note:** SCDM and TSB are equivalent.

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3.0 Aseptic Process Simulation Concepts and Principles

The validation of an aseptic process should include simulation using a microbial growth media and where necessary the use of a surrogate/placebo in place of the product (e.g., for powder filling processes). An APS normally includes exposing the microbiological growth medium to product-contact surfaces of equipment, container–closure systems, critical areas, as well as process interventions and manufacturing environment to closely simulate the same exposure that the product itself will undergo. Results are then interpreted (i.e., whether the filled media after incubation indicate absence or presence of microbial growth). CCS should be developed and implemented before executing an APS.

3.1 Prerequisites

For a new facility or production process, APSs are performed as part of the overall validation activities. Initial APSs are generally conducted after completion of:

- Equipment qualification
- Facility, heating, ventilation, and air conditioning (HVAC), and utilities qualification
- Sterilization process validation
- Implementation of environmental decontamination procedures (cleaning, disinfection and Isolator decontamination)
- Personnel training and gowning certification
- Environmental monitoring (EM) program should be established
- Standard operating procedures (SOPs) are available and personnel is trained on the applicable procedures
- Aseptic process and associated intervention risk assessment

Prior to performing APS; engineering batches and/or water trials should be performed to understand any handling issues and the interventions required for the process.

3.2 Number and Frequency of Simulations

The number and frequency of APS performed should be based on an assessment of the risks posed by the process. The initial APS supports that a new facility, line, or process is operating under the desired state of control. There is a regulatory expectation that at least three consecutive successful APSs are performed when qualifying a new facility, filling line, or process ***(1, 4)***. Using quality risk management (QRM) risk-based approaches is important as the exact number may depend on the complexity and variety of processes and may require more than 3.

While it is allowed by regulations to combine dynamic environmental qualification with APS, this is not recommended for new facilities or in general when the reliability of the controls to ensure proper environmental conditions during operation is not yet known. APS (periodic requalification) should be

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repeated twice per year (approximately every six months) for each aseptic process, each filling line, and each shift. Each operator should participate in at least one successful APS annually. Consideration should be given to performing an APS after the last batch prior to shut down, before long periods of inactivity, or before decommissioning or relocation of a line ***(1, 4).*** Additional APSs may be required based on a risk assessment to help evaluate any major changes to procedures, practices, or equipment configuration. (See **Section 11.0**) Based on risk assessment, flexibility in APS design (e.g., duration, etc.) may be appropriate for barrier systems that offer robust separation (built in by design) and a consistently heightened level of product protection; however, the underlying principles for aseptic manufacturing remain the same. (See **Section 4.3.3**)

For lines producing different container sizes, the total run number may be decided by risk assessment, bracketing, and matrix approaches. Care should be taken to make sure all container sizes and all container–closure configurations are considered in the risk assessment. It is also required to have an equivalent process and the same container–closure configuration in order to apply bracketing or matrix approaches. (See **Section 13.4)**.

3.3 Worst-Case Scenarios

The inclusion of "worst-case" situations within the APS is intended to provide assurance that sterile product will be produced when the permitted extremes of the routine process are present, specifically those conditions that pose the greatest contamination risk to the product. Examples include the maximum sterile hold times, the maximum permitted number of people present, etc. Worst-case does not include situations that should be the subject of process deviations such as equipment failure, or the inclusion of interventions that pose unacceptable risk to the process.

Worst-case conditions vary depending on the operations or risk being considered. For example, executing the APS using the maximum number of personnel is often considered the worst-case, as personnel are the greatest source of microbial contamination in an aseptic process. However, in some situations, the worst-case may include executing the process with fewer people if this results in excessive movement by the process operators, or if it complicates the process steps. Therefore, only the number of people used for the routine process should be considered for the APS for their longest duration of activities.

The worst-case approach, if used, should be duly justified, and its rationale should be documented in the APS protocol or procedure. Other examples of worst-case practices may include (but not be limited to):

- Using equipment of the longest permitted duration
- Using the slowest fill speed for the largest container (maximum exposure of vial opening to the environment)
- Using the highest fill speed for the most difficult to handle container

The worst-case conditions selected for inclusion in an APS should be predefined based on characteristics of the operation. Operational experience from routine production can provide opportunities to utilize intervention rates from different configurations, components, etc. to identify the appropriate ‘worst-case’ conditions for selection of the periodic APS study. Identifying appropriate worst-case conditions should be done by assessing the APS covering the relevant variables and their microbiological impact on the process. Such assessments can benefit from the application of QRM principles. The assessment conclusion should outline the variables selected as worst-case and the considerations or rationale for their selection. The risk assessment may also identify CAPA for reducing the "worst-case" situation such as improved operability leading to less interventions.

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Note: In cases where it may not be possible to define one or two worst-case scenarios, it may be necessary to include several (or even all) scenarios in the APS program.

3.4 Risk Assessment for Aseptic Process Simulation Design

Risk assessment is a systematic process of organizing information to support a risk decision to be made within a risk-management process. It consists of identifying hazards and analyzing and evaluating risks associated with exposure to them ***(8)***. In the context of this technical report, hazards or hazard situations associated with aseptic processing are any flaws, failures, or conditions that could compromise the sterility assurance of aseptically manufactured sterile products.

The principles of risk assessment should be considered in designing the APS. A risk assessment should be performed to determine, identify, and evaluate hazards or hazardous situations associated with the aseptic process steps and interventions that may adversely affect product sterility ***(9-11)***. Container size, configuration, line speed, batch size, and operating conditions provide an important insight into APS design. Risk assessments can also be used to determine the worst-case manufacturing scenarios related to these elements. This will be tied to the CCS where hazards are identified and the residual risk either considered acceptable or the process modified to remove/reduce the risk. It is important to consider risk to product quality and patient safety when confirming the design of the APS study.

The purpose of a risk assessment for an APS is to evaluate the process that the APS simulates and to inform the design of the resultant APS study. Only after hazards and risks associated with the aseptic process are evaluated can the APS design be deemed both representative and appropriate. Using a multifunctional team with a good knowledge of process, equipment and facility design, review the aseptic process in detail with respect to the potential for contamination ingress with attention to intervention frequency, proximity of any intervention to sterile materials and surfaces, and duration of the intervention as appropriate factors in the evaluation of aseptic risk.

The outputs of the aseptic process risk assessment used to inform the APS design should answer the following questions:

- Is your APS design appropriately representative of all stages of the manufacturing process (including assessing/providing justification as to where the APS begins) without masking any important risk factors (e.g., usage of inert gas during APS or freezing the media during lyophilization simulation might compromise the detection of contamination)?
- Is the APS mimicking the complete manufacturing process including the handling of materials, handling of media-filled units, operating equipment, and aseptic interventions from one subprocess to another?
- Is your APS design able to “challenge process capabilities” of an aseptic process and identify previously unaddressed variables?
- Are your selected and simulated worst-case manufacturing scenarios related to container size, configuration, line speed, batch size, and operating conditions appropriate, and are they able to best represent the risk to the aseptic product without adding an unnecessary risk to the APS?
- Is your APS design meeting regulatory and industry guidance requirements?

A critical success factor for such an exercise is sponsorship by the senior management of the organization to support resource allocation, appropriate team formation, any financial needs for managing the risk-assessment outcome and, most importantly, motivating the team towards the ultimate objective of protecting product quality and patient safety. It is important to note that this process can be jeopardized by individual

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subjectiveness, focus on predetermined outcomes, and the lack of meaningful participation by those most involved in the performance of the activities being assessed (e.g., line operators, mechanics). Hence, the best practice for APS design is to include people who are directly involved in the process and have a good understanding of the actual variability in the execution of the manufacturing process and of the accumulated knowledge about the product and process in the organization.

It is important to view the risk factors holistically, with their collective impact on the overall product and the process. The hazards and risks associated with an aseptic process can vary based on the specific process design and the context in which it is performed. For example, a similar human intervention of lifting a fallen vial from the track in an aseptic process could result in a different impact based on equipment design and/or air-flow design.

Human interventions in an aseptic process can be a key source of microbiological contamination of sterile products, irrespective of the type of manufacturing system used. Hence, as a part of a risk assessment, it is critical to thoroughly understand the risk contributors in each intervention while determining the level of risks and planning mitigation actions. This can help not only eliminate or reduce the risks of high-risk interventions (which may include the decision that some interventions are not allowed), it can also help in planning the APS studies. When the possibility of contamination is higher based on process design (e.g., manually intensive filling lines), a larger number of units, generally at or approaching the full production batch size, should be used. In contrast, a process conducted in an isolator can have a low risk of contamination because of the lack of direct human intervention and can be simulated with a lower number of units as a proportion of the overall operation ***(4).*** The inclusion and frequency of interventions in the APS should be considered based on assessed risks posed to product sterility ***(1)***, and the frequency with which the interventions occur during routine manufacturing. A documented approach to risk assessment of interventions can help define the size of the run, duration of the run, and frequency of the simulated interventions in an APS. (See **Section 14.1, Examples 1, 2, 4).**

Note: For very complex changes, even if the risk assessment indicates no new risks, it may be necessary to include the modified intervention (at least initially) to demonstrate there are no unexpected consequences.

Risk-assessment outcomes can be used for:

- Planning mitigation actions and setting the review process
- Risk communication
- Employee training
- Future process design
- APS design
- Investigation of Out-Of-Trend (OOT) and Out-Of-Specification (OOS) results
- Overall aseptic process improvement (e.g. Refinement of standard operating procedures for aseptic assembly of equipment and aseptic interventions)

Examples are presented in **Section 13.4**.

3.5 Ongoing Evaluation

Assessment of ongoing controls and changes to the aseptic process, personnel, equipment, computerized systems, facility, and critical utilities should be performed as part of the change-control system to determine the risk of such changes to the assurance of sterility. This assessment should be used to justify the type of response or simulation of the process change needed to ensure that the change willnot adversely affected the aseptic process. A risk assessment may be beneficial in presenting a rationale for the need and extent of an

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APS as a result of change control. The risk-assessment and risk-management decisions should be recorded, approved, and incorporated into the change-control documentation. The assessment should include periodic evaluation of the type and frequency of interventions that must be executed or simulated in the APS to ensure it remains representative of routine operations. When assessing potential improvements, the assessment of risk should include consideration for effectiveness of the change and any unintended consequences that may occur. (See **Section 4.1.3**).

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4.0 Overview of Aseptic Process Simulation for Aseptically Produced Products

This section summarizes general APS design requirements as well as considerations related to APS design with regard to products and aseptic processing technologies.

4.1 General Requirements

This section provides an overview of the various products manufactured aseptically in the pharmaceutical industry and how to perform adequate APS studies in respect to these products.

4.1.1 Scope of Aseptic Process Simulation

The APS should imitate the routine aseptic manufacturing process as closely as possible and include all critical manufacturing steps. The APS for aseptically produced sterile dosage forms and products starts at the point of sterilization of the product, for example, the sterile filtration of product solutions and the sterilization of product contact surfaces, including containers and closures, through sealing of the filled containers (e.g., vials, syringes) or other packaging units. Any aseptic activities performed during dosage-form compounding (e.g., the aseptic addition of ingredients for the preparation of a suspension), are a necessary part of the APS.

4.1.2 Design of Aseptic Process Simulation

The aseptic process is simulated through the use of production operations in which a sterile growth medium and/or placebo is handled in a manner that resembles the methods used in routine production as closely as possible. The application of this principle to a specific aseptic process or procedure may require adaptation of the methods described in this technical report to that process or procedure. However, any such adaptations should not reduce the risk (i.e., APS should not be a “best case” in comparison to routine manufacturing), or increase risk beyond normal operating parameters.

Where growth media and/or placebo is added or utilized in the APS, its sterilization need not be performed in a manner identical to that used for the product being simulated. For example, an APS does not support the validation of the sterilizing filter used in the product or process being simulated; this will be accomplished by other studies. So, differences in the filter area and filter media, for example, are acceptable providing these changes do not enhance the aseptic process or eliminate a process step that could adversely affect the sterility assurance of the product.

A company must determine the appropriate rationale and approaches applicable to their unique operations and APS design based on a sound risk assessment. Process-flow diagrams showing the routine

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manufacturing process and steps in parallel with the APS steps are very useful for an APS illustration and design assessment and to explicitly highlight the potential differences between both processes.

4.1.3 Interventions

Personnel are a significant source of microbial contamination during aseptic processes. Consequently, activities performed by personnel within the critical area, also called interventions, must be carefully designed, qualified, and controlled to ensure they do not compromise the sterility of the materials being produced. (See **Section 4.1.3.2**). Each company should have a list of "qualified interventions" for each aseptic process type.

The execution of interventions during the APS is critical to the process-capability demonstration. To demonstrate that capability, APSs should include representative interventions that occur during an aseptic filling process. While there are no truly risk-free interventions, when absolutely necessary, interventions can be considered acceptable if demonstrated to carry acceptable residual risk as evidenced by the application of appropriate aseptic technique, air-flow studies and successful APS.

It is essential to include the interventions that are known to occur during normal production runs in an APS. Interventions that are permitted in a production operation should be specifically documented and included in APSs at a similar frequency and of a similar duration based on their potential risk to product sterility. Nonqualified interventions that must be performed should be handled with the quality management system, and new interventions to be added to the intervention program should be managed through the change management system, as defined by the company. The details of the intervention conducted should be subject to risk assessment, recorded, and fully investigated under the manufacturer's pharmaceutical quality system, including a real-time assessment of the acceptance of the intervention from a risk to product/patient perspective. They should also be thoroughly assessed by the Quality Unit and considered for inclusion in the next APS based on the needs of the process or for investigational purposes. While interventions and/or stoppages are normally recorded in the batch record, the manner of documenting these occurrences varies. Line stoppages and corrective interventions should be sufficiently documented in batch records with the associated time, duration, and repetition of the event.

A thorough risk assessment of the aseptic manufacturing process is necessary to evaluate the process risk associated with each intervention. Factors such as the proximity of the intervention to critical surfaces or open containers, the duration of the intervention, the complexity of the intervention, and other parameters determined by each company may be used to evaluate the risk of each intervention. Rating the intervention risks based on objective criteria will help determine if additional actions or controls are needed to mitigate each risk and will also help determine the frequency with which each intervention should be performed during the APS. (See **Section 13.4**).

4.1.3.1 Identifying Interventions Associated with an Aseptic Process

Companies should develop and maintain an approved list of allowed and qualified interventions that may occur during production and in the APS. Companies should also establish clear instructions for the performance of each aseptic intervention with inputs from operators, line mechanics, supervisory staff, microbiology, engineering, and quality assurance. These instructions can be in the form of written procedures, visual aids, videos, other media such as virtual reality, or any combination thereof. The instructions must be periodically reviewed and updated to reflect the current processes and interventions. This ensures consistency for operators and the interventions they are required to perform.

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Companies should identify interventions and determine their frequency and duration of inclusion in the APS based on process design and historical performance documentation, such as completed batch records, batch-related documentation, SOPs, and discussions with operating personnel. The goal of this activity is to list all interventions for each product, process, container, closure, or filling line configuration. For firms with multiple aseptic operations, interventions may vary from one fill line to another, even if both are filling similar products and containers. Variations in product type may add activities that are specific to individual situations. The basis and number of the required simulated interventions must be documented.

Operators must demonstrate their ability to perform the aseptic interventions on the filling line prior to taking part in an APS or participating in a product fill (see **Section 8.0**). If multiple filling lines of comparable design are utilized for the manufacture of a product, application of a matrix approach to operator qualification might be considered. In the case of multiple primary-packaging configurations, it is technically not feasible to run every possible configuration on every filling line. Qualification of the operator on a different filling line can be done using a matrix approach only when equivalency of the intervention conducted on the other line has been established and well documented. This enables operators to simulate interventions on one filling line as part of the matrix approach in an APS, which is suitable for participation on other filling lines in routine production. The definition of worst-case configurations and matrixing therefore is a suitable alternative. For more information on qualification, please see **Section 8.0**.

Similar logic can be applied to interventions of very similar type, or interventions that could be seen as a subset of more complex operations. It is not necessary to perform every intervention individually during an APS. Depending on a risk assessment of the individual interventions and solid rationale for why a certain operation can be seen as a subset of, or sufficiently similar to, another intervention, a matrixing approach might be applicable. For example, during the removal of varying numbers of vials from a turntable due to overturned vials or glass-breakage events, the performance of the highest-risk vial-removal activity may be sufficient to demonstrate proficiency in performing lower-risk vial-removal activities.

4.1.3.1.1 Inherent Interventions

Inherent interventions are an integral part of the aseptic process and are required for setup or routine operation and/or monitoring (e.g., equipment installation/assembly, weight adjustments, closure resupply, container resupply, EM sampling). Inherent interventions are not corrections to events that occur on the filling line; rather, they are a planned and necessary part of the overall process and are performed during the APS at a defined frequency or point of the filling operation. Inherent interventions are required by batch record, procedure, or work instructions and should be executed in full during APS in accordance with the normal production process or, in part, just simulated (e.g., loading of packaging components).

Depending on the process design, the inherent interventions may pose different risks to the product. Therefore, the risk of the inherent interventions should be assessed and ranked for criticality (e.g., major, moderate, or minor). The criticality of an inherent intervention can be used to determine the frequency of its inclusion in the APS. That being said, higher-risk inherent interventions expected to take place at a relatively high frequency during production should be avoided to the extent possible, or the criticality of these interventions should be reduced by improved process, facility, or equipment design. Examples of inherent interventions are:

- Setup of direct and indirect product-contact parts vs. nonproduct-contact parts
- Change of EM plates preferably via manipulation through a glove port
- Addition of components
- Closure resupply requiring manual handling when opening the container/bag and during loading

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- For isolators and restricted access barrier systems (RABS):
  - Proximity of gloves to critical surfaces or open product during placement of hands in gloves
  - Docking and movement of materials in areas of the line with critical surfaces and/or open product

4.1.3.1.2 Corrective Interventions

Corrective interventions are performed to correct or adjust an aseptic process during its execution. There should be a list of permitted corrective interventions, each of which is fully characterized, that is, supported with a step-by-step written description, a contamination-focused risk assessment, comprehensive operator training, and detailed airflow visualization studies to confirm their suitability "by design." (See **Section 4.1.3.2**)

Corrective interventions should be challenged during the APS at a frequency (rate) similar to that in which they occur during routine manufacturing. To achieve this, the approximate rate that each permitted intervention occurs during product manufacture should be evaluated and used to inform the APS process design. For the APS to remain representative of the routine aseptic process, the cumulative rate that all corrective interventions occur during product manufacturing should also be known; the total number of corrective interventions challenged in each APS should align with this cumulative rate. The frequency with which interventions are performed during the APS should be representative of that used for production and may be adjusted periodically based on recent manufacturing data.

In practice, there are likely to be multiple permitted corrective interventions that may not occur during each production run and challenging each of them in every APS will lead to the APS becoming an un-representative challenge. Because some interventions will occur infrequently (e.g., in less than 50% of batches), the use of frequency to determine how often these interventions are challenged during APS will lead to them being overchallenged (e.g., once during each APS) and push the overall intervention rate for the APS beyond what is representative. The suggested approach is to deal with infrequent interventions in order of the risk they pose and the frequency they occur, while ensuring that each permitted intervention is included at least annually. This will mean that lower-risk infrequent interventions may be challenged annually rather than every six months.

Replicate interventions should not be performed at the same time but should be distributed throughout the APS to simulate production activities. Repeating the intervention multiple times during an APS may not represent an improved sterility assurance challenge to the process. Therefore, it should not be necessary to perform the same intervention multiple times during an APS to determine the maximum number of times an intervention may be performed in production. For example, if an intervention occurs ten times during production, it may not be necessary to replicate that number during the APS. The frequency should be representative, but does not need to equal the maximum number, especially for lower-risk interventions. The decision on the number of replicate interventions to include in the APS should be based on relative risk criteria. The inclusion of an unrealistically higher number and frequency of interventions during the APS than during the aseptic process may no longer be a true representation and simulation of the aseptic process.

Where a lower-risk intervention has an established rate, leading to it potentially being challenged multiple times during an APS, one or more replicates could be dropped to allow additional higher-risk interventions to be included more frequently (although care should be taken to ensure that the APS design does not

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become artificially such a worst-case that it is no longer representative of the process). The selection of interventions and frequency that each intervention is performed during APS should be fully justified.

Where the need to perform a new corrective intervention arises during product manufacture, a proactive assessment of risk should be performed to determine whether the intervention should be permitted and, if permitted, the risk-mitigating steps to be taken before and after it. Until that point and inclusion in a successful APS, any new intervention should be treated as a deviation from the established process with the potential for the batch to be stopped in the worst-case scenario.

4.1.3.2 Intervention Procedures and Qualification

In aseptic processing, the qualification of interventions is a critical aspect that ensures the maintenance of product sterility and overall process integrity. This section delves into the essential procedures and considerations for qualifying interventions, as well as the protocols for handling unqualified, unapproved, and disallowed interventions.

4.1.3.2.1 Intervention Qualification

Interventions should be properly qualified before they are performed in aseptic processes. The APS alone does not qualify the intervention, and it should not be considered the sole means to, or a substitute for, more-focused and effective methods to qualify an intervention. However, all interventions (inherent or corrective), after meeting other qualification criteria, should be included in the APS to evaluate whether any weakness or under-addressed variable in the process adversely affects product sterility.

The qualification of an intervention should include the following considerations:

- Risk assessment to determine the risks posed by the intervention and if those risks are adequately controlled
- Sound process design of the intervention, as well as qualification of any control measures designed to address risks posed by the intervention
- Approved procedures (as noted in formal SOPs, batch records, or other documents) describing the performance of the intervention
- Personnel training in the performance of the intervention
- Evidence that adequately trained and qualified personnel can properly perform the intervention within a consistent duration
- Evidence, inclusive of air-visualization studies, that personnel performing interventions in the critical area and in proximity to sterile product or contact surfaces can do so following good aseptic working techniques, without disrupting first air or compromising adjacent critical surfaces, e.g., by causing external “dirtier” air to enter the critical area by generating significant turbulences
- Use of ergonomics, rest periods (e.g., breaks), and other means to minimize the effects of human fatigue on aseptic process performance and risks posed to product quality (see **Section 7.9.2**).
- Oversight of intervention performance to evaluate proper execution of the intervention

A clear decision as to whether the intervention is acceptable risk, if further actions such as engineering is required or if the intervention poses an unacceptable risk

4.1.3.2.2 Unqualified or Unapproved Interventions

Interventions that are required in a production run to continue manufacturing and have not been previously qualified should be considered unqualified and unapproved. Procedures should be in place to address unqualified and unapproved interventions, including personnel and departments responsible for input and

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approval and actions required to perform the intervention if it is possible to do it with an acceptable risk. If such interventions are required or are contemplated, the company should do one or more of the following:

- Immediately contact the Quality Unit (before the intervention, if possible)
- Disallow the intervention and immediately stop production, pending corrective actions that are not normally performed during the aseptic process (through a deviation or temporary change control)
- Assess the intervention to determine the risk to product sterility posed by the intervention and control steps needed to mitigate such risk. If the assessment supports the execution of the intervention, consider whether the intervention should become part of the established process.
- Incorporate special precautions, risk mitigation, controls, and monitoring to limit the risk of product contamination, including removal of potential compromised units and disinfection or decontamination of affected areas
- Segregate and contain the affected units and define disposition criteria for the product filled after the performance of the intervention
- Evaluate the similarity of the interventions to qualified and approved interventions and activities. If the intervention has occurred before or is reasonably likely to occur again, consideration should be given to adding it to the established process as a permitted intervention once it is fully qualified (i.e., assessed and successfully challenged during the APS).
- If it is determined that the unqualified intervention will be necessary, operators should be coached on how to perform the intervention. Additionally, show the intervention be added to the qualified interventions, formal training of operators should be undertaken Document the intervention and decision criteria, including steps required to qualify the new intervention and its inclusion in future APS runs, if applicable
- Arrange for the Quality Unit to oversee the proper execution of the intervention
- Additional samples (e.g., sterility and EM taken after critical interventions) should be considered based on risk

4.1.3.2.3 Disallowed Interventions

Disallowed interventions are those that pose an unacceptable level of risk of contamination to sterile product, those that are no longer relevant or performed in production (not simulated in APS, "just in case" interventions), or those that for some other reason are no longer allowed in the production process. Disallowed interventions should be removed from the list of approved interventions, and procedures should be modified to remove these interventions from APS and normal production.

4.1.3.3 Aseptic Processing Simulation Study Design

The number, type, and complexity of inherent interventions that occur with each filling process, as well as corrective interventions and events (e.g., maintenance, stoppages, equipment adjustments) should be incorporated into the APS study design ***(12)***. When the APS procedures are consistent with those used for routine production, the simulation will be a more authentic representation of the routine process.

The purpose of the APS is to evaluate the capability of executing a process, interventions included, under specific conditions to produce units devoid of microbial contamination. During the design of the APS, it's crucial to take into account the quantity, variety, and intricacy of both inherent and corrective interventions, backed by robust scientific reasoning. Filling lines with comparable equipment setups might adopt a "worst-case" strategy for equipment arrangement and intervention to mimic the risks associated with each filling-line configuration. For example, two sets of filling parts of similar design sourced from different vendors may present a similar overall processing risk and may not require a separate APS to demonstrate the acceptability of the interventions associated with the use of each filling set.

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Interventions to be included in the APS protocol should be considered in terms of the expected frequency and duration with which they are performed during production based on process design, production logs, batch records, observations of the aseptic filling process, and the microbiological risk to the product and process as determined by a thorough risk assessment. In general, interventions that commonly occur should be routinely simulated, while those occurring rarely can be simulated periodically (e.g., annually, depending on their risk, complexity, and the frequency with which they are performed during production fills). It should be noted that rarely performed interventions may have an additional risk factor due to lower familiarity of the operator with the intervention (if it is sufficiently different from other more routine interventions). The APS should consider the anticipated number of interventions required to run and accurately simulate the aseptic process.

Proper intervention performance is dependent upon the operator training and qualification program and predicated on sound intervention design, airflow-visualization studies, procedures, and evidence supporting operator training and execution of interventions (see **Section 9.0**). Qualification studies on higher-risk complex interventions may be used to demonstrate adequate aseptic training proficiency and the ability to execute less-complex interventions during production without the operator having to perform the less-complex intervention during the APS. The execution of interventions during the APS, although important, should not be the sole indicator of an intervention's suitability or proper performance. Likewise, the ability to perform a particular intervention should not be based on the number of times it is performed during the APS, but rather the qualification and training of operators to perform these interventions. On-the-job training of higher-risk interventions using placebo (production environment) or growth promoting media (training environment), airflow visualization studies of intervention execution, virtual-reality training, and enhanced supervision of inexperienced operators may be used to ensure that operators are proficient at performing their assigned interventions.

The performance of interventions should be accomplished by qualified personnel, including those involved in maintenance activities, EM, and support personnel according to defined procedures. The ability of the operator or mechanic to intervene in the process to address mechanical failures or other infrequent interventions that may pose a significant risk to the process should be reflected in the APS. Certain corrective interventions that occur infrequently during product manufacture should be performed periodically to simulate their performance during routine operations. These include events such as equipment breakdown, part replacement, or other infrequent nonroutine corrective interventions.

Defined intervention procedures ensure controlled process activities and help avoid unintended, personnel-dependent activities that may not conform to appropriate aseptic behavior. A matrix approach (grouping interventions by similarity of execution and risk) for certain types of aseptic processing interventions may be considered for the simulation of similar interventions of comparable complexity and risk, where risk-related variables have been assessed and found to be captured in this approach. It should be noted that although some interventions may be considered within a matrix equivalent to each other, and in any one APS only a single intervention is executed the interventions should be rotated, so that over time all are included. Interventions that are demonstrated to be acceptable can normally be performed by other trained and qualified individuals.

Note: This may not be the case for highly complex specialized activities where only individuals who have performed that activity successfully (as supported by acceptable APS results) are considered qualified (see **Section 8.0**).

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4.1.3.4 Handling of Intervention-Related Units

It is common for a predefined number of units to be discarded during production runs as a risk reduction or mitigation activity following interventions, particularly after initial aseptic line setup. If the intervention procedure and batch documentation call for a minimum number of units (from specified location(s)) to be removed during or after an intervention, then those units are considered rejects associated with the intervention. Procedures should be in place to ensure that these rejected units are properly removed and discarded during production without compromising the adjacent aseptic environment and that the same procedure can be followed during the APS without incubation of rejected units.

Units rejected in response to an APS intervention, however, may provide valuable information to help the manufacturer better understand the risk that each intervention poses and may drive further risk reduction or mitigation. For this reason, manufacturers should consider incubating any integral filled and closed units (e.g., a vial with a fully inserted stopper) for informational purposes. The incubation of these units during an APS should be considered (1) if there is scientific value that can be obtained from the evaluation of these incubated units, and (2) where the operation and automation of the filling line permits collection of these units. Otherwise, the intervention rejects may be removed and noted in the APS. Rejects from higher-risk operations, for example, first-filled units following setup operations or units filled following significant interventions should be incubated if their inclusion does not introduce an artificially greater challenge to aseptic control than commercial operation (e.g., units removed and sealed by a manual vs. an automated process). In no case should more units be removed, or a larger zone be cleared during an APS intervention than would be cleared during a production run.

Where such units are incubated, they should be incubated separately. They are not to be considered part of the acceptance criteria for the APS; however, any contaminated unit should trigger an investigation for potential root cause, its impact on the overall process evaluation, and any associated corrective action. The exclusion of any units from incubation requires justification and documentation. Procedurally defined nonintegral units (e.g., cracked or unsealed containers) that are detected and discarded as part of the routine operation should not be incubated. The incubation of nonintegral units will not provide any meaningful measure of the aseptic process, since these units do not represent either acceptable production practices or acceptable container–closure integrity. Any integral units found to be contaminated should be considered in the context of potential improvement.

4.1.4 Sterile Bulk Aseptic Process Simulation

Technical Report 22 does not address APS for sterile bulk products (i.e., liquids, solids, suspensions) or for drug product, however they are described in PDA *Technical Report No. 28 (Revised 2006): Process Simulation Testing for Sterile Bulk Pharmaceutical Chemicals* (TR 28). Special considerations for the inclusion in the APS of sterile-material additions during the preparation of suspension drug products, however, is covered in **Section 4.2.3**.

4.2 Aseptic Process Simulation Design with Regard to Products

The following are aspects to consider in the design of the APS which are specific to the simulated product dosage form.

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4.2.1 Solutions

The APS should include and mimic all process steps, activities, interventions, and conditions used or present during commercial production, unless those process steps, activities, interventions, and conditions inhibit or alter the microbiological growth properties of the media. After sterilization by sterile filtration, heat sterilization, gamma irradiation, or any other validated sterilization method, the growth media should be passed through the process and equipment train as if it were an actual production batch. All routine aseptic procedures, aseptic connections, filling-system setup, sampling, sterile product and material holding periods and conditions, sterile-material handling and transfer procedures used in the manufacture of a batch performed should be simulated.

For the filling operation, the containers and closures (if necessary), product-contact equipment, and filling parts should be prepared using approved procedure. The filling machine should be operated at the predetermined fill-rate for the container size being utilized. Where identified, worst-case conditions should be used for the simulation.

The containers should be sealed, and the media-filled units collected in sequentially numbered trays or containers. If possible, the time of collection should be noted, which would allow contaminated units to be linked directly with the approximate time and activity being performed or simulated during the APS. The integral filled units (closed and sealed units) should be inverted and swirled to ensure all internal surfaces are exposed to the medium.

All integral media-filled units (passing initial 100% visual or automated inspection) should be incubated; integral, in this context, means closed and sealed units (See **Section 7.13**). Procedurally defined nonintegral units, including cracked vials, unsealed containers, or units with a defect that exposes the contents to the environment, which have been detected and discarded as part of the routine operation, should not be incubated. Units that are rejected for other reasons, including cosmetic defects or low-fill volume, which otherwise would not impact container-closure integrity and therefore sterility assurance, should be incubated and included in the study ***(13)***. (For more information on the rejection of nonintegral units, see **Section 4.1.3.4**)

To prevent contamination risk product-contact parts must be cleaned by a validated cleaning procedure to remove any growth-media residues. Cleaning may require removing gaskets and valves after an APS to avoid media contamination of commercial product. Unless otherwise noted, the recommendations for the filling of solutions are applicable to the filling of other liquid products described in **Sections 4.2.2-4.2.6**.

4.2.2 Lyophilized Products

The lyophilization APS should include and mimic all process steps, activities, interventions, and conditions used or present during routine production, unless those process steps, activities, interventions, and conditions inhibit or alter the microbiological growth properties of the media. The lyophilization process includes several unique process steps that are different from liquid fill and should be simulated to the extent feasible. Some of the steps may be performed manually, some may be performed using automated or semiautomated systems, some are performed in or through open controlled environments, some are performed using closed-environment systems, and some cannot be fully performed due to media-growth inhibition.

The conditions, interventions, and activities that pose a risk to product sterility, including shelf-loading, cycle-dwell activities and conditions, condenser operation, vacuum activation, pulsing, venting and

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introduction of gas overlay, and shelf movement should be considered for inclusion withing the APS. A risk assessment should be used to determine the worst-case conditions to be used in the simulation for each of the lyophilization process steps and to decide whether to include the step, the extent of inclusion, and the need to modify any of the following steps and conditions in the APS:

- **Filling Step:** There are various container-closure systems used, including vials with fluted stoppers, vials with a combination stopper and crimp, multichambered vials, and prefilled multichambered syringes. The containers are filled with media and partially stoppered, where required, and transferred to sterilized lyophilization chambers after filling. The approach employed for lyophilization filling-process simulation is similar to those used for solution fills except for the partial stoppering and the addition of the transport, loading, simulation of freeze-drying, full seating of the stoppers, where appropriate, and unloading steps. Because the lyophilization filling process involves partially stoppering units, simulation separately from the simulation of product that is not lyophilized may be beneficial.
- **Sterilization Hold Time:** The simulation should include the maximum holding time for all sterilized equipment and components. For the lyophilizer this is the time between the lyophilizer chamber sterilization and loading of the lyophilizer.
- **Transfer Step:** The filled and partially stoppered units are transferred from the filling line to the lyophilizer. The units are only partially stoppered or open within a sealed box with the interior of the containers and media/product exposed to the environment. The transfer is performed manually by using barrier technologies and mobile unidirectional air flow (UDAF) carriers or using an automated loading and transfer system. The environment where the transfer occurs, whether open or closed, should be controlled to Grade A conditions.
- **Loading Step:** The containers are loaded into the lyophilizer chamber at ambient temperature, either manually or using an automated loading system. Barrier systems are required or a mobile UDAF Grade A unit for manual transfer ***(1)***. The environment where the loading occurs should be controlled and should be Grade A anywhere the partially stoppered units are exposed. A risk assessment should be used to determine the number of trays, number of media units, and the positioning of the trays required to simulate the aseptic lyophilization loading and positioning process. The positioning of trays on the chamber shelves should simulate the normal loading of the unit and allow for the proper stopper-seating operation at the end of the cycle.
- **Cycle Vacuum Step:** A partial vacuum is drawn on the chamber, and this level is held for a predetermined time. A full-cycle vacuum should not be used because it could result in media boilover. This can result in insufficient media for appropriate recovery and spillage within the lyophilizer chamber.
- **Freezing Step:** The freezing step can be simulated through the activation of the condenser, the opening of valves, and other activities that occur during the freezing step. It is important not to freeze the media because that can inhibit microbiological growth.
- **Cycle Dwell Step:** Activities that occur during the cycle dwell should be included in the simulation. The duration of the dwell simulated in the APS does not necessarily have to represent the full production dwell time. The APS dwell activities and time should be determined by risk assessment and should be sufficient to include simulation of the conditions and activities that occur during the dwell time. Simulating the full duration of the dwell time may not be necessary, because the risk of contamination is the result of activities or events that occur during the cycle, including the introduction of gases (e.g., air substituted for nitrogen or other inert gases), and not necessarily the time that the units are exposed in the chamber in the absence of activity ***(14)***.
- **Full Stopper Insertion:** After the dwell period, the filled vial units are stoppered by chamber shelf compression.

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- **Unit Unloading:** The units are removed from the lyophilizer and transported to the capping station or area. The unloading and transfer can be performed manually using tools to manipulate the boxes, or barriers or mobile UDAF units. The environment through which the units are transferred should be closed or in the case of stoppered vials controlled (Grade A air supply as minimum requirement). The units are not considered to be fully sealed until they are capped.

4.2.2.1 Special Considerations Unique to the Production of Lyophilized Products

- **Freezing of Media:** In order to maximize microbial recovery, freezing or cooling of media that inhibits microbiological growth should be avoided.
- **Vacuum Levels and Duration for Traditional Lyophilization:** In the simulation of a lyophilization process, the depth of vacuum drawn on the chamber and the period of time for which this vacuum is held are important considerations. The vacuum level must not be so low as to permit the medium in the container to boil out, thereby invalidating the simulation. To avoid boiling the media, the chamber pressure during APS must not drop below the equilibrium vapor-pressure of the media (e.g., water, 32 mbar at 25 °C) at the loading temperature. The best practice is to maintain the chamber pressure between approximately 100 mbar and 200 mbar during sublimation and secondary drying simulation. Factors to consider in determining APS design, where applicable, include:
  - Using air to break the vacuum instead of nitrogen or other process gases
  - Replicating the maximum interval between sterilization of the lyophilizer and its use
  - Replicating the maximum period of time between filtration and lyophilization
  - Considering quantitative aspects of worst-case situations, for example, loading the largest number of trays or replicating the longest duration of loading where the chamber is open to the environment
- **Inert Gas Simulations:** In the production of lyophilized products, it is common practice to use sterile inert gases to break the vacuum on the chamber. These gases can remain in the product containers after sealing. The use of inert gases, such as nitrogen, in the lyophilization process does not necessarily constitute an anerobic condition that warrants the use of anerobic media. The use of an inert gas and anaerobic medium (e.g., alternate fluid thioglycollate medium) would be appropriate where the persistent presence of strict anaerobic microorganisms has been confirmed either through EM or, more likely, during product sterility testing or when the entire process is conducted under anaerobic conditions. Where strict anaerobes have not been detected in EM or sterility testing and where the entire process is not conducted under anaerobic conditions, lyophilizer APSs should be conducted under fully aerobic conditions (i.e., utilizing soybean casein digest medium (SCDM) and air in place of an inert gas to break the vacuum).

4.2.3 Suspensions

Sterile suspensions are used for the administration of insoluble sterile materials such as some antibiotics, vaccines, and corticosteroids. Because suspensions contain insoluble materials, sterilization through filtration may not be possible. Additionally, suspensions may contain heat-sensitive materials and ingredients, therefore heat sterilization may not be possible. For those reasons, sterile suspensions may involve the aseptic addition of sterilized components and ingredients as part of the compounding procedure. The APS for the filling of suspensions requires the simulation of these aseptic additions and other aseptic procedures in addition to the simulation of the aseptic filling process.

In aseptic compounding operations, the APS procedures should include the aspects of suspension manufacturing including sterilization of the vehicle, addition of the sterile component, and homogenization

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of the suspension. The most basic adaptation of the standard liquid APS is the addition of a sterile placebo powder or sterile media (instead of sterile product powder) to a tank of medium. If a sterile liquid component is added to the suspension, adding media is recommended so as not to alter the final media composition and its growth-promoting capabilities. While replacing the sterile product powder with sterile nutrient media, care should be taken to avoid excess powder in the area. A granular media helps in reducing dust formation and improving the flow. This simulates the critical difference in the production of suspensions: addition of a sterile component under aseptic conditions.

Filling operations should be carried out in a manner similar to that described for solution fills, with the introduction of any routine changes in the filling setup to accommodate suspension filling. Where recirculation lines, surge tanks, agitators, and other modifications are employed to fill suspensions, they should be employed in the simulated fill.

To effectively simulate the manufacturing process, it is crucial to follow it completely from the preparation of the sterile suspension to the end of the filling. Avoid breaking down or separating individual process operations, such as the compounding process or the aseptic addition of the suspension material. This is important because segmenting the process might lead to the omission of key process variables in the simulation. Be cautious when not formulating and filling the APS end-to-end in one continuous process, as additional steps not part of the simulated process may be required.

The one exception may be a suspension APS, where having separate simulations for the aseptic compounding procedure and the aseptic filling procedure can be beneficial. This approach can help evaluate specific aseptic process steps involved in the preparation of suspensions, identify sources of process weaknesses, and investigate process-simulation failures. While a segmented APS may be beneficial initially or during a failure investigation, there should be no need for routine separate APS for suspension products. As mentioned above, when segmenting the APS, great care needs to be applied to covering all process variable in the simulation.

The need for or benefit of segmenting the suspension APS into individual process simulations should be supported by documented justification, stating why the end-to-end process cannot occur, including a risk-based assessment that ensures all aseptic processing steps and conditions are included in the simulations, and that the combined APS runs cover the entire aseptic manufacturing process. The segmented approach should be described in a protocol that includes any steps required to ensure the adequacy of the simulation, including the overlap of process steps and handling from one segmented run to the next segmented run. The segmented runs should not create conditions that are more favorable than normal manufacturing operations.

4.2.4 Semisolids—Ointments, Creams, and Gels

For the purpose of this technial report, sterile ointments, creams, gels, and other semisolids will be referred to collectively as ointments. Producing sterile ointments involves unique manufacturing conditions and steps that may require special consideration in the APS design. The aseptic manufacturing processes can resemble either solution or suspension product processes, depending on the solubility of the active and inactive materials in the bases. The simulation should mimic the actual procedures used by the firm in their operations. This may involve the use of surrogate materials, including agents to increase the viscosity of the media needed to simulate the filling properties of the product. Where drug product processing steps and conditions may adversely affect the growth of microbiological contamination, modification of the steps or use of alternative steps may need to be incorporated in place of those steps in the APS.

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4.2.4.1 Compounding

Where the manufacturing process includes aseptic compounding (also known as formulation) operations, the procedures previously described for either solution or suspension compounding should be followed using whichever method more closely simulates the actual compounding procedure used for the product being simulated.

The product manufacturing process may also include heating steps that could reduce the growth of microbiological contamination. These heating steps should not be included in the APS if they will significantly reduce the ability to detect contamination. If heating will impact that ability, it may be necessary to modify the viscosity of the media to accommodate filling. If heating is required for the routine process, whether it will impact the growth promotion properties of the media must be evaluated and determined. It is important to ensure that the media maintains appropriate growth-promoting capabilities through all processing and handling steps, supported by the growth-promotion test after incubation.

4.2.4.2 Filling

Filling of sterile ointments generally is performed on a filling machine quite different from one employed for vials, syringes, or ampoules. The differences in equipment design and operation aside, the basic approach to conducting the fill is similar to that employed for other dosage forms. Increasing the viscosity of the media (e.g., adding agar to media) in the APS to simulate a process is not required, if the fill time of the media is the same as for the viscous product and the filler can properly operate at lower viscosities.

Depending on the capabilities of the filling equipment, increasing the viscosity may be required to enable the media to be processed without difficulty and allow an aseptic filling of media. However, the use of thickening agents should not impact the viability of the media.

4.2.4.3 Inspection

Ointments are often filled into opaque containers, tubes, and applicators. The post-incubation inspection of filled process-simulation tubes may require additional steps to view evidence of contamination. If possible, opaque containers can be replaced with similarly sized and shaped clear containers, with similar sealing systems. Where clear substitute containers are used, those containers should not run, fill, or be handled in a manner that is better than the opaque containers nor should they have properties that reduce the ability for growth of microbiological contamination.

If it is not possible to substitute clear containers, then the media-filled opaque containers should be incubated in their containers. After incubation, the media should be transferred from the opaque container to an individual glass or other suitably clear and sterile container for inspection. Care should be taken to avoid contamination from this transfer of media. It is important to culture any suspected contaminated unit as soon as practical to determine if the source of that contamination is microbiological, chemical, or from another source.

4.2.5 Powders

Aseptically produced sterile dry drug products, such as powders, microspheres, liposomes, or other solid materials, are dry nonliquid materials that are typically filled into vials, syringe cartridges, or other types of sterilized containers. These products will be collectively referred to here as “sterile powders.”

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Section 4.2.5 covers simulation of the handling, transfer, filling, and sealing of sterile powders. Some options and approaches for performing the APS will be presented, along with their respective advantages and disadvantages. Unless otherwise noted, the steps and precautions noted in the APS of solution products will be relevant to the APS of sterile-powder handling, filling, and sealing. (See **Section 4.2.1** for solution products.)

The production of sterile powders requires aseptic processes and equipment that may be quite different from those used to produce other aseptically produced liquid or lyophilized sterile dosage forms. While the filling of dry powders and materials utilizes equipment different from that used for filling liquids the APS of sterile powder-filling still involves the evaluation of sterile growth media filled in place of the powder product or in addition to a powder placebo.

The APS for sterile powders may require special processing steps for the replacement of the product and the evaluation of the media. Similarly, during the performance of the APS for a powder-filling procedure, adaptations to the filling practices may need to be employed.

Aseptic handling, filling, and sealing processes can be simulated using an appropriate placebo powder or other suitable surrogate powder material, such as sterilized Tryptic Soy Agar (TSA), mannitol, lactose, or polyethylene glycol, that mimics the sterile-powder drug product. The following bullet points need to be considered:

- Any placebo or surrogate material used should be filled and sealed in the same containers as those used in the aseptic process under evaluation.
- The placebo or surrogate material should exhibit or imitate physical characteristics similar to those of the sterile drug powder that pose a risk to product sterility.
- The placebo or surrogate material and the necessary modified process steps should not inhibit microbiological growth, impact growth-promoting properties of the media, or interfere with the ability to visualize turbidity.
- The impact of the placebo or surrogate material to inhibit the growth of microbiological contamination in the nutrient media should be assessed in microbiological growth and recovery studies.
- The placebo or surrogate material should not adversely affect the ability of the media to exhibit or show microbiological contamination.
- Where the processing steps and conditions may adversely affect the growth of microbiological contamination, steps may need to be modified or alternative steps may need to be used.Modified processing steps should not exclude processing steps that have been determined to pose a risk to product sterility.
- The APS will require two individual filling operations - one for filling of the liquid nutrient medium and one for the filling of the placebo powder or surrogate material.
- The individual contamination contribution from each of these individual filling steps may increase the overall risks of process-simulation contamination, and the process controls during the APS must take these risks into account.
- The placebo or surrogate material should be compatible with the production handling and filling systems.
- Separate simulations of individual unit operations should be avoided but, if used, they should be documented and supported by scientific or technical justification as to why these are needed. This also requires a risk-based assessment to ensure that all processing steps that pose a risk to product sterility are covered in the simulations.
- Inspectors should be trained to inspect filled media-containing placebo or surrogate material, since the material could create the appearance of turbidity that resembles microbial growth. Ideally, the

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material used should not create the appearance of turbidity; however, if that is not possible, it should be clearly distinguished from microbial turbidity, and inspectors should be specifically trained to recognize the difference between product precipitation (from the placebo) and microbial turbidity. Any questions regarding if growth is present or not requires further evaluation including isolation and testing / further incubation if required.

Sections 4.2.5.1-4.2.5.5 present some approaches for performing sterile-powder APSs. Other approaches, including the use of newer technologies that are not listed, may also be used with proper controls and consideration.

4.2.5.1 Liquid Media Filled by Powder Filling Equipment

Some sterile-powder filling machines can fill liquid with little or no modification; the main advantages and disadvantages of this are listed in **Table 4.2.5.1-1**. While these units may not fill liquids to the same degree of consistency with which they fill powders, their flexibility greatly simplifies the APS. In this procedure, the liquid medium is introduced as a direct substitute for the sterile powder in the fill hopper. The methods used to introduce the medium are different from those used when powder-filling, but that is a minor adaptation to the process when contrasted with the modifications necessary for other fillers. The conduct of the APS is essentially the same as that described in **Section 4.2**, which notes the importance of integrating all the fill-line activities during the simulation.

Table 4.2.5.1-1 Advantages and Disadvantages of Liquid Medium Filled by Powder Filling Equipment

Advantages	Disadvantages
• Only a single filling machine is required. • Simplifies the execution of the APS. • Avoids the risk of contamination from a new system. • Additional media controls for a separate liquid-fill machine are not required.	• Setup may differ from that used for powder fill. • Any variables or risks posed by the handling and filling of dry material may not be captured. • The powder-filler may be capable, but it may not be well-suited for filling liquid. • Liquid media leakage may occur. • Cleaning liquid media may be difficult.

4.2.5.2 Liquid Media Filled on Dry-Powder Filler with Supplementary Liquid-Fill Capability

Some dry powder fillers may be adapted to add a supplementary liquid-filling capability; the main advantages and disadvantages of this are listed in **Table 4.2.5.2-1**. The liquid-filling capability of these machines is not equivalent to conventional liquid fillers and is used only in APSs. In this manner, the same filling machine could be used for both the liquid- and solid-filling operation during APS.

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Table 4.2.5.2-1 Advantages and Disadvantages of Liquid Media Filled on Dry-Powder Filler with Supplementary Liquid-Fill Capability

Advantages	Disadvantages
• A single filler is used, and no additional line modifications are required.	• The filler may need to be operated at lower speeds due to its design. • Some designs may not fill liquid into every vial that receives powder. • Requires additional processing or handling steps that are not part of regular process which may add risk to the simulation that is not present in the production process. • The setup may differ from that used for powder fill. • Any variables or risks posed by the handling and filling of dry material may not be captured. • The powder filler may be capable but not well-suited for filling liquid. • Liquid media leakage may occur. • Cleaning liquid media may be difficult.

4.2.5.3 On-Line Liquid Media Fill followed by On-Line Placebo Powder Fill

In this scenario, a liquid-filling machine is added to the filling line prior to the powder filler; the main advantages and disadvantages of this are listed in **Table 4.2.5.3-1**. Use of this method depends in part on the space available on the filling line. For the APS, a volume of medium in the range appropriate for a liquid fill is added to the container, followed by a fill of sterile placebo material to the same container. When a powder and a liquid are combined, then the dispersion and dissolution of powder through the media should be confirmed. This can be achieved manually by inverting the container several times during the pre-inspection of containers prior to incubation.

Table 4.2.5.3-1 Advantages and Disadvantages of On-Line Liquid Media Fill Followed by On-Line Placebo Powder Fill

Advantages	Disadvantages
• All processes on-line, with no additional handling of containers.	• A second filler will need to be set up and qualified. • The process subjects the open units to additional exposure time. • Additional processing or handling steps that are not part of regular process are added due to liquid filling which may add risks to the simulation that are not present in the production process. • Limited space in the container may result in powder collecting on the rim of the container and affecting the seal.

4.2.5.4 On-line Placebo Powder Fill Followed by On-Line Liquid Media Fill

This method is used when the physical addition of a liquid media before the powder filler is not possible; the main advantages and disadvantages of this are listed in **Table 4.2.5.4-1** As with the process above, when a powder and a liquid are combined, then the dispersion and dissolution of powder through the media should be confirmed. This can be achieved manually by inverting the container several times during the pre-inspection of containers prior to incubation.

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Table 4.2.5.4-1 On-Line Placebo Powder Fill Followed by On-Line Liquid Media Fill

Advantages	Disadvantages
• All processes are on-line, with no additional handling of containers.	• A second filler will need to be set up and qualified. • The process subjects the open units to additional exposure time. • Additional processing or handling steps that are not part of regular process are added due to liquid filling which may add risks to the simulation that are not present in the production process. • Limited space in the container may result in media overfill or in splash collecting on the rim of the container and affecting the seal. • There is a potential for powder aspiration from the container when the liquid is filled last. • There may be an increased risk for particles impacting the inspection after incubation.

4.2.5.5 Dry Media Filled as a Powder and Reconstituted to a Liquid Form After the Fill

In this method, dry media is filled into the units using a powder filler. Sterile water is added under Grade A conditions; the main advantages and disadvantages of this are listed in **Table 4.2.5.5-1**.

Table 4.2.5.5-1 Advantages and Disadvantages of Dry Media Filled as a Powder and Reconstituted to a Liquid Form After the Fill

Advantages	Disadvantages
• No additional equipment is needed.	• Additional steps to add water are burdensome and will add significant, and potentially unacceptable, risk to the simulation, which is why this is not a recommended practice.

4.2.6 Other Dosage Forms and Drug/Device Combinations

Aseptic processing may be used for sterile drug dosage forms or healthcare products other than those addressed in **Sections 4.2.1-4.2.5**. These may include healthcare products such as inhalants, aerosols, – drug/device combinations, implants, advanced therapy medicinal products (ATMPs), and other novel sterile drug or healthcare products manufactured using aseptic processing. The methods and approaches presented in this technical report apply to these products, with consideration and modification to accommodate unique features of the products or processes. The risks posed by steps used in and conditions present during the performance of manufacturing procedures should be assessed and, where feasible, the process should be designed or modified to reduce or eliminate those risks. Where process steps and conditions remain that pose a risk to product sterility, a means to simulate the exposure of sterile product or surfaces during those steps and conditions should be designed. **Section 4.2.6** presents some of these features and considerations.

4.2.6.1 Terminally Sterilized Products

Products that are terminally sterilized or otherwise treated to destroy microbiological contamination after filling typically do not require APS because the process does not primarily rely on aseptic processing to

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ensure product sterility. While the use of APS may not be required for the terminal sterilization cycle, it may still be useful as a tool to evaluate the effectiveness in case any aseptic process steps are applied.

4.2.6.2 Combination Products

Sterile product combinations involving more than one aseptic process may require a separate APS for each aseptic process, as well as for the process of combining the products.

4.2.6.3 Aseptic Packaging and Assembly Operations

For products that are aseptically assembled and packaged, APSs may be required to evaluate the packaging and assembly processes and procedures. These simulations may include the immersion or placement of packaged or assembled products in media as part of the simulation.

4.2.6.4 Propelled Inhalants and Aerosols

These products often are packaged in metal containers that do not allow visual inspection for microbiological contamination. Unlike solutions for other liquid products, it may not be easy to completely empty the aerosol container or transfer the solution (or liquid) from the aerosol containers to a clear container. Propellants may also hamper the growth of microbiologic contamination. Growth-promotion testing and studies should be conducted to ensure that the filling conditions do not adversely affect the growth potential of the media. It may be necessary to modify the filling process and conditions, including the substitution or addition of sterilized air to the propellent to allow for adequate growth.

4.2.6.5 Implants and Aseptically Formed Products

Sterile products that are molded, created, or formed during an aseptic process may also require APS. The formation of these products may use solvents that hamper and adversely affect the microbiological growth of media. In this case, it may be necessary to substitute other surrogate materials for the formed materials to allow for simulation of the aseptic processes. The use of surrogate materials and processes should be shown not to inhibit the microbiological growth properties of the media or result in significantly less processing risk than the actual manufacturing process.

4.2.6.6 Advanced Therapy Medicinal Products

ATMPs are therapies that encompass gene therapy, somatic-cell therapy, and tissue-engineered products from animals and human origin. These products are often produced aseptically in small batch sizes. In many cases, ATMPs are prepared and filled in individual Grade A laminar flow units or biosafety cabinets (BSC) that are placed in a surrounding Grade B environments. The processes usually involve manual activities and certain processing steps may be performed in a barrier system such as an isolator. Certain closed-processing steps may be performed outside of the Grade A environment.

APS is an important element in validation of the ATMP manufacturing process and associated aseptic processes. A risk assessment should be used to determine the steps, conditions, duration, personnel participation, and other aspects of the ATMP aseptic manufacturing process that pose a risk to product sterility and should be evaluated in the APS ***(15)***.

The following are some of the key considerations for designing ATMP APS studies:

- Unless otherwise justified, the design, performance, and evaluation of the APS should follow the same steps and principles for other drug product APS.

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- The process simulation should be designed to resemble the routine manufacturing process associated with open-processing steps and any other steps and conditions that pose a risk to product sterility (including transfers through lower grade classified areas). However, not all process steps or duration may require evaluation in the APS.
- Processing activities that are closed, such as incubation or centrifuge, do not need to be simulated provided there is no open manipulation, interventions, aseptic connections, or contact with sterile product or product-contact surfaces.
- Aseptic connections to closed systems should be evaluated in the APS.
- Open addition, manipulation, or withdrawal of sterile materials associated with the process should be evaluated in the APS.
- If multiple operations are carried out together in a single room, the APS should include the worst-case area occupancy and activity levels.
- A risk assessment and approach should be used to determine the extent of operator participation in all aseptic interventions.
- Where manual aseptic filling is performed, the principles and requirements for manual filling discussed in PDA's *Technical Report No. 62: Recommended Practices for Manual Aseptic Processes* should be followed.
- Placebo/growth promoting media should contact all sterile surfaces of materials and equipment used during the manufacturing process, including waste lines and collection containers.
- The incubation of intermediate containers may be necessary if the APS is divided into sections to simulate the entire process.

4.3 Aseptic Process Simulation Design Based on Aseptic Processing Technologies

RABS, form-fill-seal (FFS), blow-fill-seal (BFS), isolation technology, and automated systems can provide a lower risk of contamination. Use of isolators and automated processes with lower-risk interventions during processing mitigate contamination risk and may allow for greater flexibility in APS design (e.g., number of vials needed, duration of fill, personnel participation, and accounting for multiple shift changes). Relative risk and unique aspects of these technologies should be taken into consideration when assessing the design of APS and the design of barrier-system studies. In general, the same criteria for a traditional APS are applicable for all aseptic filing technologies. However, the design and scope/range of the APS might be narrowed for restricted-access filling technologies based on the lower potential for contamination established by the CCS.

4.3.1 Form-Fill-Seal and Blow-Fill-Seal

FFS and BFS aseptic processes involve some unique process conditions, which should be considered in the design of the APS including:

- Ability to view contamination in translucent or opaque plastic containers
- Interventions or activities within the Grade A critical fill zone or activities in the lower classified space (Grade B or C) surrounding the critical fill zone and FFS/BFS machine Interventions involving manual manipulation in the forming, cutting, and adjustment/removal area, fill-nozzle exposure area, aseptic connections, and manipulations post-filtration
- Foaming characteristics of media that could affect the sealing of containers
- Fill volume, including lower-than-normal volume, that may affect the formation of the container
- Media exposure to heat and open-container dwell time
- Longer campaign (usually)

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- EM sampling within Grade A, if appropriate

4.3.2 Restricted Access Barrier Systems

For an APS involving a RABS, the same criteria must be applied as for other aseptic filling lines. The inherent advantage of using a RABS is to lower the risk of contamination due to the presence of physical barriers that functionally separate the product and process from operators. Opening the doors on a RABS unit should be conducted only when necessary (i.e., rare occurrence) and can only be allowed after confirmation (via airflow visualization studies) that the intervention is not allowing the intrusion of lower-grade air into the RABS enclosure. All permitted interventions are required to be simulated in the APS and based on a risk assessment. The APS should not be used to justify frequent opening of the RABS doors e.g. due to ineffective equipment design, as this would add unnecessary risk. See PDA's *Points to Consider No. 12: Restricted Access Barrier Systems* document for more information.

4.3.3 Isolator Technology

Isolators eliminate direct access of personnel into the critical environment by defined mechanical and barrier systems and are intended to eliminate human-borne contamination within the aseptic environment where sterile products, containers, closures, and equipment are exposed. Whereas glove ports are still the most common design to reduce contamination risk during interventions, gloveless isolators eliminate risks associated with human interventions. Isolators are decontaminated and material entry is accomplished using validated transfer systems.

Although the same principles of APS design apply for isolators, when evaluating worst-case conditions to simulate, the relative risk may vary, and this can lead to a different design. For example, because the aseptic process space is completely separated from the surrounding room and operators during filling, the maximum number of people allowed in the room where the isolator is located or the impact of activities occurring in those areas from operational shift changes are not relevant worst-case conditions. However, if multiple operators interact with the isolator simultaneously (using glove ports), it could represent a worst-case scenario related to pressure or air flow, for example.

Also, while isolators should still maintain a semiannual APS frequency for the line, shift structure is not a consideration, because the maximum batch size is larger, and the total filling time is often much longer than in a traditional aseptic filling line. Additionally, due to an isolator's advanced controls, it is possible to have a lower percentage of units filled with media, compared to what would be performed in a traditional aseptic line however, filling fewer than 10,000 filled integral units is not recommended, unless the commercial batch size is smaller. For small-batch sizes, the number of units should adequately represent the production run. For an APS of large commercial batches, it can be useful to periodically process empty vials through the system to more closely simulate the appropriate number of additions and the level of line activity. In this manner, the mechanical and environmental challenges to the isolator environment more closely resemble normal operations, and the ability to recover is not compromised, but the size of the APS (related to inspection, incubation, and reading) does not become excessive.

If empty vials are deemed necessary, then:

- Interventions that take place during the filling of empty vials should not count toward those required for simulation of the process during media-filled conditions.
- Any interventions that take place during this period should be executed appropriately, documented, and assessed for impact to the process.

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- Reconciliation of all units should clearly identify those filled with media and those that are empty.
- Only the media-filled units count towards the total acceptance requirements of the APS.

Processing empty vials occurs with isolators due to their larger campaign-style filling, but this is also applicable to other technologies or filling lines. All decisions to process empty vials during an APS should be justified. In recent years, the use of a "piggyback" APS (i.e., one conducted at the end of a fill or campaign) has become more widespread, and piggyback APS has gained popularity as the number of isolators has increased and campaigns have started to increase in size. Initially, piggyback APS were utilized to support the validation of an increase of campaign length in isolators. As an example, if a company had a campaign three lots in length and wanted to add a fourth, three campaigns would be conducted in which three product lots were filled, followed by a fourth lot where media would be filled instead. More recently, piggyback APS is used outside of isolators. In any case, a piggyback APS can only be considered as part of the APS strategy and not a standalone APS validation. The use of the piggyback model must be supplemented by other simulations capturing the beginning of the fill, the setup of the line, etc. (see **Sections 7.9** and **7.11**).

See *PDA's Points to Consider for the Aseptic Processing of Sterile Pharmaceutical Products in Isolators (2020)* for additional information.

4.3.4 Closed Systems

Closed systems are more commonly used for manufacturing sterile bulk products, which is covered in TR 28. Since this technology is not yet addressed by TR 28, however, and "closed systems" are defined in EU Annex 1, this section will address APS for closed systems. A closed system is a system designed to prevent exposure of the product to the surrounding environment by maintaining physical separation and ensuring microbiological integrity throughout its use. In bulk processing, this is achieved through the use of interconnected bulk product holders, such as tanks or bags, which are linked by pipes or tubing and remain closed throughout the process. In sterile manufacturing, closed systems are sterilized in place or while sealed using a validated procedure before use. They are constructed, installed, and qualified to maintain integrity across all operating conditions and the expected duration of use. Fluid transfers can occur within a closed system while maintaining asepsis, and connections to other closed systems can be made without compromising integrity through methods such as rapid transfer ports or steamed connections and other appropriate techniques. These systems are safeguarded by scheduled preventive maintenance and, where applicable, utilize sterilizing filters that are integrity tested and traceable to each product lot. By design, closed systems offer enhanced protection against contamination, and when sterile bulk pharmaceutical chemicals or biological products are processed entirely within them, the risk of contamination is negligible. The term "closed system" does not apply to RABS or isolators.

Closed systems have an advantage in that aseptic processing is possible within a background of a lower-classified environment, for example, Grade C environment. Precautions must be taken to ensure that the integrity of the closed system does not become compromised after the sterilization of product-contact surfaces is achieved. Appropriate system-integrity tests should be considered when there is a risk of compromising the integrity due to transportation, sterilization, assembly, or usage. To prevent any contamination of the interior of the sterile closed system by an ingress from the environment, closed connectors/disconnectors (e.g., rapid transfer ports (RTP)) and continuous controlled overpressure should be considered. The means by which this is achieved should be determined and captured in the CCS.

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The APS in a closed system should simulate process flow and aseptic filling into sterile containers, either by using closed connectors (e.g., single-use system (SUS)) or by direct connections, when the containers are sterilized together with the closed system (e.g., a SUS manifold system with multiple bags or fixed vessels) and are filled afterwards. Instead of product, a solution tryptic soy broth (TSB) is filled and later incubated within the containers. As always, the design and worst-case conditions should be based on QRM.

Note: Closed systems are also addressed in Section 1.4.1 of TR 28; however, that section of TR 28 should be considered different from **Section 3.4.4** in this technical report, because no aseptic filling of bulk pharmaceutical chemicals from closed systems into connected containers is intended in TR 28. Consequently, rinsing the closed system with a sterile placebo liquid to demonstrate the interior sterility of the closed system (as described in TR 28) is not sufficient. The contamination risk of aseptic filling into connected containers within a lower-classified environment than Grade A is much higher than operating a closed and well-controlled, integrity-tested sterile bulk pharmaceutical chemical system (e.g., vacuum or filter driers).

4.3.5 Single-Use Systems Technologies

SUS technologies are being used more commonly within the industry because, depending on the type of technology and its application, they may offer several advantages, such as a reduced need for aseptic manipulations, sterilization of components, and prevention of product cross-contamination. SUS are typically integrity-tested and sterilized prior to use; however, the APS can be a useful exercise to demonstrate the suitability of the entire system, including the aseptic connections to one or multiple SUS systems utilized in the process ***(16)***.

The same SUS should be used for the APS as those for the aseptic filling, except where a different system (e.g., a smaller size, different filter, etc.) is assessed and accepted through risk assessment.

4.3.6 Pre-Use Post-Sterilization Integrity Test Assembly

The objective of the APS is not to validate PUPSIT, as this is addressed within the filter Bacterial Challenge validation (refer to PDA's *Technical Report No. 26 (Revised 2025): Sterilizing Filtration of Liquids* and the *Points to Consider for Risks Associated with Sterilizing Grade Filters and Sterilizing Filtration (2020))*. All aspects of the PUPSIT procedure occurring within the aseptic process downstream of the sterile filter connection (e.g., aseptic connections, valve manipulations, integrity test conditions) shall be performed as part of the APS (refer to PDA's *Points to Consider for Implementation of Pre-Use Post-Sterilization Integrity Testing* (2020)).

Processes involving materials that come into contact with product contact surfaces but are then discarded (e.g., product flushes) and should be simulated with nutrient media and incubated as part of the APS. This requirement may be waived only if it is clearly demonstrated that the waste process does not impact product sterility.

4.3.7 Considerations for Advanced Therapy Medicinal Products

Considerations for the design and execution of an APS for manufacturing processes that employ manual operations are provided within this section. Different types of products are manufactured utilizing manual aseptic operations including many of those defined and regulated as advanced therapy medicinal products (ATMPs) which is a broad class of therapies inclusive of somatic cell therapies, gene therapies, and tissue-engineered medicinal products. While the considerations for manual operations discussed within this section

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are focused to ATMPs, these may also be applied to other product types that utilize manual aseptic processing. In addition, recommended practices for manual aseptic processes are detailed in TR 62.

Many ATMPs cannot be terminally sterilized (e.g., autologous cellular-based therapies) thus requiring aseptic processing at many steps (unit operations) of the overall manufacturing process. Additionally, these products may be produced in low batch volumes to meet specific patient needs with limitations in the application of closed, automated manufacturing systems, and thus these manufacturing processes are designed with multiple manual aseptic operations.

ATMPs are new, state-of-the-art therapies that encompass gene therapy, somatic-cell therapy, and tissue-engineered products from animals and human origin. These products are mostly produced in low batch sizes to meet specific patient needs, and the majority cannot be terminally sterilized, hence they are produced aseptically. In many cases, ATMPs are compounded and filled in constrained spaces in a hospital or laboratory setup which not only demands the careful and meticulous handling of the products but also limits the use of highly automated handling of these products from manufacturing to fill-finish.

Production of ATMPs is at a minimum done in a BSC (Grade A environment surrounded by a Grade B environment), although preferentially in a barrier system (RABS or Isolator). A risk-based design of the facility, manufacturing process, and their qualification is crucial for ensuring the final quality of the products and the safety of the operators while handling these products.

APS is an essential element of validation of the aseptic processes in this case. The design of simulation criteria should be based on the risk assessment done on the entire process. The following are key considerations for designing simulation studies:

- The APS shall be designed to resemble, as closely as possible, the routine manufacturing process associated with open-processing steps. Extended closed-processing activities, such as incubation, do not need to be simulated for the duration of the activities, provided there is no manipulation of the closed containers. All transfer steps must be included in the APS design.
- Where open product-handling is involved, the APS should simulate the entire process and should include all the interventions performed by the operators.
- Where closed systems are used, the APS must focus on the aseptic connections and system integrity.
- If there are requirements for addition and/or withdrawal of materials from the system, such processes shall be covered as a part of APS. Regulatory agencies expect used/spent processing solutions (e.g., media or wash solution) to be incubated. In the case of a facility (e.g., a hospital or laboratory setup) where multiple operations are carried out together in a single room, the APS should consider the worst-case scenario in terms of the area occupancy and its constraints during the APS. However, different stations might require separate APS based on the type of activities and risk assessment.
- A risk-based approach should be used to determine the need for operator participation in certain aseptic interventions during the APS. Appropriate aseptic-intervention training may qualify operators for multiple aseptic interventions during production, if they have been bracketed as similar interventions, without having to participate in each of those interventions during an APS and qualification such as broth filling. For example, operators can be considered qualified for a process by demonstrating that they can perform the most complex aseptic-processing steps or manipulations without contaminating the microbiological medium. Aseptic technique is necessary to assure sterility of product in a manual aseptic process and not just a passed APS run alone.
- Simulation with anaerobic conditions is not systematically required, as most cell and gene therapies are manufactured entirely in the presence of atmospheric oxygen.

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- In the case of a multiproduct facility, if a bracketing approach is used for APS, strategies like risk-based worst-case challenge combining the common processing steps or intervention types, could be adopted. However, it should be well defined through a structured risk-management plan.
- Placebo/growth promoting media should contact all sterile surfaces of materials and equipment used during the manufacturing process. The incubation of intermediate containers (e.g., bioreactor bags) may be necessary if the APS is divided into sections to simulate the entire process.
- Where there are different manufacturing steps involved for different products, a risk-based approach should be applied in the design of the matrix for APS to ensure coverage of all relevant practices.
- The number, size, and type of containers to be used in the APS need to be considered during the APS design process, and adequate incubation space needs to be planned to meet these needs. Some of the used media or wash solution that needs to be incubated in the APS runs could be large volumes and consume a large area of incubation space. Containers used for incubation in the APS runs should simulate the containers used in the actual manufacturing process when possible. Use of alternative containers should be justified.

Compounding Operations: Where the final product has no sterilization step for the compounded bulk before filling, the simulation study should include the compounding operations involving aseptic addition of the sterile input materials using sterile media and/or placebo (e.g., cell culture media demonstrated to support microbial growth).

Filling Operations: The risk involved in the filling operation is directly proportionate to the number of operator interventions involved in the process and design (open or closed line). Hence, in the case of manual filling operations, the APS should be more thorough to cover all the operations of an actual product and should be performed for a duration similar to the actual production process. Specifically, it is not unusual for ATMPs to be manually filled or pipetted into the final product unit (vial) by the manufacturing operator. In such cases where the operator serves as the "filling machine," the operator must be initially qualified to perform this activity following the requirements for qualification (at least three times for initial qualification) and requalification of manual filling (two times per year). These requirements should be isolated from the filling process and should not require that all upstream inoculation or expansion activities have the same requirements applied.

For very small batches, it may be necessary to fill more APS units than is normally done for manufacturing in order to ensure there are sufficient units for growth promotion post-incubation.

During execution of the APS, each operator working at a BSC must perform simulated aseptic manipulations; however, the APS may be performed in one or more representative BSCs as determined by a risk assessment.

A case study for the design of an APS for an ATMP (autologous gene-modified cellular therapy such as a CAR-T product) is provided in **Example 8**.

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5.0 Aseptic Process Simulation Documentation

Documentation is an important element of an APS program, since it serves as a record of the simulation's rationale and performance. Subsequently, this record can be of assistance in sterility-failure investigations as well as for review of the adequacy of the simulation by regulatory bodies ***(17)***.

Documentation provides instructions for performance of the study, criteria for acceptance of the study, historical references, study results, and proof that the studies were conducted. The documentation should be reviewed and approved by the Quality Unit and include such items as procedures, batch instructions, risk assessments, EM results, unit acceptance and rejection criteria, incubation conditions, protocol deviation investigations, final report, acceptance criteria, and positive unit investigations that are reviewed and approved by the Quality Unit.

An overall APS policy or procedure presents the requirements for scheduling, conducting, and documenting APS studies. This will include the rationale for worst-case container and line configuration (as applicable), intervention and process-step inclusion, and reevaluation frequency. A master plan may also be developed to present the requirements and rationale for conducting APS studies for a specific product, facility, or manufacturing line.

5.1 Process Definition

The processes, steps, activities, and conditions of the production process to be evaluated during the APS should be defined and documented. These are defined as any and all manufacturing steps that occur after product, equipment, and container-closure sterilization and which can adversely affect the sterility assurance of the product. In some cases, this may include process or handling steps subsequent to the sealing of the container (e.g., leak detection, automated inspection) where damage from handling can adversely affect product container integrity. Processes may include (but are not limited to) the post-sterilization handling of drug product, transfer and holding of sterile drug product, transfer and handling of sterilized container and closure, loading and removal of product from a lyophilizer, filling of product to the point the drug product is sealed and capped, and any subsequent inspection or handling steps that may affect sterility assurance of the product.

5.2 Protocol and Procedure Preparation

A formal written protocol, procedure(s), and/or batch instruction(s) describing the activities and conditions for the APS must be prepared, approved, and issued prior to the start of the study. The document must be identified for traceability and approved prior to execution by Quality Unit representatives. Other stakeholders may review and approve the document at the discretion of the company. The document should include, but not be limited to, the following information:

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- Groups responsible for execution, microbial testing, and approval of the study
- Rationale for the worst-case parameters chosen as appropriate simulation of routine operations
- Identification of the process to be simulated
- Identification of the room or rooms to be used
- Identification of the filling line and equipment to be used, including fluid-path configuration details if multiple configurations are available
- Type of container–closure to be used
- Line speed(s)
- Minimum number of units to be filled
- Number and type of interventions and stoppages
- Identification of units to be excluded from incubation and rationale
- Number, identity, and specific roles of people participating
- Media to be used
- Volume of medium to be filled into the containers
- Incubation time, temperature, and duration for the filled units
- EM to be performed
- Copy of the batch record to be used
- Accountability requirements
- Acceptance criteria for all activities
- Description of the documentation required for the final report
- Duration of the APS
- Directions on when an APS can be aborted or deemed invalid
- Duration of routine production fills being simulated
- Definition of conditions that may cause the simulations to be invalidated and decision-making authority
- Any differences between the production process and the process performed during the simulation, which may include any changes or modification to the production process or equipment required to perform the APS and the reason for the change (e.g., the replacement of an inert gas with air)
- Parameters (in the case of lyophilized products) that will be used for the simulation, including vacuum, process temperature, and dwell time

Other factors may need to be considered due to the nature of the process to be simulated. The protocol should require that, prior to execution of the APS study, critical support-system qualifications and process validations have been verified to be successfully completed and approved. The protocol should include or reference risk assessments used to support the design, performance, and evaluation of the APS.

5.3 Aseptic Process Simulation Execution Record

Execution of the protocol may be performed through the instructions noted in a batch record. The batch record gives detailed instructions on how to perform the APS. It should be written in the same format as a normal production batch record and contain all the routine data and sign-off requirements. All information applicable to the APS that would normally be attached to a batch record should also be attached to the simulation batch record, for example, cleaning and sterilization records for pieces of equipment used and release stickers for the containers and closures. All interventions, whether inherent (an integral component of the process) or corrective (required to maintain operation), and stoppages should be documented in the batch record as to the type of intervention, time the intervention occurred, aseptic operators involved, duration of the intervention or stoppage, and the number of the box or tray being filled. The batch record should include information relevant to the performance and completion of the study. The protocol and report

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should be properly filled out following appropriate good manufacturing practice (GMP) documentation practices. This record should include, but is not limited to:

- Names of individuals participating in the simulation
- Number of units filled
- Number of units incubated
- Filled-unit incubation time, temperature, and duration
- Number of positive units and box or tray number of any positive unit(s)
- Number of units rejected for cause during pre-incubation inspection (e.g., damaged container, defective seal)
- Growth promotion of medium (after incubation), see **Section 7.17**
- Filled-unit accountability
- Media sterilization method
- Filter identification and membrane integrity-test results
- Environmental and personnel monitoring results
- Record or log of all inherent and corrective interventions/ occurrences which may influence the outcome of the study with the time, duration and other details as available (including personnel in the filling room)
- Observation logs, records, and video recordings (if applicable)
- Description and resolution of any discrepancies or deviations to the protocol
- Executed batch record reviewed, signed, and dated per GMP requirements for batch production records

5.4 Final Report

The final report is an evaluation of the data from the batch record and environmental-control records. Based on this information, a conclusion is formulated regarding the acceptability of the APS as an adequate simulation of the manufacturing process.

The final report should note the results of the study, whether the study met the acceptance criteria, the resolution of discrepancies or deviations to the protocol, the conclusion of the study (e.g., success of study), and any follow-up actions (see **Section 9.0**).

Any APS run presenting one or more contaminated units should be investigated as positive and all efforts should be made to determine the root cause. This investigation and cause should be documented (see **Section 11.0**). Any aborted or discontinued APSs conducted for that line or conducted during the overall APS study should be clearly justified (see **Section 9.3**).

Note: Contaminated units that are not integral should be carefully investigated and appropriately documented to justify the determination of the root cause of the integrity failure. If the root cause is determined not to be related to the routine process, it may not necessarily indicate a failure.

The final report and completed documents should be approved by representatives of the Quality Unit. The protocol, batch record, final report, investigations, and all relevant support documentation should be retained in accordance with the firm's policies and regulatory requirements.

5.5 Aseptic Process Simulation Observation

The APS should be observed to ensure that all planned activities are properly executed and represent an appropriate challenge to process capability. Observations may also be used to augment training for aseptic

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practices and can be a valuable tool during the investigation of an APS failure. Observation should commence upon initiation of the APS, including equipment setup (and loading for isolators), and continue until the entire aseptic process has been completed (see **Section 4.1.1**). Observation of the simulation should be performed by individuals who have the knowledge and competence to assess whether participants have used proper aseptic practices, executed activities according to procedures (e.g., SOPs, batch records and protocols), and assured that aseptic interventions have been executed properly. These assessments should be documented to provide confirmation that all activities were carried out as they would be during routine production. Any observed deviation should be routed through the deviation management system.

There are several ways in which observations can take place during an APS:

- Observations may take place inside the cleanroom area as long as the person observing does not interfere with the process or the operator performing any activities, interventions, or documentation and does not provide guidance to the operator.
- Observations may take place from a window that provides unobstructed views of the filling line and aseptic process, thus reducing the number of personnel in the cleanroom area.
- Observations may be performed using a live video feed from cleanroom-compatible cameras that are strategically placed inside the cleanroom, as long as they provide clear, continuous, and unobstructed view of the aseptic activities.
- Any combination of these methods can be used for observation at different times throughout the simulation provided that the entire process is observed. Regardless of the method used to make the observations, all APS observations should be well documented. Note that this observation should not interfere with the APS procedure itself or add unnecessary risk of contamination to the process.

5.5.1 Video Recording of Aseptic Process Simulations

Video recording of the APS is recommended, although it is not currently an U.S. Food and Drug Administration (FDA) or European Medicines Agency (EMA) regulatory requirement. The decision to record an APS should align with a manufacturer's internal procedures, policies, and privacy restrictions and must include a description of the method, intended use, and retention parameters for the recording. The quality of the viewing angles, fidelity of the recording, and accuracy of date and time stamps must be evaluated to ensure they provide the best opportunity for effective evaluation. Internal policies may differ from company to company or from country to country, so it is important to comply with all local laws when recording personnel operations.

The use of video recording during an APS has several advantages:

- The process-simulation activity can be reviewed in detail after the simulation is complete to determine if lapses in technique or practice occurred. The recording should be reviewed to ensure compliance with procedures and aseptic techniques, regardless of the outcome of the APS.
- During an investigation of a process simulation failure, information from the recording can help determine or eliminate potential causes of contamination.
- The recording can be used to determine the potential for process improvements and incorporation of new or improved interventions and to assist with aseptic training programs.
- The use of cameras can be beneficial as it allows for less-restricted viewing. The implementation of cameras to avoid risk of contamination.

If used, video recording should be combined with people performing real-time observations. If used as part of an investigation, the applicable sections of the video recording should be retained just as any other GMP documentation. The video recording may then be referenced in any deviation raised. Each manufacturer

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should have a procedure that defines the purpose of the video recording, how it is to be used, and its retention period. Artificial intelligence (machine learning/deep learning) may also be used to analyze, detect, and classify interventions, and then to indicate whether or not the intervention was performed correctly.

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6.0 Environmental Monitoring

Determining and understanding the environmental conditions present during the APS study is very important. A properly defined, controlled, and executed EM program provides increased assurance of sterility by demonstrating that environmental conditions conducive to the production of sterile product are being met over the duration of the aseptic process and that appropriate systems and utilities are functioning as intended. The environmental monitoring program includes appropriately set alert levels and action limits and requires an investigation and a corrective action plan in cases where action limits are exceeded.

The elements and processes that define and detail the establishment and maintenance of an effective EM program, including sample-site selection, sampling frequency, alert and action levels, methodology and interpretation of data, can be found in PDA *Technical Report No. 13: Fundamentals of an Environmental Monitoring Program***.**

APS should be carried out using the routine EM operating procedures and sampling requirements. This should include the setup period, setup interventions, setup personnel, transfer of EM materials and samples, and performance of EM sampling. Any changes to the routine EM requirements during APS, including additional sampling or change in sampling location, should be justified. The routine monitoring program should be sufficiently robust to support commercial manufacturing and, thus, the APS. If an excursion results from a nonroutine sampling location during an APS, it should be assessed for inclusion as part of the routine EM during production. Performing less sampling than what is routine should be avoided as sampling is a routine intervention, and an APS is expected to be representative of routine manufacturing regarding interventions.

The results of the environmental and personnel monitoring are used to assess whether aseptic processing conditions were maintained and suitably simulated during an APS. Additionally, environmental and personnel monitoring results obtained during APS can aid in the identification of root cause if the APS yields any positives (see **Section 11.0**).

EM excursion investigations should be completed and approved before approving the APS report. Failure to meet established routine monitoring levels should be addressed according to routine monitoring investigation procedures, and actions should be taken according to those procedures. EM excursions are not an automatic cause to reject the results of an APS; rather, any decision should be based on the investigation results.

“Passing” an APS with EM results that exceed action limits does not mean that the aseptic process may be routinely performed in such an environment and should not be used as justification for doing so. Similarly, passing an APS simulation with EM results significantly below production limits does not establish those lower levels as the new production limits.

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7.0 Elements of Aseptic Process Simulations

This section contains general information to consider when conducting any type of APS. Issues such as operator intervention, fill volumes, line speeds, container sizes, and run duration play a key role in effectively simulating the production process.

Careful consideration should be given to each of the parameters discussed in **Section 7.1-7.18** for inclusion in the APS study design. Parameter selection and justification should be well documented and approved by the organization's Quality Unit.

7.1 Facility and Filling Machine Considerations

APS programs should ensure that each aseptic process that takes place on aseptic-processing equipment, in filling lines or rooms, or any combination thereof are evaluated every six months. Multiple APSs may be needed to address the aseptic processing that takes place on those lines and in those rooms. Where multiple container closure configurations are run on a given filling line, a bracketing or a matrix approach may be considered (see **Section 7.5.1.1**). The key consideration is that the aseptic filling or manufacturing process is being evaluated, not a specific product.

7.2 Equipment Setup

The setup for an aseptic process usually entails some manual assembly of the sterilized equipment, including items that directly or indirectly contact product, such as filler parts, closure placement and insertion systems, filter housings, stopper hopper-/bowl, and other assemblies. Because equipment setup activities may require more manipulation of critical surfaces than subsequent processing operations, the steps involved are considered inherent interventions and must be included in the APS.* The allowable adjustment or replacement of these parts during a production run should be considered as a corrective intervention and should also be included in APS runs. Personnel performing setup operations during routine manufacturing operations should perform these inherent and corrective interventions during APS studies.

***Note:** In many aseptic processing facilities, this is the most operator intensive part of the aseptic process. Greater emphasis and observation should be placed on its execution including details of transfer, handling, connection, placement, connection of parts and addition of components.

Equipment setups, including filler parts that are similar (e.g., filling needles, nozzles, pumps, valves, other spare parts) and involve the same activities or interventions, should not require inclusion in separate APS studies. However, where there are potential differences in the use or condition of these parts, for example, parts from different supplier sources or evidence of differences from process performance records, a risk assessment should be used to help determine if additional or separate APS studies are required.

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Consideration should be given to parts that are used in routine production rather than for APS studies alone, if risk related characteristics such as wear or condition vary.

Where there are differences in variables, such as the size of filler parts and closure placement and insertion systems, a risk assessment should be used to determine if an APS bracketing approach or matrix is applicable. This approach should not be based solely on the physical dimensions of the parts or lines. It should consider process variables that may or may not be included because of the bracketing approach (see **Section 7.5.1.1**). Where it is determined that not all relevant variables are captured by a bracketing approach, it will be necessary to capture such sizes and variables in multiple APS runs.

7.3 Media Selection and Preparation

The most common medium for APS is SCDM, also referred to as TSB. SCDM is a general-purpose growth medium well-suited for the recovery of aerobic microorganisms of the types commonly associated with human-borne and environmental contamination. It is identical to SCDM agar (although without the agar), which is widely used for microbial recovery in aseptic areas for the same reason. Replacement of the products, diluents, and buffer solutions with SCDM is customary when performing APS studies.

Where there is evidence that the process or environment may have anaerobic microorganism contamination that provides a risk of product contamination, the APS studies should include filling of an alternate media, such as fluid thioglycolate medium (FTM) or another suitable medium in addition to conducting an aerobic evaluation. This evidence may include aseptic processing conducted in a strict anaerobic environment, for an aerobic process if anaerobic microorganisms are consistently recovered during periodic EM (for anaerobes), or if facultative anaerobes are detected exclusively in a FTM sterility-test medium. If oxygen is excluded from processing, then such parameters as container fill-volume and inert gassing may require modification to provide a true anaerobic environment for the APS study. In the case of a purely anaerobic environment, the use of SCDM is not recommended and a purely anaerobic APS can be conducted.

Media containing animal-derived components should be sourced from non-bovine spongiform encephalopathy/-transmissible spongiform encephalopathies (BSE/TSE) origin. Alternatively, vegetable options (e.g., vegetable peptone broth) are a viable alternative. Manufacturing processes with a risk of mycoplasma or prion transmission should always utilize a vegetable-based growth medium. These media are often very similar in composition to SCDM, but with the casein component being replaced with a plant peptone (see **Section 13.2**).

7.4 Inert Gassing

Nitrogen or other inert gases are used to provide a low-oxygen environment for oxygen-sensitive products. They are also used to provide positive pressure for solution transfer. Nitrogen (or other gases) for these uses does not provide a true anaerobic environment (less than 0.1% residual oxygen is needed for anaerobic conditions). However, the use of an inert gas with SCDM, may still inhibit growth. In these instances, filter-sterilized air should be used in lieu of an inert gas for APS studies. Air should replace the inert gas and be delivered by the same delivery system, thus assuring that the purge/transfer setup-and-delivery considerations are fully considered in the simulation. If it is necessary to use an inert gas for simulation of an oxygen-free process, testing should confirm the ability of the medium selected (e.g., an anaerobic media) to support microbial growth when the inert gas is utilized. Any replacement of gases should not result in additional risk to the equipment, process, or personnel. It is important to note that the sterility of the inert

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gas system should be through filter validation, integrity testing, and sterilization of connecting lines downstream of the filter, not by means of the APS.

A risk assessment should specifically describe the aerobic or anaerobic nature of the aseptic process. The risk assessment should also include data gathered from other sources that could indicate whether anaerobic microorganisms may be present in the process or filled containers, such as sterility failures from FTM, anaerobic EM results (where applicable), or failures from routine microbiological testing of inert gases or other raw materials (i.e., powder-based materials that remain in that state for filling). The performance of anaerobic APS should be based on the outcome and data gathered regarding the anaerobic conditions during the simulated aseptic process and the appropriateness for the inclusion and frequency of strict anaerobic APSs should be justified.

7.5 Container-Closure Configuration

The container-closure configuration (e.g., glass vials with stoppers and caps, prefilled syringes and cartridges with plunger stoppers and tip locks, plastic bottles with dropper tips and caps, siliconized stoppers and vials, intravenous bags) used in commercial production on a particular production line should be replicated in the APS.

When given container-closure configurations are used on the same production line for different products (e.g., product formulations or fill volumes), the product/container combination posing the highest complexity in filling operations should be applied for the design of the APS (i.e., worst-case). After the initial qualification runs, it should not be necessary to repeat the APS for each product twice a year (approximately every six months), given that the worst-case design is still applied for the re-qualification. In those cases, worst-case and other combinations should be tested on a rotary basis. However, in most cases a single worst-case option will not exist. If the container-closure configuration for different products does involve different variables or levels of performance, interventions or activities, an assessment should be conducted to determine if those combinations require additional or separate APS studies. In these cases, a bracketing or matrix approach may be more appropriate.

7.5.1 Inclusion of Multiple Container-Closure Sizes

Where different sizes of containers and closures are used on a production line, a risk assessment that identifies the worst-case or the case that affords the best opportunity to uncover process weakness should be used to support the selection of the container-closure configuration(s) included in the APS study. This risk assessment should conclude that the container-closure evaluations, their related qualification studies, and resulting worst-case configuration APS study design addresses all relevant configuration related variables, including dimensional aspects, positioning, performance, handling, and fill-time of the container-closure. If it does not conclude this, then additional container-closure combinations should be included in the APS studies.

Evaluating more than one set of containers and closures may be necessary if the components require different production-line equipment, parts, setups, interventions, activities, or assembly. If the containers and closures exhibit different levels of performance, then additional or separate APS studies may be needed to assess those process variables. Examples of containers and closures that may pose increased levels of risk are outlined in **Table 7.5.1-1**.

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Table 7.5.1-1 Potential Risks Associated with Various Container-Closure Characteristics

Container-Closure Characteristic	Example of Potential Risk
Containers with larger openings	Greater potential for microorganism intrusion through the opening
Containers with smaller openings	Potential contact with the operator's gloved hands in the case of ATMPs with a manual fill
Larger containers	Filling of a larger volume of product may take more time, increasing the chance that contamination may be introduced
Smaller containers	Filling small volumes of product may pose a risk if the product accumulates near the top of the vial, if the container opening is close to support tracks or rails, or if containers are run at a faster speed and/or are less stable and fall over more frequently leading to higher rate of interventions
Containers and closures that exhibit higher levels of production process-related rejects and/or tend to be more difficult or complex to run	Potential for increased frequency of interventions to address container-closure handling issues

7.5.1.1 Container and Closure Bracketing

Where multiple container and closure sizes are used in manufacturing, bracketing may be an effective approach to determine the representative sizes of container-closure combinations, where process equivalence is scientifically justified. Some considerations for the use of a bracketing approach include:

- Bracketing usually involves the selection of two or more combinations of container-closures, determined to represent worst-case conditions, based on upper and lower (or other identified process variable) configuration ranges, such as container size, container-opening diameter, closure size or diameter, fill volume, or line speed. The assumption is that by running the container-closure combinations at the higher and lower ends of the range during APS studies, all container-closure combinations that fall between those sizes may also be considered qualified if supported by a scientifically sound risk assessment and rationale.
- The upper and lower bracket configurations are typically included in the total number of APS runs required to qualify a given container closure system. Configurations falling between these brackets are usually tested with a reduced number of APS runs.
- Where an APS bracketing approach is used for initial container-closure qualification, it may be prudent to continue to capture the other container-closure combinations that fall between the bracketed sizes in other APS runs or future APS runs performed for routine requalification. This allows for the capture of process variables associated with those sizes that may have been missed in the initial assessment.
- The use of a bracketing approach is not merely based on a calculation of container-size variables. The decision can be complex, and risk-based decision-making should be used to ensure that all process variables that could pose a risk to product sterility are addressed within the bracketed process conditions. The selection of bracketing sizes or configurations should be justified by risk assessment. Variables to consider include such process and component aspects as component

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material-of-construction, container height, proximity to transport rails, ease of transport, interventions required, fill-system setup, production yields, and rejection rates.

- The applicability of bracketing, where containers with differing shapes or characteristics of construction will be included in the bracketing matrix (e.g., tubular glass vials and molded glass vials), largely depends on how the difference between the container shapes effects the filling process and whether the filling of those containers pose additional risk to product sterility. A risk-based assessment should be used that considers if the containers are handled differently, are transported through the fill line differently, or are filled differently as well as if they require different types of fill-system setup or have different heights or other characteristics or conditions that may pose a risk to product sterility.
- Information and evidence of worst-case conditions can be obtained from the following: scientifically sound risk assessments, production reports, batch records, room, equipment and maintenance records, reject levels and rates, production yields, aseptic process observations, process failures, excursion investigations, and interviews with various aseptic processing personnel (including operators, quality, and mechanics who perform line set-up and normal corrective interventions).
- If worst-case container-closure configurations cannot be determined, there is uncertainty that a bracket approach will capture all relevant process variables, or there is doubt of which container–closure configurations represent the most challenging case, then the full number of APS runs determined required to qualify a given container closure configuration should be performed for each container-closure configuration.
- This approach cannot be used to justify performing APS simulations that do not represent all of the various types of aseptic processing operations performed. For example, an APS of a manual filling operation cannot be used to represent a semiautomated filling operation because manual operations represent the most challenging conditions. Semiautomated operations have unique equipment, processes, and challenges that are not represented in a manual operation and need to be challenged in an APS study. Similarly, a prefilled syringe APS cannot represent a vial filling process (or vice-versa).

7.5.2 Pigmented or Limited Observation Containers

Pigmented containers (e.g., opaque, amber, some plastic) that do not readily allow for visual examination of filled media should be replaced with clear containers that are the same size and made of the same materials as the original finished-product containers for the APS. When it is necessary for containers of a similar size and dimension to be used as the substitute, it should be determined that the substituted container encompasses all relevant process variables associated with the production container and does not perform differently than the commercial production container. If this is not possible, another option could be to open each container at the end of the final incubation period and transfer the contents of the container into a clear container to permit visual inspection of the media for potential contamination (turbidity). For some translucent containers, a preliminary inspection of the incubated container should be performed prior to this transfer to determine if visible signs of potential of potential microbial contamination (including but not limited to turbidity or presence of pelicle, sedimentation or filaments) are detectable. Extreme caution should be used when considering and performing this approach, however; the emptying of media contents from one container into another increases the risk of contamination and, if not inspected immediately, can result in positive media-filled units that may be difficult to unequivocally attribute to the emptying activity, rather than to an aseptic process failure.

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7.5.3 Components from Varying Suppliers and of Different Designs and Compositions

An assessment should be conducted of containers and closures that have the same dimensions, but different internal design or composition (e.g., synthetic or natural rubber closures, coated or non-coated, tubular or molded glass vials, EU blow-back vials) or come from different suppliers, to ensure that these components do not exhibit different levels of performance. This might include aspects such as component transport, movement, insertion methods, equipment or equipment adjustments required, and performance, or if the components require additional or significantly different preparation or handling. Where it is found that differences in procedure or performance exist, these components may need to be assessed separately and possibly included in separate APS studies. It may also be necessary to perform additional assessment and container–closure integrity test qualification studies.

7.6 Filling Speed

In general, the fill speed to be used for most containers should be set at the production-filling speed range for that size container in commercial production. Where production-filling speeds on a line vary, if a higher or lower speed results in the potential for greater interventions or other adverse impacts (such as increased product exposure to the environment), that speed can be considered worst-case and should be considered when selecting process-simulation parameters. Where a clear justification does not exist supporting a single speed, multiple speeds should be integrated into one APS or more than one APS should be conducted.

Where multiple speeds are used on a given production line, it may be possible to include running a segment of the APS at each of those speeds or bracketing the slowest and fastest speeds. If this is done, then the number of units filled at each of the speeds should be the requisite number of units required for an APS.

7.7 Fill Volume

A container need not be filled to its normal fill volume when conducting an APS. Regardless of the actual fill volume, the APS should include a fill weight/volume adjustment using methods identical to those employed during production.

While the specific amount of medium used in a partial fill may not be critical:

- There must be enough medium in the container to contact all the container–closure seal surfaces when the container is inverted and swirled. It should be confirmed that all interior surfaces are contacted.
- There must be enough medium in the container to allow for the detection of microbial growth. If there is not enough media to detect growth, it may be necessary to fill more than the production-fill volume.
- The maximum volume for each container must be defined, allowing sufficient air headspace volume in the container to permit microbial growth. This can be supported by performing a specific growth-promotion test in each container size sent for incubation where possible (see **Section 7.17**).
- The full production-fill volume should be considered if the filling of full-production volume is assessed to add risk from such characteristics as splash or foaming.

7.8 Interventions

As a general rule, interventions (inherent and corrective) should be minimized during routine aseptic processing. Interventions that would represent an unreasonable risk of contamination should not be included

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in either APS or routine production. The choice of interventions to be included in an APS should be the result of a risk assessment. Where an intervention, even if rarely performed, poses a higher risk to the product or process due to its complexity and infrequent execution, the company may consider including the intervention at a higher-than-normal frequency in the APS to allow for qualification of more than one operator.

Intervention assessments should include the activities that occur during an aseptic process, including material transfer, fill-system setup, filling, sealing, and other activities that may pose a risk to product sterility whether inherent interventions (e.g., weight adjustments and container–closure resupply) or permitted corrective interventions (e.g., equipment and container breakage, closure jams, misalignment, part replacement, post intervention critical area disinfection). See **Section 4.1.3** for more details on the inclusion of interventions in the APS.

7.9 Duration of Aseptic Process Simulations

The duration of the APS should be sufficient to thoroughly challenge all relevant production activities involved in aseptic processing. This includes simulating routine production conditions to identify potential opportunities for microbial ingress. The duration and scope of the APS should be determined through risk assessment, focusing on factors that reflect operational complexity and contamination risks. The simulation should begin with the first operational activity, such as equipment and line setup, and conclude with the final step in the routine process, ensuring all critical and routine activities are included. At a minimum, the APS should incorporate the following activities between the start and completion of operations:

- Equipment setup and preparation
- Sufficient filling and sealing of a representative number of units
- Adequate number, frequency, and types of interventions
- Transfer and aseptic connection of any surge tanks
- Maximum permitted hold times for products, components, and equipment
- Execution of routine environmental and personnel monitoring sampling
- Movement of personnel into and out of controlled environments (i.e., Grade B background for RABS) during shift changes (if applicable)
- Transfer and replenishment of components and materials, including loading of the isolator
- Replacement of filters during the aseptic process, if necessary
- Cleaning and sanitization procedures
- Visual inspection for glove integrity during use
- Consideration of unique operations or process-specific variables
- Simulation of campaign operations for extended durations

When no risk-based duration effects are identified, and when longer durations do not provide additional scientific value, the APS duration need not exceed the maximum production duration for the specific process. For operations extending over multiple days, the APS should assess the ability to maintain aseptic conditions throughout the extended period, supported by scientific justification for additional activities and personnel or material movements. For operations supporting manual aseptic filling and sealing, which are heavily dependent on operator performance, the full process duration must be replicated without truncation. The APS should be carefully designed to account for variables unique to manual operations using tools and manipulations to ensure accurate simulation of real-world challenges.

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7.9.1 Aseptic Process Validation Limitations

It is equally important to understand the constraints placed upon the APS. An APS is not designed to **establish** or **qualify** the duration of aseptic processes itself, as contamination risks are more influenced by specific events rather than the total length of the process. As mentioned above, the duration of the aseptic process should be determined beforehand, and while APS runs may match or approach the routine process duration, they do not need to exceed it as long as all critical variables are captured effectively within a shorter timeframe.

Additionally, the APS should not be used to qualify or establish holding times for sterilized materials, components, or equipment. These holding times should be independently qualified based on production requirements and environmental conditions. Once established, maximum holding times can then be periodically included in APS runs to confirm alignment with actual use conditions for routine production.

An APS is an invaluable tool for evaluating aseptic operations; however, they have specific limitations in addressing factors such as human fatigue, environmental exposure, and the qualification of maximum hold times. These considerations require alternative methods to ensure comprehensive QRM and process validation.

7.9.2 Human Performance and Fatigue

The APS is not a suitable tool for assessing the maximum time personnel can remain in the cleanroom or the effects of fatigue on performance. Risks associated with human fatigue should be mitigated through measures such as ergonomic process design, adherence to aseptic techniques, effective training, operational experience, and appropriate scheduling of breaks or rest periods. Fatigue-related risks are better managed by focusing on human-factor assessments, rather than relying on longer-duration APS runs, which may not effectively simulate these variables.

7.9.3 Environmental and Equipment Considerations

The APS is not designed to establish maximum environmental exposure times or validate the maintenance of aseptic environmental conditions. Such conditions are better assured through robust environmental system design, real-time monitoring, and microbiological assessments, rather than through the simulation itself ***(18)***. Similarly, the APS is not intended to establish maximum equipment usage times. Equipment condition and performance should instead be verified through proper selection, qualification, cleaning validation, and preventive maintenance programs.

7.9.4 Qualification of Maximum Hold Times

Maximum hold times for sterilized or sterile materials should be established based on production requirements and supported by assessments of packaging integrity under expected conditions. Protective barriers such as wrappings, bags, or containers must be qualified to demonstrate their ability to maintain sterility throughout the maximum hold duration, while accommodating the handling, transfer, disinfection, and operational needs of the process. Periodically, the APS should include materials held for the maximum qualified time to ensure that no duration-related variables are overlooked ***(13)***; however, APS runs should not be used as the sole method to qualify maximum hold times.

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7.10 Number of Units

The number of units filled for an APS should be sufficient to adequately challenge the entire aseptic process. The number of units needed to assess aseptic process events, interventions, and activities can be based on an assessment of the relative risk of those aseptic process steps. Units should be filled with media following interventions and activities to assess the effect of variables and conditions resulting from those interventions and activities. Steps, events, activities, and interventions to be assessed for inclusion in the APS include those associated with, but not limited to:

- Transfer, assembly, and setup of direct and indirect product-contact equipment and surfaces, such as filtration, filling, and sealing systems
- Movement of personnel into and out of the aseptic processing area (APA), including the entry and use of operators' hands in the critical area through fixed glove ports
- Movement of materials into the supporting area, as well as the transfer and replenishment of containers, closures, or parts into the critical area
- Sanitization, disinfection, and preparation of fixed parts and areas where interventions were performed or contacted critical surfaces
- Performance of other aseptic process interventions, groups of interventions, or activities that occur in the critical zone and APA areas that are deemed to pose a risk to the microbiological quality of the product
- Execution of all routine setup, in-process, and post- process environmental and personnel monitoring

As noted in EU Annex 1, the number of units processed (filled) for APS should be sufficient to effectively simulate all activities that are representative of the aseptic manufacturing process ***(1)***. Justification for the number of units to be filled should be clearly captured in the CCS. Typically, a minimum of 5,000 to 10,000 units are filled. For smaller production batch sizes (e.g., those under 5,000 units), the number of containers for APS should at least equal the size of the production batch. However, these minimum numbers may not be appropriate for all filling operations, technologies, and APS studies. Each company should determine the rationale and approaches applicable to determine the minimum number of units to be filled during their APS.

7.11 Means to Achieve Full or Longer Duration APS Fills

One challenge with running long-duration APS fills is the need to incubate and inspect a large number of filled units. This can present a burden to those performing the APS due to limitations of incubation space and the resources and time needed to inspect a large number of units. The scenarios listed below present the corresponding rationale and related concerns that longer-duration APS runs are specifically designed to address, highlighting process risks that may only emerge over extended operational periods. When determining the duration of the APS, the process variables selected to challenge the conditions should be clearly identified. The chosen method should ensure that these variables are captured with a scientifically valid approach and appropriate regulatory requirements taken into consideration.

Scenario 1: Alternate Between Media-Filled Units and Idle Time

The line first fills media intermittently throughout the run, alternating with periods of idle time. Planned interventions and other critical activities are performed during the media-filling portion only. Media is filled periodically throughout the process, including at the beginning and end of the routine process duration, as well as during and immediately after any planned interventions.

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Rationale: Under this approach, the aseptic personnel, procedures, and processing environment continue to be evaluated, but the number of media-filled units produced is reduced.

Caution: The time the machine remains idle does not present the same level of challenge (to the process, personnel, or environment) as the actual production. This option is of limited benefit because only interventions and critical activities are challenged during the media-filling portion.

Scenario 2: Alternate Between Media-Filled and Empty Units

The line first fills media for a portion of the run and empty units for another portion of the run or intermittently throughout the run. Planned interventions and other critical activities are performed during the media-filling portion only. Media is filled periodically throughout the process, including at the beginning and end of the routine process duration, as well as during and immediately after any planned interventions.

Rationale: Interventions and critical activities are challenged during the media-filling portion. The line is allowed to run for a longer duration without the need to incubate and inspect a larger number of media-filled units. Under this approach, the aseptic personnel, procedures, and processing environment continue to be evaluated, but the number of media-filled units produced is reduced. The running of those units increases the vial-handling challenge for the machine and can lead to additional interventions and increased activity and personnel presence.

Caution: This option is of limited benefit because the processing of empty units during an APS does not allow for the evaluation of those units for potential contamination. Therefore, any process variables, adverse events, or compromise of the aseptic process during the periods of time when the empty units are being processed will go undetected. The only units and period that can be evaluated will be the portion filled with media.

Scenario 3: Alternate Between Water for Injection-Filled and Media-Filled Units

This method is similar to the approach in Scenario 2, except units are filled with water for injection (WFI) when not filling with media.

Rationale: Liquid media or WFI are filled, which may capture variables associated with actual filling otherwise missed by running empty units.

Caution: This option is of limited benefit because the filling of units with WFI rather than media still does not allow for adequate evaluation of possible contamination happening during the filling of vials with WFI. Further, the impact of media dilution with WFI, which could alter the growth-promoting characteristics, must be considered. Adequate qualification of the growth-promoting characteristics of the media should be demonstrated when alternating with WFI.

Scenario 4: End of Production-Fill Aseptic Process Simulations (or Piggyback Campaigns)

In this case, the APS is run after a full- or longer-duration production fill in an identical configuration without intervening decontamination / sanitization of the APA.

Rationale: This is an evaluation of units filled with media, occurring at the end of an aseptic processing operation of long duration. Therefore, variables associated with the length of the filling operation may be captured and evaluated in the APS, without the necessity of running, incubating, and inspecting a large number of media-filled units.

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Caution: This option is of limited benefit because running an APS after a production lot will not necessarily capture all relevant process variables associated with process start-up, including such activities as certain material transfer, fill-system setup, equipment assembly, and others occurring during the early parts of the fill. Therefore, to capture those variables, this approach should be periodically coupled or alternated with simulation of the normal production setup, start-up, and commencement of filling. Scheduling and sequencing piggyback APS studies should be supported by risk assessment.

In addition, the product fluid pathway should be flushed with a sufficient quantity of media to remove drug product that may dilute or otherwise affect the growth-promoting ability of the media. The flush volume should be specified and shown to result in adequate growth-promoting characteristics of the media. For inhibitory products, the product fluid pathway contact points should be changed with newly prepared and sterilized parts or equipment prior to conducting the APS. Where the process allows collection of flushed media, the flushed media should be collected and incubated for evaluation, unless it can be clearly demonstrated that this waste process would not impact the sterility of the product. Any contaminated flush material should be investigated.

7.12 Campaign Operations

The campaign approach is increasingly common with FFS, BFS, and isolation technology (primarily isolators). During a campaign, environmental conditions are essentially constant and are expected to be unchanged throughout the campaign duration. Decontamination of the isolator and disinfection of RABS and/or removal, cleaning, and sterilization of critical surfaces is not normally performed between batches within a campaign. However, clean-in-place or sterilize-in-place (CIP/SIP) of the direct product-contact parts and filter changes of the product pathway may be performed between production batches. In addition to other activities, the APS for campaigns should include, in addition to other activities, start-of-campaign and end-of-campaign studies in a manner similar to that described for large batches.

In campaign modes, the following situations may be possible and supportable in an appropriately designed APS:

- Manufacturing of multiple-product lots of the same formulation, product, or product family
- Changing configurations or fill volumes
- Campaign lengths substantially longer than one day
- Changing product formulation and/or concentration if cleaning and re-sterilization of the delivery system are performed aseptically
- Breaks such as days or shifts without production are acceptable (i.e., production may not need to be continuous over the campaign period)

Initial and periodic demonstration of campaign duration (total time or batches) should be based on an assessment of the operational elements which is an appropriate activity for the application of QRM approaches (see **Section 3.4**).

7.13 Pre-Incubation Container Inspection

The normal product inspection process is qualified for the removal of nonintegral containers (e.g., missing or misaligned closures, cracks in glass, poor crimps, empty containers) due to the possible breach of product sterility, and other non-container integrity related defects (e.g., presence of particulate, fill volume, cake shape in lyophilized products, cosmetic defects). The routine inspection process should be maintained for APS-filled units, with only nonintegral APS units removed during the preincubation inspection. Alternative

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inspection processes may also be used as long as the inspectors have been appropriately qualified to remove only nonintegral units. The removal of nonintegral units is appropriate because the evaualtion of these units in the APS would yield no valuable scientific information. For the purpose of the APS, cosmetic, particulate, and fill-volume defects should be ignored, and such units should be incubated and included in the APS evaluation and contamination rate. The inspection should be conducted as soon as possible to prevent delay of incubation of the units. (See **Section 4.1.3.4** for the handling of intervention-related units.)

7.14 Incubation Conditions

Prior to incubation, filled APS units should be inverted and manipulated to ensure contact of the medium with all internal surfaces of the closure system before they are incubated. APS units should be incubated for a minimum of 14 days unless supported by another qualified duration or method.

The chosen temperature should be based on its ability to recover microorganisms normally found in the environment or in the product bioburden. Any incubation temperature or temperatures in the range of 20-35 °C may be used if data is available to show the suitability of the selected incubation temperature to support growth, including recovery of facility isolates and/or molds (as indicated by EM data) ***(19)***.

If multiple temperatures are selected, a minimum of 7 days of incubation must be done at each temperature (e.g., seven (7) days at 20-25 °C followed by another seven (7) days at 30-35 °C) ***(19)***. Data should be available to show the suitability of the selected incubation temperature to support growth, including recovery of molds. The selected temperature(s) should be controlled and monitored continuously throughout the incubation period. When more than one temperature is utilized, the lower temperature should be used first, and the time to ramp up to the second temperature should be determined and considered to allow for full incubation period at the second temperature. A maximum time limit should be considered to ensure that inspection occurs as close to completion of incubation as possible, so neither the detection of potential turbidity nor the ability to subculture the potentially contaminated unit is compromised.

The conditions for incubation should be qualified, including the conditions, timing and activities, such as sample inversion, after the period of processing prior to and during incubation ***(1)***. Incubation of filled units should begin without undue delay following completion of the simulated batch, to ensure that any potential contamination can be detected under conditions that reflect the true state of the process. Where the visual inspection process entails a lengthy manual operation, it should be assessed to understand if it can be considered as part of the initial ambient temperature incubation (20-25ºC) or not.

7.15 Post-Incubation Inspection

Firms may choose to do an additional inspection of units part-way through the incubation period (often referred to as an interim read). This can be performed either by lower manipulation of the units (looking for gross contamination) or with a full inspection. Regardless of whether or how it is performed, care must be taken in handling so as not to compromise the units under incubation, and time out of the incubator should be considered in the overall incubation time.

At the end of the incubation period, visual inspection of all APS units for growth is performed to determine the outcome of the APS. The inspection process should be performed by qualified inspectors who have demonstrated the ability to detect both low- and high-level microbial-growth patterns. Oversight should be provided by personnel adequately trained in microbiology and quality assurance.

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The analysis of filled containers for visible particulates may be used to gauge overall process control for intrinsic and extrinsic particle contamination. However, the presence of visible particulate matter should not result in APS failure unless microbiological contamination is detected.

Since the preincubation inspection is expected to remove any units with container–closure defects, if such a defect is detected during the post-incubation inspection, it must be appropriately investigated for cause and corrective action(s) taken. Unless the investigation concludes otherwise, any contaminated units are considered positive regardless of the absence of integrity post-incubation.

7.16 Unit Accountability and Reconciliation

As the target acceptance criterion for an APS study is zero contaminated units, a precise chain of custody for all APS units and accountability is necessary. Accurate counts (i.e., units rejected during filling process, filled integral units, filled nonintegral units rejected during inspection) should be performed at each major process step (i.e., media volume involved, filling, lyophilization, crimping), preincubation inspection, and post-incubation inspection. At the conclusion of the post-incubation inspection, filled units are recounted to verify pre- incubation and post-incubation accountability, with a target accountability of 100%. In the event of a discrepancy, an investigation should be performed to determine the source of the variance and the potential impact on the validity of the APS study.

The reconciliation should include:

- Filled units rejected at filling line (includes capping, if appropriate), including automatic rejections and manual rejections (must be categorized further, e.g., broken vials, empty, not stoppered, not capped, units removed as part of line clearance)
- Filled, setup units (stoppered/integral) for incubation in APS
- Quality control samples (destructive samples, not integral and not incubated)
- Filled units (automatic counting – leaving the fill/capping machine)
- Units counted at preincubation visual inspection
- Units rejected at preincubation visual inspection (nonintegral)
- Units with cosmetic defects at preincubation visual inspection (integral and inspected)
- Units received and placed into incubation*
- Inspected units after 7 days of incubation*
- Inspected units after 14 days of incubation*

***Note:** It is critical that the number of units incubated remain identical for items in listing with a "*".

7.17 Growth Promotion/Positive Control

The growth-promotion properties of the incubation media should be evaluated using pharmacopeial methods. The inclusion of tests for key environmental isolates or those isolated from sterility-test positives are recommended. Growth-promotion tests are commonly performed after 14 days of incubation using negative media-filled units.

The growth medium selected for an APS must demonstrate that it can support the growth of such microorganisms such as bacteria, yeast, and molds should they be present in the manufacturing process or surrounding environment. The media must be formulated properly according to a batch record and include validated mixing times and temperatures to ensure that the media retains its ability to support growth. Media should be of appropriate quality; a vegetable-based medium is the standard because it has the advantage of being mycoplasma, BSE- and TSE-free.

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7.17.1 Sample Collection

Demonstration of the growth-promoting properties of the media used for APS should occur at meaningful steps throughout the study. Samples should be collected throughout the batch after the entire incubation period is complete. The samples should represent the batch by randomly selected containers and also include worst-case. This is typically achieved by taking samples from the beginning, middle and end of the APS run. The worst-case is typically considered to be at start and end:

- Beginning of batch: filter may take more nutrients out at the beginning
- End of batch: the duration of exposure may reduce ability to recover growth

The number of units will depend on how much media is filled into each unit, but the sample volume should adequately represent the batch. This is performed to demonstrate that handling or processing has not compromised the media's ability to support microbial growth. This is often referred to as a positive control. The sampling of media should be performed with care using appropriate aseptic techniques. Sample locations must be justified as appropriate to the specific process and may require additional sample locations and timing. For example, samples can be collected on the bulk-formulated media just prior to use and tested for growth promotion. This can be used to demonstrate that the media, as prepared, is suitable and to support potential contamination investigations.

7.17.2 Method Criteria

Whether compendial indicator microorganisms are sufficient or if facility isolates should be supplemented should be determined. Justification for the types of microorganisms used in the growth-promotion study should be provided, consideration should be given to the use of gram-positive, gram-negative, yeast, and mold samples. The decision to use isolates from EM, sterility-test failure, and/or other microbial recoveries from the process must be justified. The microorganisms should be the same for both the bulk sample and post-incubation test samples. Units should be inoculated with a <100 colony forming units (CFU) challenge for each and repeated for each of the microorganisms required ***(20)***.

7.17.3 Acceptance Criteria

Growth of the indicator microorganism must be visibly present in each of the media samples. If the growth-promotion testing fails (i.e., growth is not demonstrated), the test fails, and the simulation is rendered invalid. If there is growth present other than the indicator microorganism, a lab investigation is required to determine the source of the microorganism and may require retesting. If the growth-promotion fails but contamination is found during the final inspection, the APS should be considered a failure ***(21)***. An investigation is required to determine the cause of any failure, corrective actions must be implemented, and the APS must be repeated.

7.18 Post-Simulation Cleaning

Defined procedures should be in place, and cleaning validation should account for the effective removal of residual media from direct and indirect product-contacting equipment surfaces. Procedures on how to handle spillage during filling should be in place to avoid media entering difficult to clean areas. Where it is not possible to clean surfaces that may have been contacted by media, such as for gaskets and seals, these should be replaced.

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8.0 Personnel Qualification

The qualification program should be risk-based and commensurate with the complexity and the level of personnel skill required for a specific role in the aseptic manufacturing process. Qualification of personnel for critical aseptic interventions (e.g., machine parts assembly, filling-nozzle adjustment, or manual aseptic compounding), might require the highest level of qualification efforts, starting with selection, training, supervised execution, demonstration and, finally, participation in a successful APS.

A phased approach to qualification is useful in breaking down this complex qualification process into different stages, which allows the personnel to advance from lower-risk activities to higher-risk activities, based on meeting different levels of qualification criteria. The need for direct supervision should be commensurate with these levels, with more direct supervision required initially and, when competence is demonstrated, less supervision required as they progress through levels of qualification.

A typical qualification program for beginners or new employees could include stages to qualify for entry into the aseptic cleanroom, followed by qualification for performing less-critical tasks under supervision and, finally, qualifying for critical interventions (see **Figure 8.0-1**). In every stage of qualification, there is a need for an appropriate mix of training, practicing, supervised execution, and final assessment of performance. For the existing employees in the organization, a mechanism should be established to reassess their performance at regular intervals to ensure their qualification status. This should be a formally documented qualification program to ensure that those personnel can perform their tasks effectively, reliably, using appropriate technique, methods, and behaviors that minimize the potential risk of microbiological contamination to product. Company-specific qualification acceptance criteria should be established in a phased qualification process.

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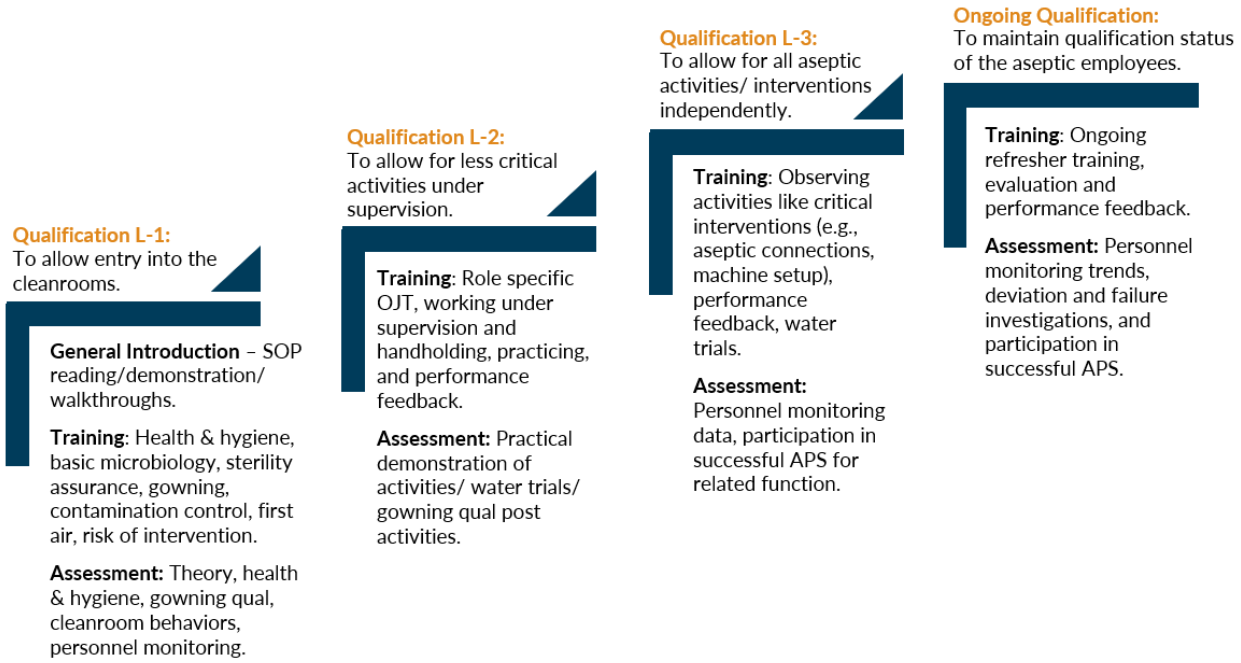

Figure 8.0-1 APS Personnel Qualification Criteria

Note: The level of training and assessment may vary based on qualification, prior knowledge and experience, role to be played, and the technology in use.

8.1 Qualification Level 1

Qualification Level 1 (L-1) represents the initial stage of personnel qualification and grants access and entry into the cleanroom environment to new or entry-level. This foundational training is designed for persons new to the cleanroom and focuses on:

- General health and hygiene
- Basic Microbiology
- Sterility assurance principles including contamination control ***(18)***
- Gowning
- Cleanroom behaviors
- First air principles ***(4)***
- Impact to patients from a potential contamination

These critical topics are required for personnel operating in an aseptic environment, which is in alignment with current regulatory expectations. To ensure comprehension at this initial stage, the program should include theoretical instruction delivered in a classroom or computer-based immersive setting, followed by proficiency-based assessments. Then, gowning qualification should adhere to a structured procedure, beginning with a visual inspection of the individual's ability to properly don sterile garments and concluding with a microbial assessment to verify compliance with aseptic standards. This comprehensive approach ensures personnel are adequately prepared for working in controlled environments while also meeting regulatory and operational requirements.

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8.2 Qualification Level 2

Qualification Level 2 (L-2) builds upon the foundational training of Qualification L-1, enabling personnel to perform less critical activities while still under supervision within the cleanroom environment. Training at this level includes role-specific on-the-job training (OJT) while working under direct supervision and with guided practice towards a specific set of technical skills, accompanied by performance feedback to ensure gradual skill development. Competency-based assessments focus on practical demonstrations of assigned activities, including water or media trials, and are designed to evaluate the trainee's ability to perform tasks accurately and consistently while maintaining proper aseptic technique. Additionally, gowning qualification is reassessed post-activity for further assurance of cleanliness and aseptic standards. This stage of qualification ensures that personnel can contribute to cleanroom operations under supervision while gaining the experience and competence required for more advanced responsibilities.

8.3 Qualification Level 3

Qualification Level 3 (L-3) represents the advanced stage of personnel training and qualification, preparing individuals to independently perform critical aseptic procedures and interventions with minimal supervision. This stage builds upon the foundational and intermediate qualification levels (Qualification L-1 and L-2) and focuses on mastering high-risk tasks that need specific skills to operate. Training at this level includes role-specific, hands-on activities designed to develop competence in executing critical interventions, such as aseptic connections, equipment setup, and/or manual aseptic manipulations. Personnel are required to participate in simulated trials, such as water fills, that mimic actual production conditions. These simulations, combined with regular feedback from supervisors, enable personnel to refine their technique and bolster confidence. The qualification process at this stage involves demonstrating consistent compliance with aseptic behavior, positive trends in personnel monitoring results, and finally successful participation in an APS run in which all personnel perform the same roles as required in routine production. Qualification L-3 ensures employees are fully equipped to operate independently in high-risk aseptic environments.

8.4 Ongoing Qualification

Ongoing Qualification (at least annually) is a critical component that confirms maintenance of competence and readiness of personnel working in aseptic environments. It ensures that aseptic processing personnel continue to meet the high standards required for contamination control and sterility assurance. Regular refresher training sessions are essential to reinforce fundamental aseptic techniques, contamination control principles, and current regulatory requirements. Additionally, ongoing performance evaluations and feedback from supervisors and aseptic observers help identify and address any skill gaps. The assessment component of ongoing qualification involves analyzing personnel monitoring trends to identify abnormalities, determining root causes, and implementing corrective actions, as well as regular participation in APSs to ensure continued proficiency in critical aseptic techniques under production conditions. A structured ongoing qualification program ensures that aseptic processing personnel remain proficient, reliable, and compliant and contribute to product quality and patient safety.

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9.0 Aseptic Processing Simulation Acceptance Criteria

The acceptance criteria for the number of positives in any APS should be zero, regardless of the number of units filled during the APS. Meeting the acceptance criteria alone does not definitively validate the acceptability of the aseptic process: a comprehensive evaluation of the design of the process, the robustness of the controls, and its continuous monitoring are necessary to ensure the process suitability to prevent product contamination.

Failure to meet the acceptance criteria requires a thorough documented investigation to determine the cause of the positive unit(s), and what steps should be taken to address the cause (see **Section 10.0**). However, if the outcome of the investigation concludes that the root cause is unrelated to the aseptic process being simulated, the result may be acceptable.

Failure to identify the cause of the positive unit(s) does not mean that the aseptic process is acceptable; in this case, a most probable root cause(s) should be identified and addressed before repeating the APS run, with considerations for impact on the production batches manufactured prior to the APS (see **Section 10.0** for details).

9.1 Recommendations

In addition to the zero positive units acceptance criteria, several other factors should be taken into consideration when considering the overall suitability/success of an APS ***(25)***. These factors can include:

- First consideration is product and patient safety. Hold all potentially impacted product and evaluate the need for a recall based on the outcome of the investigation.
- Consideration should be given to the use of proper aseptic behavior and/or technique as part of the overall assessment of the suitability of the APS.
- APS should run as intended and follow the requirements specified in the APS protocol or procedure and should simulate the process as closely as practical.
- Aseptic process should run as intended, with any deviations and excursions investigated and resolved, including observation of unacceptable aseptic behavior or technique.
- Media should be shown to support anticipated levels of microbiological growth.
- Rationale for the APS study procedure, limits, and worst-case conditions should be justified, documented, and based on assessments of the relative risks to the aseptic process.
- The investigation must determine one or more root or plausible causes of the positive unit(s). It is not sufficient to continue performing APS runs without determination and correction of the cause of the failed APS.
- Following the investigation, corrective and preventive actions (CAPA) should be determined and implemented to eliminate the cause or potential causes.
- The investigation, conclusion, and CAPAs required to eliminate the cause of the contamination should be supported by scientific evaluation and risk assessment.

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- Where these CAPAs change the aseptic process, those changes should be qualified, reflected in changes to relevant SOPs and training and, if required, validated.
- The APS should then be repeated to help determine if the CAPA changes achieved the intent of preventing the initial contamination, and to show that there are no unintended consequences of the CAPA changes that adversely affect the aseptic process.
- The number of APS runs, subsequent to the failure investigation and CAPA changes, should be risk-based and may depend on the impact of the changes and clarity in the identification of the root cause. Typically, three (3) runs are performed, unless there is clear evidence that the positive unit was not the result of a failure of the aseptic process.

9.2 Invalidating an Aseptic Process Simulation Run

Invalidating an APS run should be well documented and investigated within the Quality System, typically through a deviation or nonconformance investigation, depending on the reason for invalidation. An invalidated APS is one that is completed but is not evaluated as part of the overall APS study. An APS can be invalidated if it has been compromised by conditions not normally a part of the aseptic process being simulated which render the results of the APS run invalid or unusable.

The incubation and inspection of an invalidated APS does not have to proceed, and the media-filled units may be discarded without inspection and evaluation.

The event(s) or conditions that will invalidate the APS should be documented in a procedure, and the reason for the invalidation of a particular APS should be justified and documented. Examples of reasons to invalidate the APS, include:

- Definitive evidence that the media used for the APS was not sterile (e.g., sterilizing filter integrity-test failure)
- Improper preparation of media
- Failure of growth-promotion of media after incubation, providing there are no positive units in the APS, if inspection of the media-filled units has occurred
- Discovery of other conditions or use of materials that would negate the results of the APS

However, if units are incubated and evaluated, then any positive units should be investigated and may constitute an APS failure depending on the outcome of the investigation.

9.3 Aborting or Discontinuing an Aseptic Process Simulation

An APS may be aborted or discontinued under the authority of Senior Production and Quality Management or equivalent and only if conditions are present that would result in the stoppage of an actual production run. The conditions under which production must be stopped should be clearly defined in a procedure and any such events must be documented within the Quality System.

The event(s) or conditions that will cause the discontinuation of an APS or production batch should be identical and documented in a procedure. The reason for discontinuing a particular APS should then be justified and documented as part of the APS study report, with the results of the filled-unit incubation and evaluation from that APS. Examples of reasons to discontinue an APS run include:

- Production equipment failure, malfunction, or prolonged stoppage
- Power outage
- Air-handling stoppage
- Material shortage

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- Product line or vessel leakage or disruption
- Other conditions that would not allow the APS or production batch to continue

Any units that have been filled prior to the event that led to the discontinuation should be incubated and evaluated. If positive units are found (and are unrelated to the discontinuation event), then the APS may be considered as failed, regardless of the number of units filled and would need to be investigated as such. However, if an aborted APS shows no positive units, then the aborted APS should not be considered as a passing, unless the requisite number of units have been filled and all other conditions and requirements of the APS have been met.

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10.0 Considerations for Investigation

Some key elements to include in the investigation for the presence of contaminated units in an APS are described below ***(21)***. Additional points to consider can be added, based on the knowledge of the manufacturing process in question:

- How far back to go to assess potential impact can depend on the individual circumstances surrounding the failure and the identified root cause. While it is common to look back to the previous successful APS, there are circumstances that can dictate a smaller or larger time frame for evaluation of potential impact. For example, if an APS failure is correlated to a mechanical malfunction of a specific piece of equipment, and the failure mode is easily detectable in routine maintenance, the time frame can be set to the previous maintenance event that demonstrated the equipment was operating within specification. Alternatively, if the failure is found to be due to an intermittent equipment fault, one that would not have been detected by maintenance or routine operation, it may be necessary to go back and evaluate all manufactured batches within expiry. In such a case, going back to the previous successful APS is not meaningful, as the fault was already determined to be intermittent and difficult to detect.
- When evaluating the potential impact on previously manufactured batches, not only must the time frame be defined, but also the scope. The fact that an APS failure occurred in a single line, for a specific process, does not immediately confine the evaluation of that line or process. For example, if the failure is attributed to an issue related to a lyophilizer, then it must be assumed that it could impact any product that was processed within that unit (regardless of in which line it was filled). Conversely, any products filled on the impacted line that did not utilize the lyophilizer would be excluded from potential impact. In each case, it is important to justify the time frame and the scope selected for the evaluation of impact to previous batches.
- A detailed history of the investigation should be maintained.
- When positive units are encountered, all possible sources of contamination should be investigated.
    - All positive units should be identified at a minimum the species level (strain typing or similar is recommended). A comprehensive sampling and identification scheme is critical in the investigation and determination of the contaminant source. In cases of widespread contamination, it is not practical to identify each individual positive vial. Therefore, a statistically significant plan that is representative of the aseptic process is acceptable.
    - The identification of the contaminating microorganism should be compared to the database of the microorganisms identified through the EM program (i.e., air viables, equipment surface, personnel).
    - The mass spectroscopy/ biochemical and/or genetic profile of the contaminating microorganisms should also be compared to that of microorganisms obtained from testing programs, including testing of sterility and bioburden (e.g., water, gases, raw materials, drug substance, drug product), as well as any prior APS recovery, to help identify the potential sources of the contaminant.

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  - Isolates should be checked for possible identification matches, especially from areas that exceed their limits or are trending upward.

- It is important not to limit the review to the immediate filling room; all airlocks and supporting areas should be included in the assessment. For this reason, it is helpful to retain any EM samples that exhibit growth until the incubation of the APS units is finalized. In addition, literature references describing possible sources of contaminating microorganisms may be helpful in locating the point of entry into the process.
- A batch-production record, similar to that for routine production, should exist for each APS.
- Deviations, downtimes, and repairs, before or during the filling, should be carefully evaluated.
- If available, a video recording of the APS should be carefully reviewed to identify any potential events that could have impacted the batch. If not, notes from APS observers can also yield information regarding unexpected circumstances or observed behavior that can be useful in narrowing down or eliminating potential root causes.
- Operations and quality assurance shop-floor personnel should be interviewed as soon as possible to determine if any unusual circumstances occurred.
- Maintaining traceability of APS containers, so they can be traced to approximate fill times and even bracket particular interventions, can be extremely useful during an APS failure investigation. Therefore, every effort should be made during any after-filling step (e.g., capping) and subsequent visual inspection to maintain the traceability of the container-filling order. It is important to note that the use of accumulator tables anywhere in the process can significantly impact container order, and that must be taken into account when attempting to set a time bracket for the event that could have led to the contamination.
- Filter integrity-testing results and all sterilization records associated with product components and equipment should be reviewed.
- Cleaning and sanitization records should be reviewed, including type of agent, formulation of the agent (if required), and when, where, and how it was used, ensuring contact time was met.
- Critical systems (e.g., HVAC, compressed air/gas, water, steam) should be reviewed for documented changes and requalification or acceptance criteria for those changes.
- Calibration records should be checked.
- HEPA filters in the filling area should be inspected and recertified, if warranted.
- Training and personnel qualification records for all individuals (e.g., production, maintenance, EM, cleaning) involved in the fill should be reviewed to identify any possible gaps.
- Change management and validation records related to the specific APS should be reviewed for any design, procedure, or process changes that have the potential to affect the aseptic process.
- All deviations from the original validation should have an associated justification for not performing a new validation. The justification for acceptance of initial or on-going validation deviations impacting the aseptic process should be reviewed.
- Based on the outcome of the investigation, an assignable plausible cause of the failure must be determined, and corrective action(s) need to be taken and documented. Where an assignable cause cannot be determined, the investigation should continue until one or more plausible causes can be determined. The aseptic process should not proceed without a cause assigned and corrected.
- Observers should be periodically observing routine operations, as well as APS. These observers should have specific expertise in aseptic processing and knowledge of the APS procedure to assess whether participants have used proper aseptic practices, behave as required and executed activities according to standard operating procedures, batch records, and protocols. Any inappropriate practices, if detected, should be addressed.
- If an APS fails, potential batch impact to previously manufactured batches must be evaluated, as well as any that may have been manufactured after the APS.

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The investigation report should contain:

- Summary of the occurrence
- List of all the systems investigated, not just those tied to the failure, along with a conclusion regarding whether the system contributed to the failure (with supporting documentation in each case)
- Identify the microorganisms to at a minimum the species level (strain typing or equivalent is recommended)Nature of the microorganism(s) identified and any relevant factors with regards to the microorganism(s) (e.g., whether it is a waterborne microorganism or a spore former)
- Conclusion as to root cause(s) and supporting documentation (if discovered) from an investigation completed in a timely fashion
- Potential impact on previous batches produced (justifying the scope and time frame for the review)
- Corrective action(s) taken and supporting documentation
- Justification for the number of additional APS to be completed after appropriate CAPA completion
- Outcome of additional APSs, if performed
- Assess the need for regulatory notification
- Appropriate signatures of the investigators of the individual systems and, at a minimum, the Production and Quality Units

This list is not intended to be all-inclusive. Additional elements may need to be added depending on the process. If the investigation determines that the cause of the contamination is not related to the performance of the aseptic process, for example, the contaminated units are a result of laboratory error or the use of nonsterile media, then the aseptic process has not been determined to be unacceptable. However, the APS run should be considered invalidated and replaced with a single APS.

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11.0 The Use of Aseptic Process Simulation in Ongoing Process Evaluation

The aseptic process should be evaluated on an ongoing basis to determine if:

- The state of control for all processes and systems that impact aseptic processing has been maintained since the last assessment.
- Changes to or variations in the aseptic process have not adversely affected the ability of the aseptic process to prevent microbiological contamination of sterile product.

11.1 The Role of Aseptic Process Simulation in Ongoing Process Evaluation

The performance of ongoing APS studies, in addition to periodic requalification tests, is used to help demonstrate that the aseptic process remains in a state of control. Each firm should determine the frequency of and the interval between ongoing APS for each process including consideration of regulatory requirements, additional risk-based assessments, processing-line controls, and performance, including:

- Changes to the process
- EM data (including utility systems, facility systems, and personnel)
- Sterility-test data (including increases or changes to prefiltration bioburden)
- Sterilization validation data
- Process-related failure and deviation investigations
- Updated risk assessments
- CCS

11.2 Periodic (Time-Driven) Ongoing Aseptic Process Simulation Assessment

Regardless of any events that could result in the need for the performance of an event-driven APS, a semiannual interval (twice per year, approximately every six months) between APSs should be performed to help determine if process variables have adversely affected the ability of the aseptic process to control or prevent microbiological contamination. The periodic APS time-interval range and criteria and the response for any time-interval deviation should be risk-based and defined in a procedure. If no process changes are noted and unless there is evidence or reason to the contrary, a single APS run should be sufficient.

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11.3 Event-Driven and For-Cause Ongoing Aseptic Process Simulation Assessment

The APS for ongoing evaluation should not be strictly driven by time. Performance of APSs prior to or in addition to the scheduled reassessment may be necessary following a process change or other event of such scope that previous simulation studies would no longer be representative or applicable, or where such events are assessed to add risk to the process. The criteria for performing event-driven APS runs should be documented in a procedure, typically part of the change-control system, including what types of changes, the extent of those changes, and the activities that necessitate the run. Examples of events that may result in additional APS studies (whether to complete validation of the changes or to confirm effectiveness of implemented corrective action) include those associated with:

- Changes to process or process controls and facility or equipment
- Extensive maintenance events or equipment modifications (including improvements)
- Before and/or after a facility shutdown, as warranted
- After extended periods of inactivity
- Prior to decommissioning or relocating a line
- Following APS or other process-performance failures, significant deviations, or excursions

In such cases, the number of APSs may vary, depending upon the extent of the event.

11.4 Other Considerations

There may be several different permutations of a filling process that take place on a given filling line. For example, if a risk assessment indicates that a bracket approach of given process aspects (e.g., speeds, container configurations, etc.) is warranted, then the inclusion of different bracketed processes or component configurations or system parts may be periodically included in APS runs. This approach may not be applicable to dissimilar equipment and systems (e.g., displacement pumps and peristaltic pumps). In such cases, one approach may be to perform APSs for these processes on a rotational basis, with each process challenged at least annually. Depending upon individual circumstances, however, more frequent APSs may be necessary. The rationale for the rotation or alternation of different permutations should be documented in the overall APS approach of the company or site, for example, in the Site Validation Master Plan.

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12.0 References

1. European Commission. *Annex 1: Manufacture of Sterile Medicinal Products, EudraLex – Volume 4 – EU Guidelines for Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use*; European Commission: Brussels, 2022.
2. Parenteral Drug Association. *Technical Report No. 13-2: Fundamentals of an Environmental Monitoring Program Annex 1: Environmental Monitoring of Facilities Manufacturing Low Bioburden Products*; PDA: Bethesda, Md., 2020.
3. International Organization for Standardization. *ISO 14644-1:2015 Cleanrooms and Associated Controlled Environments — Part 1: Classification of air cleanliness by particle concentration*; ISO: Geneva, 2015.
4. U.S. Food and Drug Administration. *Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing—Current Good Manufacturing Practice*; U.S. Department of Health and Human Services: Rockville, Md., 2004.
5. U.S. Food and Drug Administration. *Guidance for Industry: Process Validation: General Principles and Practices*; U.S. Department of Health and Human Services: Silver Spring, Md., 2011.
6. U.S. Food and Drug Administration. *Guidance for Industry: Quality Systems Approach to Pharmaceutical CGMP Regulations*; U.S. Department of Health and Human Services: Silver Spring, Md., 2006.
7. Parenteral Drug Association. *Technical Report No. 61: Steam In Place*; PDA: Bethesda, Md., 2013.
8. International Council for Harmonisation. *Quality Guideline Q9(R1): Quality Risk Management*; ICH: Geneva, 2023.
9. Parenteral Drug Association. *Technical Report No. 44: Quality Risk Management for Aseptic Processes*; PDA: Bethesda, Md., 2008.
10. Parenteral Drug Association. *Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations*; PDA: Bethesda, Md., 2012.
11. Parenteral Drug Association. *ANSI/PDA Standard 03-2025: Standard Practice for Quality Risk Management of Aseptic Processes*; PDA: Bethesda, MD, 2025.
12. Baseman, H, et al. A Line of Sight Approach for Assessing Aseptic Processing Risk: Part III. *PDA Letter* **2016**.
13. Parenteral Drug Association. *Points to Consider No. 1: Aseptic Processing (Revised 2023)*; PDA: Bethesda, Md., 2023.
14. Hamilton, D, et al. A Better Approach to Aseptic Process Simulation for Lyophilized Products. *Pharmaceutical Online* **2021**.
15. Association, P D. *Points to Consider for Microbial Control in ATMP Manufacturing*; PDA: Bethesda, MD, 2022.
16. Parenteral Drug Association. *Technical Report No. 66: Application of Single-Use Systems in Pharmaceutical Manufacturing*; PDA: Bethesda, Md., 2014.

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17. Prout, G, et al. *The Use of Process Simulation Tests in the Evaluation of Processes for the Manufacture of Sterile Products: No. 4: Technical Monograph*. The Parenteral Society: Brazil, 1993.
18. Parenteral Drug Association. *Technical Report No. 90: Contamination Control Strategy Development in Pharmaceutical Manufacturing*; PDA: Bethesda, Md., 2023.
19. Whyte, W; Eaton, T. Microbial Risk assessment in Pharmaceutical Cleanrooms. *Eur J Paren Pharm Sci* **2004,** *9*, 16-23.
20. U.S. Pharmacopeial Convention. General Chapter <71> Sterility Tests. In *USP 41–NF 36*, USP: Rockville, Md., 2018; p 5984.
21. Parenteral Drug Association. *Technical Report No. 88: Microbial Data Deviation Investigations in the Pharmaceutical Industry*; PDA: Bethesda, Md., 2022.
22. Parenteral Drug Association. *Technical Report No. 1 (Revised 2007): Validation of Moist Heat Sterilization Processes: Cycle Design, Development, Qualification and Ongoing Control*; PDA: Bethesda, Md., 2007.
23. Parenteral Drug Association. *Technical Report No. 26 (Revised 2025): Sterilizing Filtration of Liquids*; PDA: Bethesda, Md., 2025.
24. Parenteral Drug Association. *Technical Report No. 69: Bioburden and Biofilm Management in Pharmaceutical Manufacturing Operations*; PDA: Bethesda, Md., 2015.

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13.0 Appendix 1: Media and Placebo Considerations

In the conducting of aseptic process simulation (APS) for suspensions, ointments, creams, and dry powder fills, the use of a sterile placebo or surrogate material in addition to growth-promoting media is common. The placebo or surrogate material should closely resemble or mimic the physical characteristics of the product being simulated without appreciably adding or reducing the level of risk associated with filling or processing these materials. Surrogates or placebos may include such materials as dry powders, thickening agents, buffers, water for injection (WFI), or other materials that mimic product.

Material utilized as a placebo or surrogate should be packaged and handled as closely as possible to the material being simulated, and the justification for its appropriateness as a substitute for the simulation should be documented. Any differences in the packaging and handling of the placebo should be assessed for potential impact to the process and the need for additional cleaning and/or dedicated cleaning validation.

13.1 Selection of Placebo or Surrogate Material

The selection of placebo material to be used in APS should consider several factors: the placebo or surrogate material must be sterile; it must be as similar as possible to the product characteristics; and it must support anticipated levels of microorganism contamination and growth.

Due to the poor flow properties, the selection of dry sterile media for APS has been proven to be less effective for suspensions, ointments, creams, and dry powder fill operations. The use of lactose, mannitol, polyethylene glycol 6000, and sodium chloride for APS in these operations has been found successful, providing it does not adversely affect the growth of microorganisms. The chosen material should be easily sterilizable (using a validated method), dispersible or dissolvable, not inhibit growth (at minimum), and it should be easily handled in the mock-formulation processes.

13.2 Sterilization of Placebo or Surrogate Material

The material should be sterilized prior to the APS using a validated method. Methods usually employed include irradiation, gas, dry heat, or filtration. The sterilization process should be shown to have no significant effect on the media's growth-promotion or the surrogate's physical properties. Along with the placebo material prepared for use in the filling trial, additional material in separate containers can be used for negative-control sterility testing, if necessary.

13.3 Growth-Promotion Testing of the Placebo or Surrogate Material

Growth-promotion testing of the surrogate and media combination should be performed using pharmacopeial methods. Consideration should be given to include in-house isolates recovered from EM, personnel monitoring, and sterility-test failures. If the initial concentration of the growth-promoting media is

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different from the end concentration (i.e., double strength because of subsequent dilution steps), the initial concentration should also be challenged for growth promotion (see **Section 7.17**).

13.4 Solubility Testing of the Placebo or Surrogate Material

The solubility of the placebo materials at the desired concentration is determined in the test medium. The amount of agitation required to dissolve or disperse the material, as well as the time and extent of dissolution, should be noted. If the material fails to dissolve or disperse fully, it can be retested at a lower concentration or replaced.

13.5 Media Preparation and Sterilization

The three critical factors in media preparation for APSs are that the media is sterile, is growth-promoting, and does not require handling procedures that materially change (or add steps to) the aseptic process being simulated. Media formulation and sterilization are very different exercises from product compounding and filtration. Media used for APSs may be liquid or powder, depending on the type of filling process to be simulated.

Soybean-casein digest medium (SCDM), also referred to as tryptic soy broth (TSB), is often used to recover and promote the growth of bacteria and fungi. SCDM is a broad-spectrum medium suitable for the recovery and growth of the typical human flora that predominates in the aseptic processing environment. SCDM is also capable of recovering spore-bearing microorganisms, gram-negative bacteria, and fungi. In recent years, plant-based media (with almost identical formulations but with plant peptones in place of casein) have become increasingly popular due to the similar growth promoting capabilities and assurance of BSE/TSE free. In addition, plant based media is not a potential of mycoplasma either.

Media may be supplied from a vendor as presterilized and ready to use, or it may be prepared and sterilized at the manufacturing site. The sterilization of SCDM prior to filling may be accomplished through a variety of methods, most commonly, by moist heat, radiation, or filtration. The type of sterilization potentially impacts the growth properties of the media so it must be chosen and confirmed as being able to be used after the specific sterilization process is completed. Where mycoplasma is determined or suspected to be a contamination risk from the media source, the use of heat or radiation sterilization can be used to eliminate potential mycoplasma contamination that may not be eliminated by filtration. When filtration is chosen as a method, a special consideration for using mycoplasma-retentive 0.1µm filtration may be needed. The moist-heat sterilization process in autoclaves is very reliable, especially where relatively small quantities of media are required (e.g., APSs completed to evaluate clinical scale or other low-throughput operations) ***(22)***. However, in larger process-simulation batches required in support of commercial-scale manufacturing, it may be difficult to produce the required large volume of media in an autoclave. Media for such larger batches may be sterilized in place in a suitable vessel through a validated process. It is important to note the manufacturer's recommendations regarding sterilization time and temperature to ensure the sterilized media retains its requisite growth-promotion capabilities. F_0 values (parameter that quantifies the lethality of a sterilization process) in the range of 15 to 20 minutes are generally sufficient. Longer or hotter sterilization cycles delivering higher levels of lethality may cause a Maillard reaction (including caramelization) of the medium and adversely affect growth promotion. It can also be useful to combine sterilization methods, for example, purchasing irradiated powder media (to ensure no mycoplasma) and generating a very low bioburden solution by dissolution in WFI, followed by a final sterile filtration prior to filling.

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The filtration system used to produce sterile products is validated independently of the APS and does not require further validation by virtue of an APS ***(23)***. However, any post-filtration aseptic manipulation of products, such as disconnection, reconnection, and transfer (e.g., using a transfer loop) shall be covered under APS. Wherever possible, the same filtration system and vessels employed for product manufacturing should be used, even if the media is purchased ready to use; however, in many cases, this is not technically feasible or practical. The sterilizing-grade filter may not be identical to the one used for the product being simulated, but it should be sized properly for the preparation of the required volume of media. The use of prefilters may be required as media may contain a substantial number of fine particles that could clog the primary sterilizing-grade filter.

The hold time between compounding the media and filtration of the media into a sterile vessel should be minimized. Media held in less than fully sterile conditions will immediately begin to support the growth of any microorganisms that may be present. Bioburden issues can be mitigated to some degree by preparing the media using hot water ***(24)***. Water temperatures above 60 °C will reduce the risk of growth of vegetative bacteria and fungi and ensure that the media dissolves more completely, reducing the likelihood of filter blockage. When appropriate, use of a prefilter to remove material that does not go into solution will reduce the likelihood of clogging the final sterilizing-grade filter, which usually has a 0.2µm pore-size rating. It is preferable to sterile-filter the media into a sterile holding tank prior to the start of the fill rather than to hold nonsterile media for several hours. Media should be allowed to cool to <35 °C before use in an APS.

Powdered media employed to challenge powder-filling or aseptic-compounding operations is sterilized by radiation. The radiation sterilization process should be validated, and each lot should have dosimetry information included in its certificate of analysis. It may be advisable to sample and evaluate media for reconstitution and growth-promotion prior to use in an APS.

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14.0 Appendix 2: Aseptic Process Simulation Performance Sequence

The activities in a typical APS generally occur in the order presented in **Table 14.0-1**.

Table 14.0-1 Aseptic Process Simulation Performance Sequence Checklist

Activity	Additional Information*	Completed
1. Prior to Initiating Aseptic Process Simulation Confirm Satisfactory Completion of the Qualification, Validation, and Smooth Operation of the Aseptic Process and its Support Systems, including (see Section 3.1):		
Sterilization Process Validation	• *Technical Report No. 1: Validation of Moist Heat Sterilization Processes: Cycle Design, Development, Qualification and Ongoing Control* • *Technical Report No. 3: Validation of Dry Heat Processes Used for Depyrogenation and Sterilization*	☐
Standard Operating Procedures (SOPs) are Available	N/A	☐
Personnel Training on the Applicable Procedures and Qualification Status	N/A	☐
Personnel Aseptic-Gowning Certification	• **Section 8.0 Personnel Qualification**	☐
Disinfectant Qualification	• *Technical Report No. 70: Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities*	☐

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Activity	Additional Information*	Completed
Implementation of Environmental Decontamination Procedures (cleaning, disinfection and isolator decontamination)	• *Points to Consider for the Aseptic Processing of Sterile Pharmaceutical Products in Isolators* • *Technical Report No. 34: Design and Validation of Isolator Systems for the Manufacturing and Testing of Health Care Products* • *Technical Report No. 70: Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities*	☐
Facility Cleaning and Decontamination Program	• *Technical Report No. 70: Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities*	☐
Equipment Cleaning Validation	• *Technical Report No. 29: Points to Consider for Cleaning Validation Revised* • *Technical Report No. 49: Points to Consider for Biotechnology Cleaning Validation*	☐
Product, Container-Closure, and Equipment Sterilization	• *Technical Report No. 1: Validation of Moist Heat Sterilization Processes: Cycle Design, Development, Qualification and Ongoing Control* • *Technical Report No. 3: Validation of Dry Heat Processes Used for Depyrogenation and Sterilization*	☐
Manufacturing and Fill-System Qualification	• *Manufacturing Technology Guide No.1: Aseptic Filling, Engineering, and Operation*	☐
Container-Closure Integrity	• *Technical Report No. 27: Pharmaceutical Package Integrity*	☐
Environmental Controls (e.g., air flow, high efficiency particulate air (HEPA) filtration, temperature and humidity control, and viable and total particulate control (Environmental Monitoring Performance Qualification (EMPQ))	• *Technical Report No. 13: Fundamentals of an Environmental Monitoring Program*	☐
Airflow Visualization Studies	N/A	☐

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Activity	Additional Information*	Completed
Utility Gases Sterilization (product-contacting)	• *Technical Report No. 40: Sterilization Filtration of Gases*	☐
Materials Disinfection and Transfer Controls	• *Points to Consider No.1: Aseptic Processing*	☐
Filter Integrity for Sterilizing Filters as well as Vent/Gas Filters (where applicable)	• *Points to Consider for Implementation of Pre-Use Post-Sterilization Integrity Testing* • *Points to Consider for Risks Associated with Sterilizing Grade Filters and Sterilizing Filtration* • *Technical Report No. 26: Sterilizing Filtration of Liquids*	☐
Definition of Environmental Monitoring (EM) Program	• *Technical Report No. 13: Fundamentals of an Environmental Monitoring Program*	☐
Equipment Qualification (especially runnability studies)	N/A	☐
2. Define the Scope of the APS to Include Routine Aseptic Process Conditions and Variables Such As:		
Aseptic Formulation/Compounding Process, Equipment, and Operations	N/A	☐
Aseptic Filling Process, Equipment, and Operations	• **Section 7.0 Aspects of Aseptic Process Simulation**	☐
Operation Conditions (line speed, pressure, and other critical process variables)	• **Section 3.4 Risk Assessment for Aseptic Process Simulation Design** • **Section 5.2 Protocol and Procedure Preparation** • **Section 7.6 Filling Speed** • **Section 7.7 Fill Volume** • **Section 7.10 Number of Units**	☐
Operation Hold Times Impacting Sterility Assurance	• **Section 7.9.4 Qualification of Maximum Hold Times**	☐

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Activity	Additional Information*	Completed
Number of Operators (maximum room occupancy) or Maximum Simultaneous Isolator-Glove Usage	• **Section 4.2.6 Other Dosage Forms and Drug/Device Combinations** • **Section 4.3.7 Considerations for Advanced Therapy Medicinal Products** • *Points to Consider No. 12: Restricted Access Barrier Systems* • *Points to Consider for the Aseptic Processing of Sterile Pharmaceutical Products in Isolators*	☐
Process Setup, Interventions (inherent and corrective), Stoppages, and the Frequency with which they need to be executed and/or simulated	• **Section 4.1.4 Interventions** • **Section 5.2 Protocol and Procedure Preparation** • **Section 5.3 Aseptic Process Simulation Execution Record** • **Section 7.8 Interventions**	☐
Maximum Time Operators are in Clean Room	• **Section 7.9.2 Human Performance and Fatigue**	☐
Process Length (including breaks)	• **Section 7.9 Duration of Aseptic Process Simulations**	☐
Container-Component Configuration to be Utilized and Target Fill-Volume	• **Section 7.5 Container-Closure Configuration** • **Section 7.7 Fill Volume**	☐
Stress Conditions on the Routine Aseptic Process	• **Section 7.9 Duration of Aseptic Process Simulations**	☐
Environmental Conditions (air flows, temperature and humidity, and pressure-differential requirements)	• **Section 7.9.3 Environmental and Equipment Considerations**	☐
Aseptic Process and Associated Intervention Risk Assessment	• **Section 7.8 Interventions**	☐

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Activity	Additional Information*	Completed
3. Develop the Aseptic Process Simulation Plan		
Develop a protocol that includes the scope, rationale, justification, and acceptance criteria that establishes APS conditions necessary to qualify normal aseptic operations. The protocol is reviewed and approved by the Quality Unit.	• **Section 5.2 Protocol and Procedure Preparation**	☐
Develop an APS batch record that defines scope and execution requirements and captures the actual execution variables. The APS batch record is reviewed and approved by the Quality Unit. If the commercial-process batch record is electronic, the APS batch record should also be electronic.	• **Section 5.3 Aseptic Process Simulation Execution Record**	☐
4. Conduct the Aseptic Process Simulation		
Conduct and monitor the APS to ensure it is executed in compliance with the APS protocol and batch-record requirements. Filled volume should be sufficient to wet all internal container surfaces and to allow for identification of contamination, but sufficient head space must be ensured to avoid impacts on the growth of potential aerobic contaminants.	• **Section 5.5 Aseptic Process Simulation Observation** • **Section 7.7 Fill Volume**	☐
Intervention-related vials may be rejected (or incubated separately), but ONLY if the same approach is taken during routine manufacturing. Additionally, the number of units rejected must also match (or be lower than) those that would be rejected during routine production.	• **Section 7.16 Unit Accountability and Reconciliation**	☐

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Activity	Additional Information*	Completed
Perform qualified container–closure integrity inspection, retain acceptable units, and initiate accountability, rejecting only those units with container–closure integrity defects.	• **Section 7.13 Pre-Incubation Inspection** • **Section 7.15 Post-Incubation Inspection** • **Section 7.16 Unit Accountability and Reconciliation**	☐
Prior to incubation, invert and swirl APS units to ensure media contact with all internal surfaces.	• **Section 7.14 Incubation Conditions**	☐
Incubate acceptably filled units under defined temperatures and duration. If two temperatures are used for the incubation of the media filled units, the units should be incubated for at least 7 days at each temperature (starting with the lower temperature) ***(1)***.	• **Section 7.14 Incubation Conditions**	☐
Perform inspection (utilizing qualified inspectors and validated inspection conditions) for microbial growth and reconcile the count with units placed into incubation. Segregate any units with positive microbiological growth and submit them for microbial identification. Identify any positive units to the species level and investigate to establish contamination root cause for any acceptably filled units exhibiting positive microbial growth.	• **Section 7.13 Pre-Incubation Inspection** • **Section 7.15 Post-Incubation Inspection** • **Section 7.16 Unit Accountability and Reconciliation**	☐
Perform growth-promotion tests on APS media in post-incubation media-filled units.	• **Section 7.17 Growth Promotion/Positive Control**	☐

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Activity	Additional Information*	Completed
5. General Documentation Requirements		
Document APS results, evaluation, and conclusion in a report approved by the Quality Unit.	• **Section 5.4 Final Report**	☐
Identify and document operators who participated in the APS by the executed APS and the term of their qualification.	• **Section 5.3 Aseptic Process Simulation Execution Record**	☐
Identify and document interventions that were successfully demonstrated by the executed APS. If necessary, update appropriate procedures.	• **Section 5.3 Aseptic Process Simulation Execution Record**	☐
*TR 22 Section or Relevant PDA Technical Document N/A = no relevant PDA Technical Documents for this activity		

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15.0 Appendix 3: Practical Examples of Using Risk Assessment in Aseptic Process Simulation

The practical examples of a risk assessment provided herein are offered to stimulate thought and help those engaged in such evaluations develop and use assessment approaches. The examples are not intended to offer definitive prescriptions, establish flawless criteria and values for the industry, or to suggest a required format or approach. The examples are meant only to provide an approach or format that a company may consider.

These examples are not universally applicable. They may not include ***all*** aseptic process steps, APS considerations, or risk factors. They do ***not*** represent a PDA-endorsed method or the only method that can or should be used. The reader is cautioned not to dwell on the specific aseptic processing details of the assessment because the examples may not present circumstances applicable to the reader's specific aseptic process. Any actual assessment of aseptic processing risk is a combination of factors affecting aseptic process risk and must consider the specific circumstances and challenges under which the process is performed and assessed. Companies should have their own logic for selecting these criteria or risk-analysis methods based on their experience and specific process knowledge.

For more details and assistance on the use and development of approaches using risk assessment formats, see **Section 3.4**. Additional information may be attained from PDA technical documents including, *Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations* series, *Technical Report No. 90: Contamination Control Strategy Development in Pharmaceutical Manufacturing*, *Points to Consider for the Aseptic Processing of Sterile Pharmaceutical Products in Isolators* and the *PDA/ANSI Standard 03-2025: Standard Practice for Quality Risk Management of Aseptic Processes*.

15.1 Example 1: Applying Risk-Based Aseptic Process Simulation Frequency

This example is related to **Section 3.2.** It considers the overall risk of an aseptic process in terms of its complexity, novelty, frequency of operators performing it and the experience of those operators, degree of automation, degree of protection (barrier system), recent changes and history of compliance regarding EM, APS failures, sterility-test failures, deviations, and audit findings.

An overall risk score is then calculated and used to define the appropriate frequency of APS runs (see **Table 15.1-1**). In most cases, the minimum regulatory frequency (twice per year) will apply. For high risk that cannot be mitigated due to highly manual steps in ATMPs, a higher APS frequency will be required to re-qualify each operator every 6 months rather than every 12 months. This will be indirectly reflected on the total number of APS runs per year, example: a company that has two operators will be required to conduct at least 4 APS runs per year for the same process to comply with EU Annex 1 regardless of risk level

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obtained from the example. In other cases, when the risk is low, such as when using an isolator, a lower frequency of shift simulation can be justified.

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Table 15.1-1 Applying Risk-Based APS Frequency

Risk Factor	Risk Level (Score)			Assigned Risk Score
	High (5)	Medium (3)	Low (1)	
Process Complexity	High complexity, large number of aseptic assemblies, large number of inherent interventions	Medium complexity, medium number of aseptic assemblies, medium number of inherent interventions	Low complexity, low number of aseptic assemblies, low number of inherent interventions	
Process Novelty[1]	<2 years	2-5 years	>5 years	
Process Frequency[2]	<Once every 3 months	Quarterly to Weekly	Weekly to Daily	
Process Automation	Low (e.g.,high number of manual steps)	Medium (e.g., automated filling line with medium inherent interventions)	High (e.g., dependent on robotics)	
Degree of Critical Zone Protection	Low (e.g., conventional unidirectional air flow unit/ biosafety cabinet)	Medium (e.g., oRABs or cRABs without vapor phase hydrogen peroxide (VPHP) decontamination cycle)	High (e.g., isolator or cRABs with VPHP decontamination cycle)	

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Risk Factor	Risk Level (Score)			Assigned Risk Score
	High (5)	Medium (3)	Low (1)	
Degree of Experience of Technicians in Aseptic Processes[3]	Low: <2 years	Medium: 2-5 years	High: >5 years	
Environmental Monitoring Compliance History	>1 OOL* per month	1 OOL every 1 to 6 months	<1 OOL every 6 months	
Process Compliance History APS Failure/ Sterility Test Failures in Past 2 Years	>1 OOS* per 6 months	1 OOS every 6 to 12 months	<1 OOS every 12 months	
Aseptic Process Compliance History (Process-Related Major Deviations/Audit Findings/ 483s)	>3 major deviations/audit findings/Regulatory observations per year	1-2 major deviations/audit findings/Regulatory observations per year	<1 major deviation/audit finding/Regulatory observation per year	
Recent Major Changes in Facility/Equipment/Key Personnel	>3 per year	1-3 per year	<1 per year	

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Assigned Risk Level Total	>35	15-35	<15	
Recommended APS Frequency	Mitigate risk level of the process. If mitigation is not feasible with current technologies, and the process is fully manual (e.g., ATMPs), personnel participation in APS should occur every 6 months (+/- 1 month) per shift, rather than increasing the overall frequency of APS runs.	Conduct APS every 6 months, +/- 1 month for each shift.	Lower frequency per shift (e.g., every 12 months) may be justified, e.g., in case of well-established closed aseptic process inside gloveless isolator with no human intervention and with no history of OOS results.	N/A

[1] Author's note: Example: A company historically aseptically filling liquid products using stainless steel systems and recently deciding to use single use systems for the first time in their facility.

[2] Author's note: The process-frequency factor risk-scoring measures the level of proficiency that operators are expected to have with the process, given how often it is being executed.

[3] Author's note: Reflects the overall operator proficiency in general aseptic processes (not necessarily just this one under consideration).

* OOL = Out-of-Limit

* OOS = Out-of-Specification

While this example focuses on results above action limit, similar considerations should be given to significant negative trends of results above alert level.

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15.2 Example 2: Intervention Simulation Frequency Based on Application of Intervention Risk Evaluation & Management Model

This example is related to **Section 3.4**. One objective of a sound aseptic process design is to minimize the requirement to perform interventions and to reduce the risk involved as much as possible. The inclusion of a high-risk intervention in an APS does not mitigate its risk during aseptic processing. Hence, in this case study, after completing an intervention risk evaluation & management model (IREM) exercise, each intervention is assessed to determine if a procedural or engineering solution would reduce the risk level and then the process is reassessed for impact of the reworked intervention (***2, 3, 4, 5***). Next, the frequency and number of interventions to be included in the APS are defined based on the preceding year's trend of actual commercial production, and takes into consideration the new risk level of each intervention based on the IREM.

Table 15.2-1 represents a risk-based APS intervention simulation plan that was developed to help determine the frequency of the inclusion of interventions and to ensure that simulations are representative of all actual aseptic manipulations performed during regular aseptic operations (during online filtration, filling, stoppering, and sealing). Interventions that commonly occur would be routinely simulated, while those occurring rarely can be simulated periodically. A grouping logic can be established for the similar nature of interventions, provided they have similar risk factors like duration, proximity, and complexity. Interventions design aspects such as intervention location, tools used, and/or air flow effects may also be considered in establishing a grouping logic.

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Table 15.2-1 Intervention Simulation Frequency-Based on Application of Intervention Risk Evaluation & Management Model

	Line Type – Aseptic Vial Filling Line with open Restricted Access Barrier System placed in Grade B external environment										
No.	Intervention Description	Intervention Category	Type of Intervention	APS Simulation Plan		Risk Categorization after IREM* Evaluation	Existing Risk/Risk Mitigation Actions	Final Risk Categorization after IREM* Reevaluation	Revised APS Plan		
				Total No. of Interventions	Simulation Frequency				Revised No. of Interventions	Revised Simulation Frequency	Rationale for Change
1	Filling machine in-process-control (IPC) calibration	Inherent	Open door	3	Every APS	High Risk	Intervention of filling machine IPC calibration was performed after completion of the machine parts assembly. It is now performed before decontamination procedure and assembly.	N/A	-	N/A	No aseptic process intervention involved, hence, not evaluated in the APS for the same.[1]
2	Group-A Stopper lock adjustment Stopper plate adjustment Improper stoppers for vials	Corrective	Open door	10	Every APS	Very High Risk	Activity was performed with open door due to infeasibility with existing glove port. Additional glove port provided to perform this activity with gloved hand in closed door condition without breaking first air over critical surfaces.	Medium	6	Once per year	Number of interventions is revised from 10 to 6 based on the trend of interventions in the commercial batches. Risk of intervention mitigated.
3	Fill volume adjustment and initial in process checks	Inherent	Closed door	1	Every APS	Low Risk	No human intervention is performed during fill-volume adjustment and initial in-process checks. This activity is done by the operator entering the command on the machine panel.	N/A	-	N/A	No aseptic process intervention involved. Removed from APS list.

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	Line Type – Aseptic Vial Filling Line with open Restricted Access Barrier System placed in Grade B external environment										
No.	Intervention Description	Intervention Category	Type of Intervention	APS Simulation Plan		Risk Categorization after IREM* Evaluation	Existing Risk/Risk Mitigation Actions	Final Risk Categorization after IREM* Reevaluation	Revised APS Plan		
				Total No. of Interventions	Simulation Frequency				Revised No. of Interventions	Revised Simulation Frequency	Rationale for Change
4	Group-B Filling machine setup: Machine parts assembling (stoppering station, gassing station, and filling station)	Inherent	Open door	1	Every APS	High Risk	Machine set up with open door by process design. Risk is recognized as "High" and, hence, shall be simulated in every APS. Risk mitigation action is to adopt new procedure as follows: 1. There is no open product-contact part exposed to the environment during machine setup. 2. All product-contact surfaces are wrapped, and those are opened only after door closure. 3. Operators trained on new procedures.	High Risk	1	Every APS	No change considering filling setup is an integral part of every batch processing (inherent intervention).
5	Vial breakage at in-feed turntable of filling machine	Corrective	Closed door	5	Every APS	Low Risk[2,3]	Risk is accepted. Procedure is in place to remove other empty vials in the turntable. Operators are trained to perform the intervention aseptically.	Low Risk	3	Once per year	Number changed based on the trend. Frequency of simulation changed based on risk level.

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	Line Type – Aseptic Vial Filling Line with open Restricted Access Barrier System placed in Grade B external environment										
No.	Intervention Description	Intervention Category	Type of Intervention	APS Simulation Plan		Risk Categorization after IREM* Evaluation	Existing Risk/Risk Mitigation Actions	Final Risk Categorization after IREM* Reevaluation	Revised APS Plan		
				Total No. of Interventions	Simulation Frequency				Revised No. of Interventions	Revised Simulation Frequency	Rationale for Change
6	Group-C Vial Jam Vial overload/ breakage Sensor-format parts adjustment at in-feed turntable Adjustment of in-feed star wheel of filling machine	Corrective	Open door	5	Every APS	Very High Risk	Procedural mitigation implemented to discard all surrounding exposed vials and sanitization of inner surfaces of Restricted Access Barrier System (RABS) door. Operators trained to perform the intervention aseptically. Controls are in place to detect microbial excursion in the environment during handling of intervention.	Medium Risk	2	Once per year	Number of interventions changed based on trend. Simulation frequency changed based on mitigated risk level.

Note: Other types of interventions that can be evaluated utilizing a similar approach are, for example, stopper-track minimum accumulation, loading of the stopper in the hopper through a rapid transfer port (RTP), an object falling on the in-feed turntable filling machine, and an empty-run machine-run check for vial transfer.

[1] Author's note: This Intervention simulation is eliminated after mitigation actions since this is no longer done as a part of the aseptic process in regular production.

[2] Author's note: The risk is rated low in this particular case. However, the risk level can increase depending on the frequency and extent of vial breakage and the procedure adopted to discard the adjacent empty vials.

[3] Author's note: During the evaluation of this example, it was suggested that the installation of a physical barrier and removal of all vials in the same sector be considered as a mitigation to prevent the risk of glass particles and other contamination from contacting open vials or other product-contact surfaces.

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15.3 Example 3: Use of QRM to Define Inclusion and Frequency of Interventions During Aseptic Process Simulation

This example is related to **Section 4.1.3**. The purpose of this example is to show how the risks associated with each intervention can be assessed and used to define when and how many times to simulate intervention during an APS. The risk of intervention is defined considering its complexity, duration, proximity to product and product-contact surfaces, frequency, results of air-flow visualization, and degree of protection (barrier system used) (**Table 15.3-1**). A score is assigned for each risk factor, and then the overall intervention risk level can be assigned and used to identify frequency of intervention simulation and the number of times it must be simulated during an APS (**Table 15.3-2**) with examples given (**Table 15.3-3**).

Note: The scoring assigned in **Table 15.3-1** is based on specific interventions for a specific line/product. The scoring assignment can vary widely depending on the particulars of the intervention, the line design, and the process/product. The intent is not to provide examples of specific scoring, but rather to show how the tool can be used.

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Table 15.3-1 Use of Intervention Risk Criteria with Scores to Define the Overall Intervention Risk Level (Example 3)

Intervention Risk Criteria	Risk Level (Score)			Assigned Risk Score
	High (5)	Medium (3)	Low (1)	
Complexity	High (e.g., machine setup)	Medium (e.g., removing a jammed rubber stopper)	Low (e.g., removing a toppled vial from conveyor belt)	
Duration	>10 minutes	2-10 minutes	<2 minutes	
Proximity to Product/Contact Surfaces	<15 cm	15-45 cm	>45 cm	
Frequency	>4 times per shift	2-4 times per shift	≤1 time per shift	
Air Flow Visualization (AFV)	Disruption of first air with something that is not in a sterile state (e.g., sleeve of gown); additional controls required	Disruption of first air with a sterilized tool or utensil	No disruption of first air	
Barrier	Conventional Filling Line*	RABS	Isolator	
Overall Intervention Risk Level Total	>20	11-20	<11	

*A conventional aseptic filling line is one without a RABS or isolator barrier. These are legacy systems and are in the process of being phased out of the industry.

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Table 15.3-2 Recommended Frequencies and Number of Interventions for Each Risk Level (Example 3)

Overall Intervention Risk Level	Recommended Frequency of Simulation	Recommended Number of Simulations per APS Run
High	Biannual*	At least the same number as in routine production
Medium	Biannual*	May be less than the number in routine production
Low	Yearly	May be less than the number in routine production
*Biannual: twice a year (approximately every 6 months)		

Table 15.3-3 Risk Assessment of Possible Interventions for Vial-Filling in Restricted Access Barrier Systems without Built-In In-Process Control (Example 3)

No.	Intervention	Complexity	Duration	Proximity to Product/ Contact Surfaces	Frequency	AFV	Barrier	Overall Intervention Risk Level Total
1	Equipment aseptic assembly (open-door intervention)*	5	5	5	1	3	5	24
2	Removal of vials for IPC using glove ports	1	1	3	5	1	3	14
3	Introducing EM plates (open-door intervention)*	1	3	3	1	3	5	16

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No.	Intervention	Complexity	Duration	Proximity to Product/ Contact Surfaces	Frequency	AFV	Barrier	Overall Intervention Risk Level Total
4	Changing EM plates using glove ports	3	1	3	3	1	3	14
5	Change filling needle (open-door intervention)*	5	3	5	1	3	5	22
6	Removing toppled open vial using glove ports	1	1	5	3	1	3	14
7	Removing jammed rubber stopper near stoppering station using glove ports	3	3	5	3	3	3	20
8	Removing toppled closed vial near exit using glove ports	1	1	1	3	1	3	10

*Open-door Interventions should be avoided when possible.

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Explanation of Interventions included in **Table 15.3-3** and **Table 15.4-3:**

- Interventions 1 & 5 can be simulated every 6 months with at least the same number occurring during routine production.
- Interventions 2, 3, 4, 6, & 7 can be simulated every 6 months; however, the number may be less than that occurring during routine production.
- Intervention 8 can be simulated every year, and the number may be less than that occurring during routine production.

Note: Use of a filling line with built-in IPC within an isolator instead of oRABS (thus executing all interventions with closed doors) can lead to:

- Eliminating Intervention 2 from routine process and accordingly from APS.
- Reducing overall risk level of Interventions 1 & 5 from "High" to "Medium". Accordingly, they can be simulated every 6 months; however, the number may be less than that occurring during routine production (if applicable).
- Other interventions will have lower overall risk scores; however, they will remain in the same group of overall risk level. Such risk reduction also has a high value, as adding design controls is the priority risk control in EU Annex 1, followed by procedural controls. APS as a part of the monitoring system is the lowest priority in reducing aseptic processes risks.

15.4 Example 4: Use of Quality Risk Management to Define Inclusion and Frequency of Interventions during Aseptic Process Simulation

This example is related to **Section 3.2** and **Section 7.5.1.1**. It shows how to define the required number of APS for lines where different container–closure sizes with the same configuration and similar process are filled. This applies to the number of APS runs for each container size in initial validation, ongoing verification, and upon introducing a new container–closure size using bracketing and/or matrixing approach (**Table 15.4-1**, **Table 15.4-2**, and **Table 15.4-3**).

This cannot be used for different container–closure configurations, different processes, or when other factors necessitating three (3) APS runs (e.g., different machine parts such as pumps or closure sorters are used and have impact). Apart from container size, other factors that can drive worst-case selection criteria shall be considered in the risk assessment to decide the number and frequency of runs.

Note: The scoring assigned in **Table 15.4-1** is based on specific interventions for a specific line/product. The scoring assignment can vary widely depending on the particulars of the intervention, the line design, and the process/product. The intent is not to provide examples of specific scoring, but rather to show how the tool can be used.

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Table 15.4-1 Use of Intervention Risk Criteria with Scores to Define the Overall Intervention Risk Level (Example 4)

Intervention Risk Criteria	Risk Level (Score)			Assigned Risk Score
	High (5)	Medium (3)	Low (1)	
Complexity	High (e.g., machine setup)	Medium (e.g., removing a jammed rubber stopper)	Low (e.g., removing a toppled vial from conveyor belt)	
Duration	>10 minutes	2-10 minutes	<2 minutes	
Proximity to Product/Contact Surfaces	<15 cm	15-45 cm	>45 cm	
Frequency[1]	>4 times per shift	2-4 times per shift	≤1 time per shift	
Air Flow Visualization (AFV)	Interrupt first air (must be corrected first)	Minor flaws but no ingress of less clean air	No flaws	
Barrier[2]	Conventional Filling Line*	RABS	Isolator	
Overall Intervention Risk Level Total	>20	11-20	<11	

[1] Author's note: Frequency may or may not be a risk factor. For this example, it was chosen to be a factor of consideration.

[2] Authors' note: Using oRABS does not mean that all risks are the same since very few interventions are open-door interventions (e.g., during machine set-up) and they cannot have the same risk level of other interventions. Even if the assumption is that all interventions will be closed door interventions, this risk factor his has to be considered because it will lead to the need for different frequencies of the same intervention when done through oRABs than the frequency needed when conducted using another line with an isolator.

*A conventional aseptic filling line is one without a RABS or isolator barrier. These are legacy systems and are in the process of being phased out of the industry.

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Table 15.4-2 Recommended Frequencies and Number of Interventions for Each Risk Level (Example 4)

Overall Intervention Risk Level	Recommended Frequency of Simulation	Recommended Number of Simulations per APS Run
High	Biannual*	At least the same number as in routine production
Medium	Biannual*	May be less than the number in routine production
Low	Yearly	May be less than the number in routine production
*Biannual: twice a year (approximately every 6 months)		

Table 15.4-3 Risk Assessment of Possible Interventions for a Vial-Filling in Open Restricted Access Barrier Systems without Built-In In-Process Control (Example 4)

No.	Intervention	Complexity	Duration	Proximity to Product/ Contact Surfaces	Frequency	AFV	Barrier	Overall Intervention Risk Level Total
1	Equipment aseptic assembly (open-door intervention)	5	5	5	1	3	5	24
2	Removal of vials for IPC using glove ports	1	1	3	5	1	3	14
3	Introducing EM plates (open-door intervention)	1	3	3	1	3	5	16

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No.	Intervention	Complexity	Duration	Proximity to Product/ Contact Surfaces	Frequency	AFV	Barrier	Overall Intervention Risk Level Total
4	Changing EM plates using glove ports	3	1	3	3	1	3	14
5	Change filling needle (open-door intervention)	5	3	5	1	3	5	22
6	Removing toppled open vial using glove ports	1	1	5	3	1	3	14
7	Removing jammed rubber stopper near stoppering station using glove ports	3	3	5	3	3	3	20
8	Removing toppled closed vial near exit using glove ports	1	1	1	3	1	3	10

Note: See the explanation of Interventions in **Table 15.3-3** for more information.

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15.5 Example 5: Aseptic Process Simulation Design for Lines Producing Single Dosage Form with Different Fill Volume

This example is related to **Section 3.2** and **Section 7.7** and provides guidance regarding lines with the same dosage form with multiple container sizes or fill volumes. In case there is a vial filling line on which 2, 4, 8, and 10 mL vials will be filled and none of them is assessed to be a worst-case, at least three (3) runs from each of minimum and maximum vial sizes should be done, in addition to at least one run from each other vial size (**Table 15.5-1**).

Table 15.5-1 APS Design for Lines Producing Different Products/Dosage Forms (No Worst-Case Vial)

Vial Amount (mL)	Initial Number of Runs
2	3
4	1
8	1
10	3

However, in the event one of these vials sizes (based on a science-based risk assessment; (see **Section 3.4**)) is a worst-case (e.g., 10 mL), at least three (3) runs of this vial size plus at least one run of each of the other vial sizes should be done in the initial validation study (**Table 15.5-2**) (see **Section 7.5.1.1** for additional information on bracketing criteria).

Table 15.5-2 Aseptic Process Simulation Design for Lines Producing Different Products/Dosage Forms (Worst-Case Vial)

Vial Amount (mL)	Initial Number of Runs
2	1
4	1
8	1
10	3

In the examples in both **Table 15.5-1** and **Table 15.5-2**, if a new vial size (e.g., 6 mL) needs to be introduced, it should be assessed as to whether it represents a worst-case condition. If so, at least three (3) runs of the new vial will be required; otherwise, at least one (1) run will be required as it lies between the minimum and maximum vials, each of which had three (3) successful consecutive runs. During ongoing (continued) process verification, rotation of all vial volumes should be represented over a risk-based schedule to cover all of them (e.g., over two years). When risk-based evaluation indicates a worst-case container size, its APS frequency should be higher than other container sizes.

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15.6 Example 6: Aseptic Process Simulation Design for Lines Producing Different Products or Dosage Forms

This example is related to **Section 3.2**. **Table 15.6-1** shows how to assess the number of APS runs needed for a line used to fill different dosage forms and how, in some cases, if supported by a risk assessment, it can be possible to design one APS run that simulates and covers the worst-case scenario for more than one dosage form manufactured on the same line. Understanding the process and associated risks is crucial to ensure that the APS design will represent worst-case scenarios for all dosage forms. In the example, only differences in process steps were considered with other factors fixed. When applying the concept, other factors should also be considered that could impact aseptic process and may necessitate more APS runs.

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Table 15.6-1 Assessing Number of Aseptic Process Simulation Requirements for a Line Used to Fill Different Dosage Forms

Process Step	Dosage Forms Manufactured at Line			Inclusion of APS for Line
	Vials filled with Solution	Vials filled with Suspension	Lyophilized Vials	
Preparation in Grade C	Yes	Yes	Yes	Not an aseptic process step and no need to typically simulate while preparing media
Sterile Filtration	Yes	Yes	Yes	Yes (for aseptic assembly but not necessarily using same filter)
Aseptic Addition	N/A	Yes	N/A	Yes
Homogenization	N/A	Yes	N/A	Yes
Filling	Yes	Yes (may include suspension stirring and recirculation to avoid sedimentation, and may also include automatic rejection of first-filled units after machine stoppage)	Yes	Yes (using worst-case scenarios regarding equipment configuration for suspension dosage forms in terms of suspension stirring and recirculation to buffer the vessel, and without rejecting first-filled units after machine stoppage
Stopper Insertion	Yes (complete insertion)	Yes (complete insertion)	Yes (half insertion of lyophilization stopper)	Yes*
Capping after Stopper Insertion	Yes	Yes	N/A	Yes+

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Process Step	Dosage Forms Manufactured at Line			Inclusion of APS for Line
	Vials filled with Solution	Vials filled with Suspension	Lyophilized Vials	
Lyophilizer Loading	N/A	N/A	Yes	Yes*
Lyophilizer Cycle	N/A	N/A	Yes	Yes (exposed to partial vacuum and vacuum release)*
Lyophilizer Unloading	N/A	N/A	Yes	Yes*
Capping of Lyophilization Vials	N/A	N/A	Yes	Yes*

*It might be possible to utilize the lyophilizer configuration as worst case for the line given that it includes all the steps for liquid fill, plus the lyophilization specific steps. If this approach is taken, it must be fully documented and justified in the APS program (specifically for the potential difference in the container-closure system). Alternatively, both configurations will need to be simulated in the APS program.

†The capping process, whether for liquid or lyophilized configurations should be practically identical, as in both cases fully seated stoppers are being capped.

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The type of media used may also impact the number of runs and frequency of APSs. For example, if there are types of products (e.g., oxygen-sensitive) exposed to anaerobic conditions (less than 0.1% residual oxygen levels) during the entire process (e.g., nitrogen purging of WFI before preparation, nitrogen blanket in vessels, and nitrogen purging before, during, and after filling until the container is sealed) and other products filled under aerobic conditions, three (3) APSs using each type of media may be needed in the initial study, which may be followed by alternating biannual APS runs between two media types. This concept applies even if the two products share the same equipment and the same process steps.

For lines that handle different dosage forms using different container-closure systems (e.g., equipment used to fill different nested primary-packaging materials, such as vials, prefilled syringes, and cartridges), a separate APS will be required for each.

15.7 Example 7: Campaign Strategy Example for an Isolator Line

The following describes a manufacturing campaign that is produced under isolator conditions (**Sections 4.3.3**, **7.11**, **7.12**). Examples will be provided for the initial startup of a completely new isolator line and, upon successful initial validation, subsequent revalidation of the line under routine conditions.

Note: This description is for a particular process, and it is not meant to represent the only way to run a campaign.

Initial New Line Setup with Four (4) Batches in Campaign

In this case, a campaign is defined as four consecutive batches of material filled following established and validated supporting processes.

The high-level overview of the production setup in this example is:

- The filling line is high-speed and high-volume (~600 units/min) filling high volumes (> 200.000 units per batch)
- Decontamination of the isolator takes place prior to the first batch and does not occur again prior to subsequent batches.
- Product delivery is via single-use tubing, installed through an RTP to a filling manifold.
- All wet parts may remain in place, with a product flush and a new product tank connected to the filling manifold between each batch.
- The line will be cleared only in the same manner as for commercial product fills, such as product spills or glass breakage.
- All indirect product contact parts are sterilized and mounted on the line before the decontamination and campaign filling.
- EM of air takes place continuously throughout, culminating in surface sampling at the end of the fourth batch only.

The “campaign-style” APS is repeated three times for initial validation of the aseptic process on the new isolator line. Activities that have been assessed to pose a risk to product sterility are described in **Figure 15.7-1**, including those that may be dependent on duration. Two approaches for the APS of this type of isolator campaign product-fill are schematically presented below:

Example 1A: Initial APS Validation – The Conservative Approach

This approach is recommend for a facility with little to no experience or historical data for production in isolators. In this case, the APS with media is executed for the same duration of time that it takes to execute

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four consecutive product batches, with the same established and validated supporting processes. This means four consecutive media-filled batches. The campaign may be simulated only by the filling of media separated by batch changeovers. The interventions and activities are based on a risk assessment of presumed activities (e.g., from engineering runs, water batches, training sessions) and are fully simulated. This must \be repeated three times for initial validation.

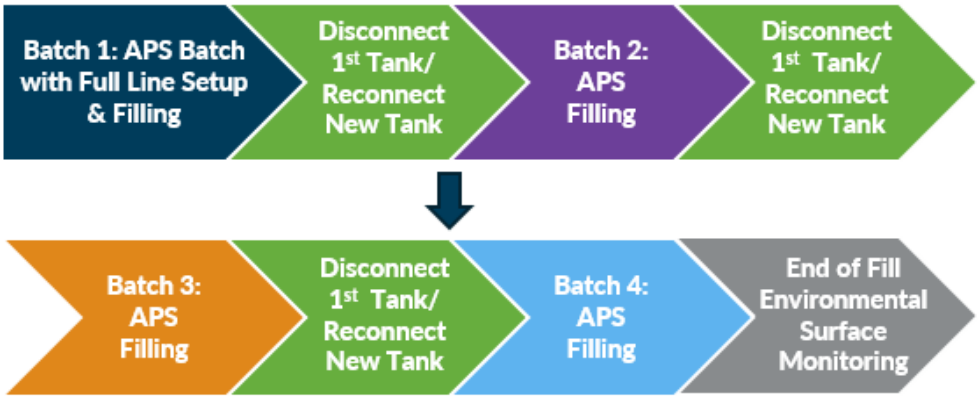

Figure 15.7-1 Initial APS Validation: Conservative Approach

Example 1B: Initial APS Validation – A Risk-Based Approach

The approach illustrated in **Figure 13.4-2** can be applied to a facility with experience in isolator technology and campaign-filling supported by adequate historical data. The campaign may be simulated first by performing a single APS on the isolator line to capture the setup process and the complete filling process. This is followed by another campaign to fill the first four batches of the campaign with product, followed by a media-filled batch. The filling of both media batches includes interventions/activities that are representative worst-case scenarios, based on a risk assessment, in which data from similar lines can be included and assessed (e.g., type/frequency of interventions, EM data). In this way, the number of media-filled units and the time spent on filling media can be reduced. This must be repeated three times for initial validation.

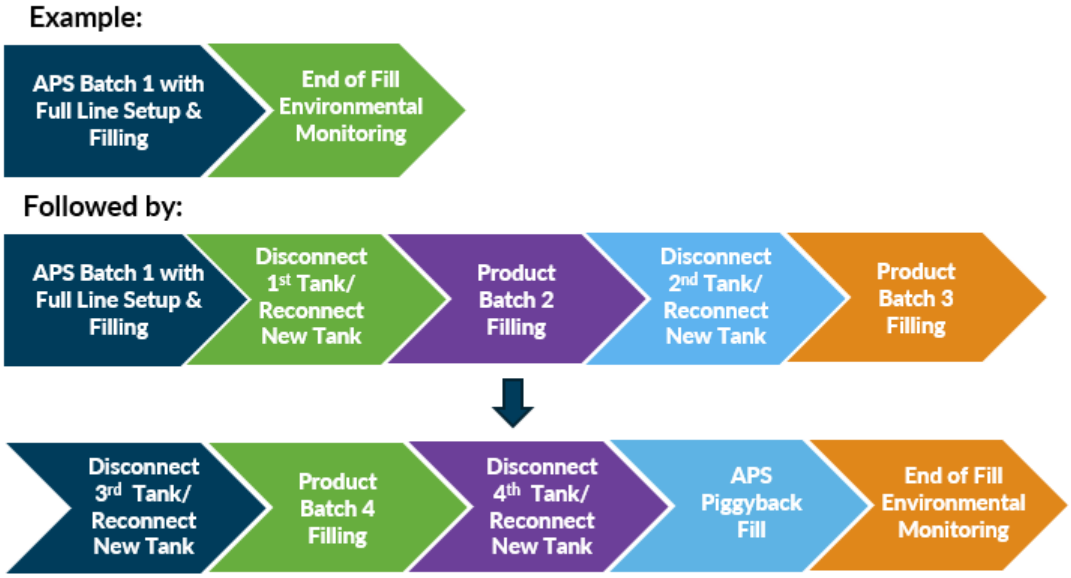

Figure 15.7-2 Initial APS Validation: Risk-Based Approach

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Note: This validation plan does not include reverting to the previous state due to irreversible changes or a completely new line. If the initial APS is triggered by a change, at least one APS for decommissioning the previous design should be completed before the design change. Product batches run in a campaign ending with media fill cannot be released until full validation (three times) is concluded.

Concerns and Considerations

This campaign example presented here is intended for isolator conditions, where the risk for contamination is minimized due to (1) the decontamination process and (2) the barrier system, including exclusive use of glove ports for interventions. The following are aspects to consider in a risk assessment for selecting which approach is acceptable and the appropriate number of batches for the process being evaluated including (but not limited to):

- Size of the batch and speed of the filling technology
- Filling of same product/formulation (unless CIP/SIP is in place, or changeout of the filling path) and/or container size
- Historical data about the process and any process failures (e.g., APS, sterility testing)
- Intervention criticality and (expected) frequency
- Duration of batch and campaign size
- Integration of SUS and disposable tubing when batches are changed over
- Ongoing quality risk management (QRM) of events during the product batches to ensure the quality of the environment is maintained
- Experience of the staff with isolator technology, including the operators' familiarity and training in running isolator lines
- Integrity of the barrier technology, including gloves, as well as integrity-test results/trends, including (but not limited to):
    - Visual inspection of glove ports during each intervention
    - Glove integrity-testing at risk-based intervals, either in between batches or every other batch, depending on the total number of batches in the campaign and based on risk.
    - Visual inspection of RTP canisters and integrity testing of the filter
    - Robustness of docking systems and transfer of product and material through relevant transfer ports
- Maximum time from end of VPHP treatment and end of last fill
- Supporting data from EM performance qualification (EMPQ) and trending from a robust, risk-based routine EM program, including:
    - Viable-and nonviable-particulate monitoring for the entire duration of the fill (to include all batches)
    - End-of-campaign surface samples

Periodic Revalidation and Bracketing Worst-Case Conditions

In this case, a campaign is defined as four batches; an APS is performed as a combination of a single batch, including full setup process and end of the simulation, and a campaign with a media-batch as a piggy-back to the end of four product fills to mimic worst-case conditions with respect to duration. (see **Sections 7.11**, **7.12**) Both APS studies are repeated every six months in the periodic revalidation (see **Section 3.2**), with a total of four APS studies for this line annually. In this way, the "start and end" of the campaign is simulated, and a number of units and the duration of the APS can be kept at a reasonable level. All activities, including

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setups and interventions normally performed, are represented in those two media batches of typically 10,000 units each.

Periodic revalidation will be the same regardless of how the company established their initial validation setup, as described in the initial validation setup of **Example 1A** and **Example 1B**.

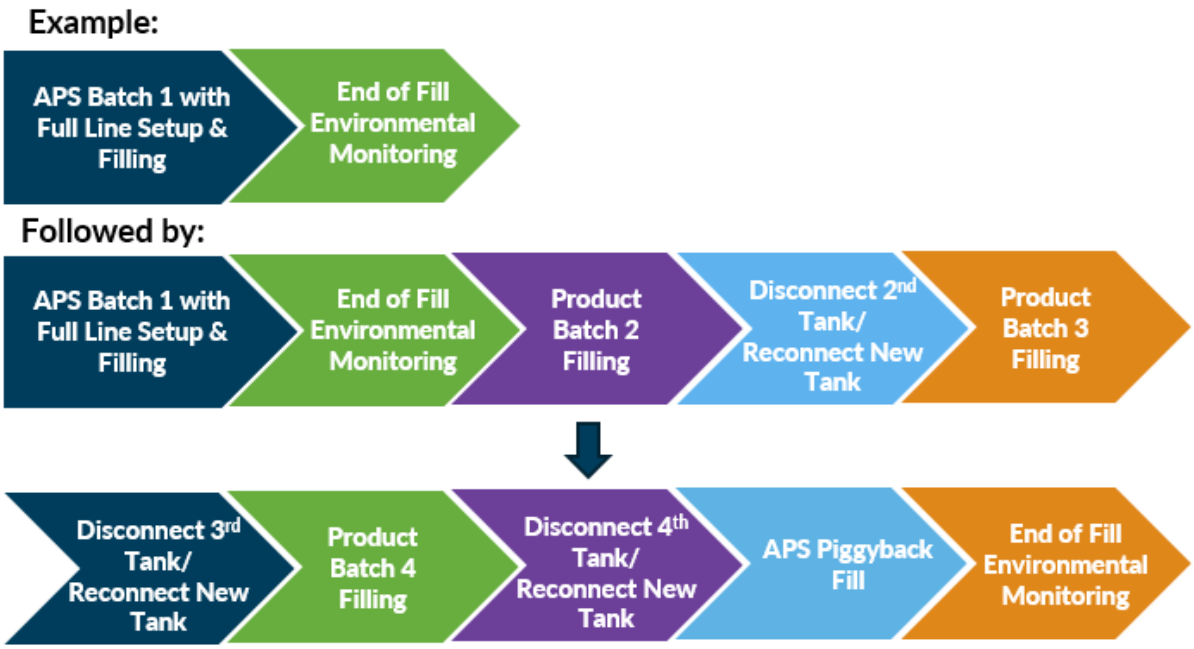

Figure 15.7-3 Periodic APS Revalidation Setup Once the Full-Validation Criteria are Met

Concerns and Considerations for Periodic Revalidation

Typically, a risk assessment can determine the type of activity that may introduce the highest risk (e.g., the setup, which involves many steps, the transfer of materials into the line and through the line by use of the gloves). Corrective interventions done via glove ports and the transfer of items through RTP ports minimize the contamination risk considerably compared to an open RABS. The target size of the APS can vary and will depend on the number of units needed to perform the planned activities, interventions, etc. (**Section 7.10**). Should any of the media-filled campaigns yield a failed batch, then options for returning to a more conservative initial validation must be evaluated.

15.8 Example 8: Design of the Aseptic Process Simulation for an Advanced Therapy Medicinal Product

The scenario presented is for the risk-based approach to the design of an APS for an advanced therapy medicinal product (ATMP) ***(6)*** (see **Sections 4.1.2** and **4.3.7**). The product in question is an autologous gene-modified cellular therapy named "GMCT-A," a product similar to many chimeric antigen receptor t-cell (CAR-T) products as well as other cellular therapies including T-cell receptors, tumor-infiltrating lymphocytes (TILs), and regulatory T-cells. An APS should be conducted once an acceptable manufacturing process and corresponding interventions have been designed and established. In this case, the manufacturing process for GMCT-A is being transferred from the manufacturing site where early clinical trial material was

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produced ("sending site") to a cGMP-compliant manufacturing facility ("receiving site") for the purposes of later-stage clinical and potential commercial drug product manufacturing.

GMCT-A is manufactured using a process performed predominantly using aseptic processing and sterility assurance is of the utmost importance since the drug product is administered to patients by infusion. A process flow diagram of the major unit operations of the manufacturing process is shown in **Figure 15.7-4**.

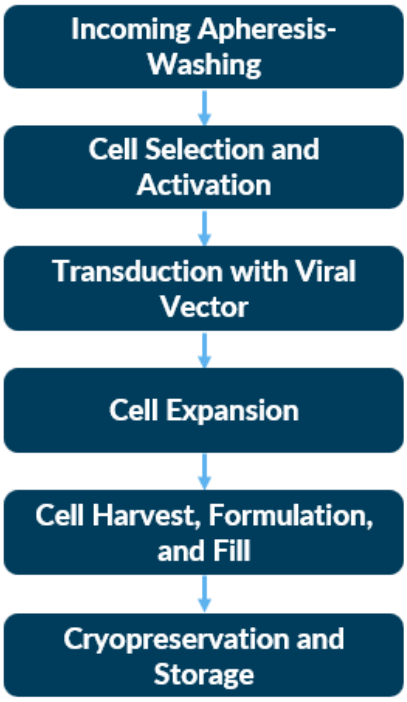

Figure 15.7-4 Process Flow Diagram of the Major Unit Operations of the Manufacturing Process

Risk Assessment of the Aseptic Process

As with any quality risk assessment, per ICH *Quality Guideline Q9(R1): Quality Risk Management*, the methodology begins at the risk initiation stage of the lifecycle by first defining the risk question. In this case, the question can be captured as simply: "What is the relative risk of contamination from each individual manipulation performed within the aseptic boundary during the manufacture of GMCT-A?"

In this case, a modified version of the IREM was applied for the assessment. The modified version is named the Aseptic Risk Evaluation Model (AREM) and has been optimized to apply to the many unique aseptic processes encountered with ATMPs (beyond fill and finish operations). The AREM, like the IREM, leverages collective product knowledge and process understanding, is compliant with the evolving global regulatory landscape, and is based on industry best practices. The steps of the AREM are detailed in **Figure 15.7-5**.

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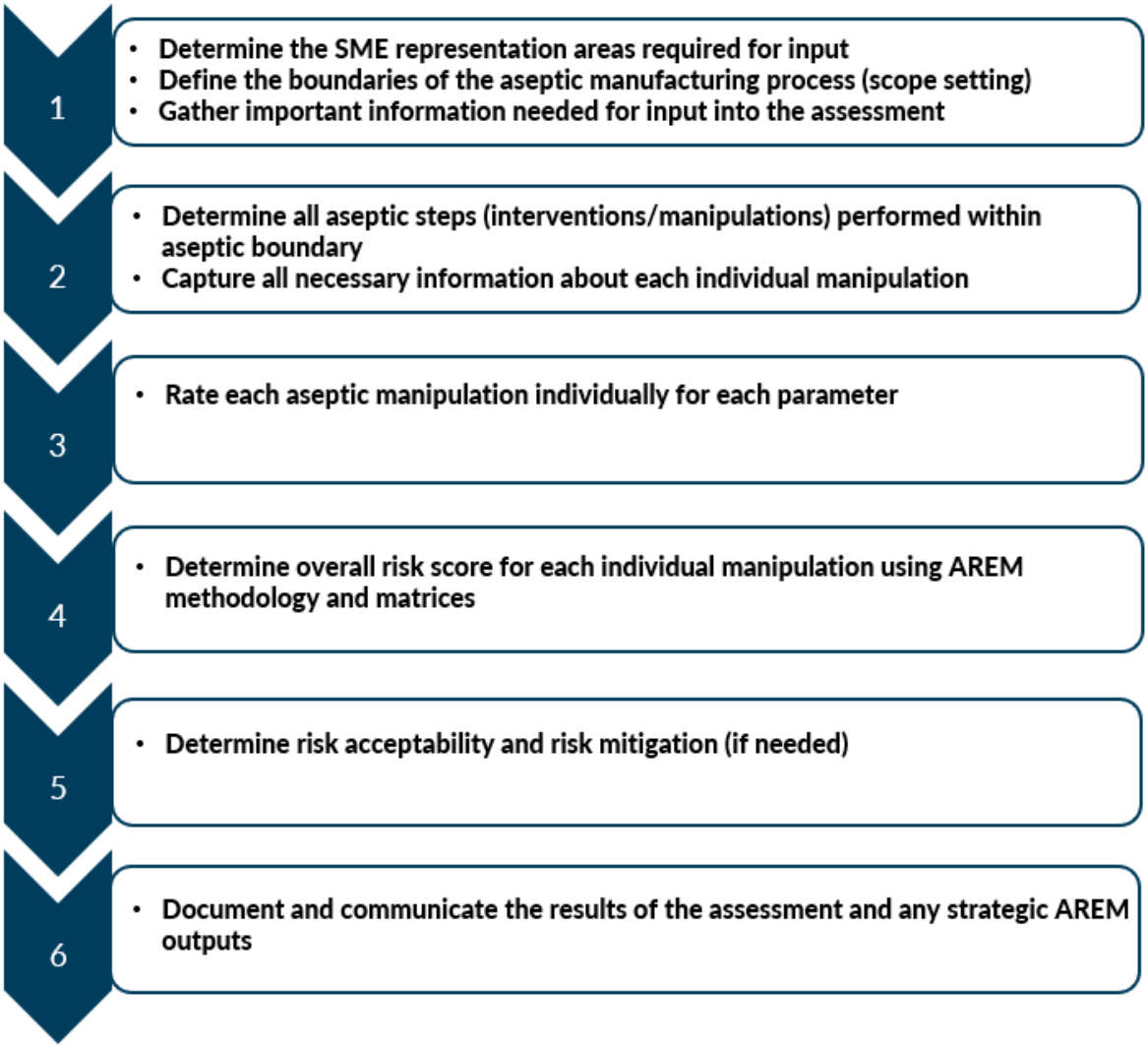

Figure 15.7-5 Aseptic Risk Evaluation Model – Steps in the Risk Assessment Process

Step 1

The "AREM Team" was assembled including all relevant subject matter experts from Manufacturing/Production, Manufacturing Science & Technology, Process Development, Quality Control, and Quality Assurance. Additionally, all relevant knowledge management documentation was gathered (e.g., process description, CCS, production batch records, and SOPs).

Step 2

The scope of the AREM was determined by defining the boundaries of the aseptic process. All unit operations with aseptic processing were defined and then the individual aseptic steps within each unit operation were identified. The AREM team should perform this task by reviewing each batch record (if available) step by step to determine each individual aseptic manipulation performed.

Step 3

Each of the individual aseptic steps (i.e., manipulations) is rated based on three factors - Complexity, Duration, and Proximity - as defined in **Table 15.7-1**.

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Table 15.7-1 Definition of the Aseptic Manipulation Factors

Factor	Definition
Duration	The time that it takes to perform the overall aseptic procedure when the product or product-contact surfaces are exposed to the surrounding environment. The typical or standard times for these aseptic procedures for the processes evaluated was determined to be within a range of 1–10 minutes. Any aseptic procedure less than 1 minute was considered as lower in terms of relative risk, while any aseptic procedure greater than 10 minutes was considered of greater relative risk.
Complexity	The more aseptic process steps within the procedure that are performed, the more complex the overall risk is. The aseptic procedure starts with the initial exposure of the product or a product-contact surface and ends with the final closure (or closing) of the product within the container. Any procedure that impacts or breaks the sterile boundary or any direct product-contact surface is considered for assessment in the AREM.
Proximity	Proximity is determined in relation to the operation with respect to first air within the environment (e.g., biosafety cabinet). For any aseptic procedure, it is first determined whether first air is broken or not. In the case that it is, then whether first air is broken with a sterile entity or with a nonsterile entity is determined.

The relative ratings for each of these three factors, based on the input from this AREM Team, were defined as shown in **Table 15.7-2**.

Table 15.7-2 Identification of the Risk Ranking for Each of the Factors

Risk Ranking	Duration	Complexity	Proximity
High	>10 minutes	More than 5 steps; highly difficult to perform manual manipulations	Operator breaks first air with a nonsterile entity (e.g., the sanitized isolator glove)
Medium	1-10 minutes	2–5 steps; moderate difficulty to perform manual manipulations	Operator breaks first air with a sterile entity (e.g., the sterilized forceps)
Low	<1 minute	1 step; easy to perform manual manipulation	Operator does not break first air

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Step 4

Upon determining the risk-ranking of each factor for an aseptic procedure, a preliminary matrix was used to determine the initial Risk Class, as shown in **Table 15.7-3**.

Table 15.7-3 Preliminary Matrix Used in Determination of Initial Risk Class

	Duration		
	Low	Medium	High
High Complexity	Risk Class 2	Risk Class 2	Risk Class 1
Medium Complexity	Risk Class 3	Risk Class 2	Risk Class 1
Low Complexity	Risk Class 3	Risk Class 3	Risk Class 2

A secondary matrix is then used to determine the overall risk score using the previously determined Risk Class and the proximity, as shown in **Table 15.7-4**.

Table 15.7-4 Secondary Matrix for Determination of the Overall Risk Score

	Proximity		
	Low	Medium	High
Risk Class 1	Low	Medium	High
Risk Class 2	Low	Medium	High
Risk Class 3	Low	Low	Medium

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Step 5

Upon the determination of the overall risk score, the aseptic procedures are ranked by risk, and the following actions are to be taken, as shown in **Table 15.7-5**.

Table 15.7-5 Action Level Table based on Overall Risk Score

Risk Ranking	Action
High	Unacceptable level of risk. A high risk should be reduced through the reduction of one or more risk factors (proximity, duration, and/or complexity) taking into account the effect of residual risk or unintended consequences of action.
Medium	Actions should be taken to reduce the risk to a lower level through the reduction of one or more risk factors, taking into account the effect of residual risk or unintended consequences of action. If risk cannot be reduced practically, it may still be accepted; however, ways to reduce the risk should be considered as a part of periodic risk review.
Low	Acceptable risk. No additional action necessary.

Step 6

The outputs from the risk assessment were documented by a risk assessment facilitator. An example of the AREM output from one of the unit operations of the GMCT-A manufacturing process, the transduction of the target cells (recovered post-selection and activation) using a viral vector for purposes of gene transfer is provided in **Table 15.7-6**.

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Table 15.7-6 Aseptic Risk Evaluation Method for the GMCT-A Process (Example: Unit Operation of Viral Vector Transduction)

Description of Aseptic Steps/ Manipulations	Complexity	Duration	Risk Class	Proximity	Overall Risk	Mitigation/ Actions Taken
De-crimp multiple vials of the viral vector and remove the cap of each. Transfer contents of each vial to a single (pooled) conical tube using a pipet.	High	Medium	Risk Class 1	Medium	High	Alternate supply of the viral vector filled in a container that can be aseptically connected to the cell suspension (to be assessed for future use). Additionally, a closed, automated transduction process (to be evaluated as a potential long-term optimization). Risk is accepted at this time based on a Risk-Benefit evaluation approved by the Quality Unit.
	Each transfer constitutes an aseptic step with up to ten (10) vials of viral vector to be used	Up to (but no more than) ten (10) minutes to pool maximum number of vials of viral vector	Risk Class 1	Operator breaks first air with a sterile entity (e.g., the sterilized forceps)		

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Description of Aseptic Steps/ Manipulations	Complexity	Duration	Risk Class	Proximity	Overall Risk	Mitigation/ Actions Taken
Gently swirl conical tube with pooled viral vector solution. Use a syringe and needle to collect pooled viral vector solution. Remove needle from syringe and then connect syringe containing the viral vector solution to the product bag with the target-cell suspension (prior to vector addition) using the Luer-lock connector. Dispense pooled viral vector into product bag.	Medium Three (3) aseptic steps are performed	Medium 3–5 minutes based on simulation performed by multiple operators	Risk Class 2 Risk Class 2	Medium Operator breaks first air with a sterile entity (e.g., the sterilized forceps)	Medium	The mitigation noted above is also applicable to this step. Risk is accepted.
Gently mix the product bag now containing target cells and pooled viral vector. Remove a sample (for quality control testing) using a syringe via the Luer-lock connector.	Low A single (1) aseptic step to connect the syringe via Luer-lock connector	Low Less than one (1) minute for aseptic step to be performed	Risk Class 3 Risk Class 3	Low Operator does not break first air	Low	Risk is accepted. This is a common aseptic step performed throughout the manufacturing process for the purpose of obtaining a product sample.

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Design of the Aseptic Process Simulation Based on Outputs of the Risk Assessment

The outputs and learning from the AREM for the GMCT-A manufacturing process are then used to develop and focus on proper process design and on the training for the aseptic personnel/operators who will be responsible for performing and supporting this manufacturing process.

Based on the relative risk of the aseptic process identified for this unit operation, per the AREM, these aseptic steps will focus on the proper performance of higher-risk activities and the effectiveness of the training of aseptic operators. In addition, successful demonstration of these aseptic steps is required for the qualification of the operators prior to their participation in cGMP manufacturing.

APS batch records are constructed based on the output of the AREM. As part of the successful technology transfer of GMCT-A, three consecutive APS batches that meet all acceptance criteria are to be completed (and documented within an APS protocol).

Additional considerations in the design of the simulation for this unit operation include:

- The boundaries of the aseptic process are defined from the initial introduction of the patient-specific leukapheresis (starting material) in a single-use bag, through the subsequent unit operations, until the final fill and closure of the product container representing the drug product, as shown in the process flow diagram (**Figure 15.7-4**).
- Each of the unit operations have been divided into individual unit operations for the purposes of the APS. The viral vector transduction unit operation described here is performed as one subset of the overall APS.
- The inputs to the viral vector transduction unit operation are simulated using an appropriately qualified microbiological growth medium (MGM) supplied in both cryovials (similar to the viral vector containers) and single-use bags (similar to the target cell suspension bags). MGM volumes have been selected to be representative of the manufacturing process and sufficient to cover all internal (product-contact) surfaces of the containers. The maximum number of viral vector containers will be simulated.
- The interventions associated with the handling of these containers has been identified as worst-case conditions through the AREM analysis. Higher-risk activities are simulated in the APS to identify any unresolved weakness or variability in the process. Failure of the APS attributed (through investigation) to these activities would require process modification/improvement or steps to further ensure proper performance of the process.
- The outputs to the viral vector transduction unit operation, which include the single-use bag containing the simulated transduced cell suspension (post-vector addition) and the quality-control sample removed for microbiological evaluation, are collected post-simulation. These outputs are incubated and must pass acceptance criterion of “no growth” for successful simulation. No waste solution was identified for this unit operation.
- Upon review of process-related deviations reported from the manufacturing history (obtained from the sending site of the process as part of the technology transfer), a corrective action where the viral vector is spilled within the BSC and the cleanup procedure (per SOP) is performed, has been included in this unit operation.

More detailed information on aseptic processing risk assessments is available in PDA *Technical Report No. 44: Quality Risk Management for Aseptic Processing*, PDA *Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations*, and the PDA *ANSI/PDA Standard 03-2025: Standard Practice for Quality Risk Management of Aseptic Processes*.

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16.0 Appendices' References

1. European Commission. *Annex 1: Manufacture of Sterile Medicinal Products, EudraLex – Volume 4 – EU Guidelines for Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use*; European Commission: Brussels, 2022.
2. Baseman, H, et al. *A Line of Sight Approach for Assessing Aseptic Processing Risk: Part I*. PDA Letter 2016.
3. Baseman, H, et al. *A Line of Sight Approach for Assessing Aseptic Processing Risk: Part II*. PDA Letter 2016.
4. Baseman, H, et al. *Interventions Risk Evaluation and Management in Aseptic Manufacturing - A Case Study*. PDA J Pharm Sci Technol 2022, 76 (6), 485-96.
5. Baseman, H; Chakraborty, S. *Interventions Risk Evaluation and Management in Aseptic Manufacturing - Part 2*. PDA J Pharm Sci and Tech 2024, (3).
6. Baker, T; Pillsbury, D. A Novel Approach to Managing Risks in Aseptic Processing of Cell and Gene Therapies. Cell & Gene 2024.

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